Regime-Aware Tactical Asset Allocation for Equities and Fixed Income: Independent and Coupled Jump-Model Labels

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Abstract

We study whether forecasted equity and bond return regimes improve monthly allocation across equities, bonds, and cash after transaction costs. The empirical design estimates realised regimes, forecasts next-month regimes, and maps forecast probabilities into portfolio weights. Unsupervised jump-model teachers estimate equity and bond return environments from realised return-based features. Student models are supervised only by these teacher labels and forecast next-month regimes from lagged macro-financial predictors; the COUPLED student also uses current-regime indicators and regime run length. The resulting forecast probabilities are converted into regime-conditional mean–variance allocations with transaction costs and turnover penalties. We compare two regime representations. The INDEPENDENT specification estimates separate two-state favourable and unfavourable labels for equities and bonds. The COUPLED specification estimates a three-state joint stock–bond label with calm, normal, and stressed regimes. Both representations distinguish the central portfolio mechanism: bonds support equity drawdowns in disinflationary episodes such as the dot-com bust and the global financial crisis, while inflation-driven sell-offs such as 1994 and 2022 weaken the bond hedge. In the student stage, regularised logistic regression and random forests produce informative next-month regime probabilities. In the allocation stage, selected dynamic strategies improve on the 60/40 benchmark along distinct margins, including lower drawdowns, higher Sharpe ratios, and higher terminal wealth, depending on risk aversion and turnover control.

1 Introduction

Balanced portfolios depend on how equity and bond returns interact across market environments. In disinflationary equity sell-offs, bond returns often rise and reduce portfolio losses. In inflation-driven sell-offs, equities and bonds can both lose value. A tactical allocation rule therefore needs information about the prevailing stock–bond environment when choosing next-month equity, bond, and cash exposures.

This paper studies monthly regime-aware allocation for a broad equity leg and a broad fixed-income leg. The empirical objects are asset-class return environments. The equity leg represents the stock component of a balanced portfolio, and the bond leg represents the fixed-income component. The allocation problem is whether information available at the end of month t supports a change in next-month exposure after transaction costs and turnover penalties.

The empirical design has three stages. First, unsupervised jump-model teachers estimate realised regime labels from equity and bond returns without external regime annotations. Second, student models forecast the next-month teacher labels from lagged macro-financial predictors; the COUPLED student also uses current-regime indicators and regime run length.

The student stage is supervised only by the teacher labels. Third, the allocation stage maps the forecast probabilities into regime-conditional return moments and monthly equity, bond, and cash weights. The out-of-sample test evaluates the economic value of those forecasts after transaction costs.

We compare two regime-label structures. The INDEPENDENT approach estimates separate binary favourable-state labels for the equity and bond legs. The COUPLED approach estimates one joint stock–bond label with calm, normal, and stressed states. The independent structure gives asset-level signals; the coupled structure gives a direct signal for the joint diversification environment.

The rest of the paper follows the same sequence. Section 2 describes the return series and predictors. Section 3 reports the realised regime labels. Section 4 reports next-month forecast performance. Section 5 evaluates the allocation results. The appendices report formulas, grids, diagnostics, and sensitivity checks.

2 Data

The empirical analysis uses monthly excess returns for the equity and bond legs; cash is the zero-excess-return asset in the allocation stage. The equity leg is the S&P 500 total return. The bond leg is an aggregate bond-index return. The one-month risk-free rate defines excess returns,

$$r_t^{\text{eq}} = \text{equity return}_t - rf_t, \qquad r_t^{\text{bd}} = \text{bond return}_t - rf_t.$$

Teacher features. The INDEPENDENT equity teacher uses daily equity exponentially weighted returns and Sortino ratios over 21, 63, 126, and 252 trading days. The INDEPENDENT bond teacher uses monthly bond exponentially weighted returns, downside risk, Sharpe ratios, and volatility over one- to twelve-month horizons. The COUPLED teacher uses monthly equity and bond Sortino, drawdown, and trend features. Appendix A reports the formulas, horizons, and transformations.

Student panel. The student panels are monthly and lagged. The INDEPENDENT panel combines equity, Treasury, credit, commodity, volatility, macro, sentiment, and stock–bond correlation predictors, with causal winsorisation, rolling z-scores, and exponential smoothing. The COUPLED panel uses lagged macro-financial, Treasury, commodity, and regime-context predictors for the joint stock–bond label, and also includes current-regime indicators and regime run length. Appendices A and G report the predictor definitions and transformations.

3 Teacher: regime labelling with statistical jump models

The teacher stage defines the realised regimes that the student models later forecast. The jump-model teachers are unsupervised: they estimate regimes from return-based features without manual or external regime labels. A regime label must be persistent and must separate different return outcomes. Persistence measures whether a state represents a sustained environment. State-conditional returns show whether the states distinguish favourable from adverse outcomes. Crisis-window diagnostics compare the labels with known stock–bond episodes.

3.1 The jump model

A jump model assigns each observation to a finite set of states while penalising frequent state changes. For a standardised feature vector x_t, state s_t, and state centres $\{c_k\}$, the state sequence solves

$$\min_{s_1,\dots,s_T,\,\{c_k\}} \sum_{t=1}^{T} d(x_t, c_{s_t}) + \lambda \sum_{t=2}^{T} \Lambda(s_{t-1}, s_t), \tag{1}$$

where $d(\cdot, c_k)$ measures distance from state centre c_k, Λ penalises switches, and λ controls persistence. Higher values of λ produce longer spells. Given the centres, the state sequence is obtained by dynamic programming. Given the state sequence, the centres are updated. These two steps are repeated from several initialisations.

3.2 Two regime parameterisations

Both regime specifications use the jump-model penalty, but they define different state spaces. Both teachers are refit causally. For a label assigned to month m, the fitted teacher uses only observations dated at or before m; no future returns or features enter the teacher label for month m.

independent: two independent binary labels. The INDEPENDENT teacher estimates separate two-state jump models for the equity and bond legs. For asset $a \in \{\text{eq}, \text{bd}\}$, feature vector x_i^a, and teacher member p, the state sequence solves

$$\min_{s_1^a,\dots,s_{T_a}^a,\,\{c_0^a,c_1^a\}} \sum_{i=1}^{T_a} d_p\left(x_i^a, c_{s_i^a}^a\right) + \lambda_p \sum_{i=2}^{T_a} \mathbf{1}\{s_i^a \neq s_{i-1}^a\}. \tag{2}$$

The equity teacher uses daily observations. The bond teacher uses monthly observations. The equity model starts after a 12-year daily training window and is updated at month ends. The bond model starts after a 144-month training window and is updated monthly. At each update, the available training features are standardised using the training sample available at that date.

The teacher ensemble combines L^2 and L^1 jump-model members across persistence penalties. The L^2 members use squared Euclidean loss and mean centres; the L^1 members use absolute loss and medoid centres. Appendix B reports the persistence grids and ensemble construction.

After each fit, states are ordered by realised in-sample Sharpe ratio. Let $g^{a,p}$ denote the higher-Sharpe state for asset a and teacher member p. This state is labelled favourable. The monthly label from member p is

$$S_m^{\text{eq},p} = \mathbf{1}\left\{\frac{1}{n_m}\sum_{i\in m}\mathbf{1}\{s_i^{\text{eq},p} = g^{\text{eq},p}\} \geq \frac{1}{2}\right\}, \qquad S_m^{\text{bd},p} = \mathbf{1}\{s_m^{\text{bd},p} = g^{\text{bd},p}\}. \tag{3}$$

The equity label is the majority daily state within the month. The bond label is the monthly state itself.

coupled: three joint regimes with a coupling probability. The COUPLED approach represents the stock–bond environment with three joint regimes. Each leg has a benign or adverse bit, and the admissible joint states are

$$\mathcal{S} = \{\,\text{calm} = (0,0), \quad \text{normal} = (1,0), \quad \text{stressed} = (1,1)\,\},$$

where the pair is the equity bit and the bond bit. Calm means that both legs are benign, normal means that equity is adverse while bonds remain benign, and stressed means that both legs are adverse. The benign-equity/adverse-bond state is excluded in this specification.

For a joint state j with bits (j_E, j_B), the per-month cost combines the two leg-specific distances with a coupling term:

$$e_t(j) = d\big(z_t^{\text{eq}}, \mu_{j_E}^{\text{eq}}\big) + d\big(z_t^{\text{bd}}, \mu_{j_B}^{\text{bd}}\big) + \eta\Big[\,\kappa_t\,\mathbf{1}\{j \in D\} + (1-\kappa_t)\,\mathbf{1}\{j \in A\}\,\Big], \tag{4}$$

where A denotes the agree states, calm and stressed, and D denotes the disagree state, normal. The scalar η controls the strength of the coupling term. The variable $\kappa_t \in [0,1]$ is a learned coupling probability: low values favour the disagree state, while high values favour the agree states. Within each monthly refit, given the provisional state path, κ_t is fitted by an L^2-penalised logistic regression of the leg-agreement indicator $y_t = \mathbf{1}\{s_t^{\text{eq}} = s_t^{\text{bd}}\}$ on equity–bond correlation, equity and bond trend, and equity and bond drawdown:

$$\kappa_t = \sigma\big(\beta_0 + \beta^\top x_t^{\text{mkt}}\big)\,, \qquad x_t^{\text{mkt}} = \big\{\rho_{12,t}^{\text{eq,bd}},\ \text{trend}_t^{\text{eq}},\ \text{trend}_t^{\text{bd}},\ \text{dd}_t^{\text{eq}},\ \text{dd}_t^{\text{bd}}\big\}. \tag{5}$$

The coupling regression is estimated on the expanding history available through the labelled month, so the fitted κ_t uses only market features dated at or before that month. The transition penalty is applied separately to the equity and bond bits, $\Lambda(i,j) = \lambda\,\mathbf{1}\{i_E \neq j_E\} + \lambda\,\mathbf{1}\{i_B \neq j_B\}$. The coupling term acts as a soft bias in the state cost. It raises the cost of states that conflict with the fitted stock–bond dependence signal and lowers the cost of states supported by that signal.

Relationship. The two parameterisations define different regime objects. The INDEPENDENT labels keep equity and bond regimes separate, so the final state can take any of the four favourable–unfavourable combinations. The COUPLED label is a single joint state. Calm corresponds to favourable equity and favourable bonds, normal to unfavourable equity and favourable bonds, and stressed to unfavourable equity and unfavourable bonds. The INDEPENDENT representation gives two asset-level regime signals. The COUPLED representation gives one regime signal for the joint stock–bond environment.

3.3 Ensembling and voting

The INDEPENDENT teacher combines fitted paths separately for equities and bonds by majority vote across teacher members. The COUPLED teacher combines 22 joint-regime members, formed from feature families crossed with the L^1 and L^2 specifications. The resulting vote shares measure member agreement. Appendix B and Appendix D report the vote formulas, transition matrices, and member-level diagnostics.

3.4 Label results

The two teacher specifications produce persistent regimes with economically distinct return profiles. Figure 1 compares the resulting label histories. The INDEPENDENT panel shows equity and bond excess-return indexes with favourable-regime months marked for each leg. The COUPLED panel shows the same stock–bond history using the calm, normal, and stressed joint regimes. The coupled ensemble vote shares and coupling-probability path are reported in Appendix D.

Figure 1: Causal teacher-label representations of the stock–bond history. Both panels report labels formed using information available through the labelled month. Left (INDEPENDENT): equity and bond excess-return indexes with favourable-regime months marked for each leg. Right (COUPLED): cumulative excess-return indexes shaded by calm, normal, and stressed joint regimes.

State economics. Table 1 reports label-conditional return statistics. In the INDEPENDENT panel, favourable states have positive Sharpe ratios and unfavourable states have negative Sharpe ratios for both equity and bonds. In the COUPLED panel, calm is the strongest equity state, normal combines positive equity returns with bond support, and stressed records losses in both risky legs.

Table 1: Label-conditional economics. The table reports same-month annualised excess return, annualised volatility, Sharpe ratio, sample share, and average spell duration. The INDEPENDENT panel conditions on each leg's binary label. The COUPLED panel conditions on the three joint regimes. Panel headers report the causal label sample used for each representation.

independent: two binary labels (1989–2024)						
Class	Regime	Share	Ann. ret.	Ann. vol.	Sharpe	Avg. dur.
Equity	Favourable	76.3%	+13.8%	12.1%	+1.14	14.3
Equity	Unfavourable	23.7%	−15.4%	19.8%	−0.78	4.6
Bond	Favourable	67.1%	+5.7%	3.2%	+1.81	13.8
Bond	Unfavourable	32.9%	−4.1%	5.1%	−0.81	6.8
coupled: three joint regimes (1988–2024)						
Class	Regime	Share	Ann. ret.	Ann. vol.	Sharpe	Avg. dur.
Equity	Calm	13.8%	+21.0%	8.1%	+2.60	4.7
Equity	Normal	77.1%	+9.9%	14.8%	+0.67	12.2
Equity	Stressed	9.0%	−40.9%	13.7%	−2.99	2.2
Bond	Calm	13.8%	+0.8%	3.5%	+0.22	4.7
Bond	Normal	77.1%	+3.9%	3.9%	+1.00	12.2
Bond	Stressed	9.0%	−8.4%	5.0%	−1.69	2.2

Persistence. Persistent labels make the forecast horizon economically usable: a one-month-ahead signal has allocation value only if regimes last long enough for portfolio weights to adjust without being dominated by turnover costs. Table 1 reports average spell durations, and Appendices B and D report the transition matrices and switch-count diagnostics.

Crisis windows. Table 2 compares the labels across historical stress windows. The comparison separates equity drawdowns with bond support from episodes in which equity and bond returns fall together.

Table 2: Crisis-window behaviour under both labellings. The INDEPENDENT columns report the bond label's favourable share and Sharpe ratio. The COUPLED columns report the joint regime mix within each window and the equity and bond Sharpe ratios. Sharpe ratios use same-month annualised excess returns over each window.

	INDEPENDENT bond		COUPLED regime share			COUPLED Sharpe	
Window	Fav.	Sharpe	Calm	Normal	Stress.	Eq.	Bd.
1990 recession	100.0%	+1.33	0%	100%	0%	+0.05	+1.41
1994 bond shock	8.3%	−1.55	8%	33%	58%	−0.48	−1.61
Dot-com bust	87.5%	+1.81	0%	97%	3%	−1.07	+1.97
Global financial crisis	72.2%	+0.74	0%	94%	6%	−1.58	+0.74
Euro sovereign 2011	100.0%	+3.46	12%	88%	0%	−0.61	+3.69
Covid crash	100.0%	+2.25	0%	100%	0%	−0.88	+2.76
Inflation 2022	0.0%	−1.82	0%	33%	67%	−0.93	−1.90

The crisis evidence links the regime labels to the allocation question. Disinflationary equity sell-offs favour bond exposure, while inflation-led sell-offs call for lower exposure to both risky legs.

Feature separation. Feature-separation diagnostics identify the inputs that define the teacher states. The INDEPENDENT equity labels load most on medium-horizon Sortino ratios and return trends; the bond labels load most on Sharpe ratios and downside-risk measures. In the COUPLED teacher, equity trend distinguishes calm months, and bond trend and bond Sortino measures distinguish stressed months. Full rankings are reported in Appendices B and D.

4 Student: forecasting the next-month regime

The student stage turns realised teacher labels into out-of-sample forecast probabilities. At the end of month t, the predictors X_t are observed and the target is the teacher label in month $t+1$. The INDEPENDENT student estimates two binary favourable-state probabilities, one for equity and one for bonds:

$$\hat{p}^a_{t+1|t} = \Pr\big(S^a_{t+1} = 1 \mid X_t\big)\,, \qquad a \in \{\text{eq}, \text{bd}\}. \tag{6}$$

The COUPLED student estimates the corresponding three-class probability vector over calm, normal, and stressed joint regimes. The student models use regularised logistic regression and random forests; Appendix E and Appendix G report the predictor designs, validation protocol, model grids, and full forecast diagnostics.

Forecast quality. Table 3 reports out-of-sample forecast performance. The INDEPENDENT forecasts are evaluated as two binary favourable-state predictions. The COUPLED forecast is evaluated as a three-class probability forecast over the joint stock–bond regimes. AUC values are therefore compared within panels, not as direct cross-panel levels: the INDEPENDENT tasks are binary per-leg forecasts, while the COUPLED task is a three-class joint-regime forecast. The allocation section tests whether these probability forecasts improve net portfolio outcomes.

Table 3: Out-of-sample student forecast quality. The INDEPENDENT panel reports binary diagnostics for the equity and bond favourable-state forecasts. The full independent predictor-design results are in Table F.20. The COUPLED panel reports diagnostics for the joint three-class forecast; AUC is the macro one-vs-rest average, and Brier and log loss are reported in their multiclass forms.

independent: per-leg binary forecast
OOS 1996 to 2024

Label	Model	AUC	Bal. acc.	Brier	Log loss
Equity	Logistic regression	0.929	0.846	0.088	0.296
Equity	Random forest	0.903	0.831	0.107	0.361
Bond	Logistic regression	0.892	0.831	0.128	0.462
Bond	Random forest	0.873	0.800	0.147	0.471

coupled: three-class forecast
OOS 1996 to 2024

Model	AUC	Bal. acc.	Brier	Log loss
Logistic regression	0.770	0.677	0.541	1.062
Random forest	0.776	0.595	0.383	0.686

The INDEPENDENT results favour logistic regression on the headline binary metrics. The COUPLED results show a trade-off between class accuracy and probabilistic scoring. Full predictor-design comparisons and probability-path diagnostics are reported in Appendices F and G.

Predictor relevance. Predictor-relevance diagnostics are reported in Appendices F and G. Equity-regime forecasts rely more on equity cross-section, industry, and style information. Bond-regime forecasts use a broader mix of equity, Treasury, and credit information. The COUPLED forecast draws on a more dispersed set of regime-context, commodity, Treasury, and macro-financial blocks, consistent with a joint stock–bond target.

5 Regime-aware allocation

The allocation stage tests whether forecast probabilities improve monthly equity–bond–cash choice after transaction costs. At the end of month t, the student forecasts are mapped into next-month conditional return moments, and realised portfolio returns are evaluated net of one-way transaction costs against a monthly rebalanced 60/40 benchmark.

For the INDEPENDENT allocator, $\hat{p}^a_{t+1|t}$ is the end-of-month-t probability that asset $a \in \{\mathrm{eq}, \mathrm{bd}\}$ is favourable in month $t+1$. State-specific means and variances are estimated from past returns assigned to each state and shrunk toward pre-out-of-sample unconditional moments with weight $n^a_{s,t}/(n^a_{s,t}+36)$. The probability-mixture mean is

$$\hat{\mu}^a_t = (1-\hat{p}^a_{t+1|t})\hat{\mu}^a_{0,t} + \hat{p}^a_{t+1|t}\hat{\mu}^a_{1,t}. \tag{7}$$

The variance uses the corresponding second-moment mixture. The covariance matrix uses an expanding stock–bond correlation estimate shrunk toward the pre-prediction stock–bond correlation. Appendix H reports the full probability-to-moment and realised net-return formulas.

For the INDEPENDENT allocator, the monthly portfolio is long-only in equity, bonds, and cash and solves

$$w_t = \arg\max_{w\in\mathcal{W}} \left\{ w^\top \hat{\mu}_t - \gamma w^\top \hat{\Sigma}_t w - \tau c \|w - w_{t-1}\|_1 \right\}, \tag{8}$$

where $\mathcal{W}$ is the long-only equity–bond–cash simplex, γ is risk aversion, c is the one-way transaction cost, and τ scales turnover control. Appendix H reports the independent grid, selected-strategy settings, additional benchmarks, and parameter sweeps.

The COUPLED allocator uses the predicted probabilities of calm, normal, and stressed joint regimes in the same portfolio objective. Its conditional return moments are estimated from past returns assigned to the corresponding joint regimes, using either the most likely regime or a probability mixture, and its covariance matrix is estimated with an exponentially weighted stock–bond covariance with a 12-month half-life. Appendix I reports the coupled allocation details, benchmark rules, cost sensitivity, and parameter sweeps. The headline results for both approaches are evaluated against the 60/40 benchmark. The comparison is benchmark-relative within each approach; Appendices H and I report the approach-specific allocation grids, covariance estimators, and turnover-penalty conventions.

5.1 Allocation trade-off

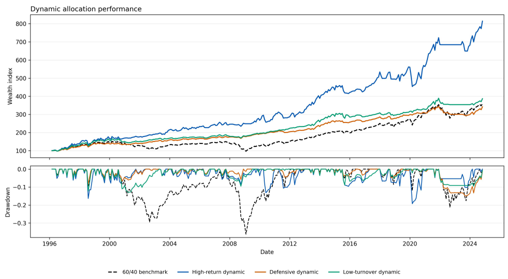
The allocation results show that regime forecasts have different portfolio roles across the selected dynamic strategies. Table 4 reports three representative roles from each allocation grid: a defensive allocation, an intermediate allocation, and a higher-return allocation. The selected rows summarise the main trade-off between drawdown control, trading intensity, and terminal wealth. Appendices H and I report the selection protocol, full allocation tables, cost sensitivity, and parameter sweeps. Additional independent benchmarks, including equity-only, bond-only, and volatility-targeted 60/40 allocations, are reported in Appendix H.5.

Table 4: Selected allocation results, out of sample. Each panel reports three representative dynamic allocation roles and the corresponding 60/40 benchmark. Returns are annualised excess returns net of transaction costs. "Final" indexes wealth to 100 at the start of the evaluation sample. Both panels use long-only equity–bond–cash weights, monthly rebalancing, and 10 bps one-way transaction costs. The appendices report the approach-specific moment estimates, grid choices, and turnover-penalty conventions. The INDEPENDENT allocation sample runs to 2024-10; the independent student forecast diagnostics run to 2024-12. The COUPLED sample is 1996-02 to 2024-10.

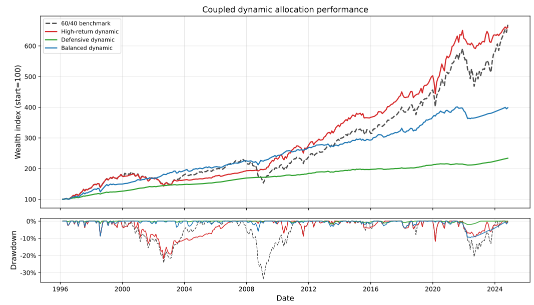
Role	Strategy	Ann. exc.	Vol.	Sharpe	Max DD	Final
independent: selected dynamic allocations and benchmark						
Defensive	Random forest, smoothed predictors	+4.4%	5.8%	0.77	−14.9%	341.9
Low-turnover	Logistic, z-score + smoothed z-score	+5.0%	7.2%	0.69	−13.0%	386.6
High-return	Logistic, smoothed predictors	+7.9%	10.6%	0.74	−19.3%	814.2
Benchmark	60/40	+4.5%	9.6%	0.47	−36.2%	359.4
coupled: selected dynamic allocations and benchmark						
Defensive	Minimum variance, probability mixture	+0.80%	1.2%	0.64	−2.0%	234.0
Balanced	Mean–variance, probability mixture	+2.74%	4.4%	0.63	−9.6%	398.5
High-return	Mean–variance, probability mixture	+4.70%	7.6%	0.62	−21.9%	661.2
Benchmark	60/40	+4.92%	9.6%	0.47	−33.9%	668.9

Table 4 shows that the selected strategies improve on the 60/40 benchmark along different margins. Defensive allocations mainly reduce drawdowns and raise Sharpe ratios. Low-turnover and balanced allocations preserve part of this risk control with less aggressive rebalancing. High-return allocations accept larger drawdowns in exchange for greater participation in favourable regimes. The full metric set is reported in Tables I.41, H.24, and H.25.

Figure 2 reports the corresponding wealth and drawdown paths.



(a) INDEPENDENT



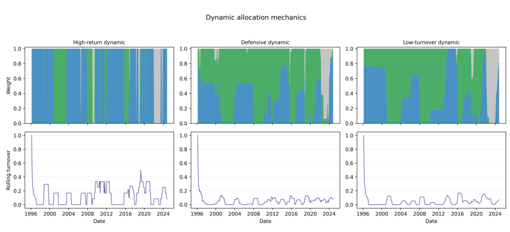
(b) COUPLED

Figure 2: Dynamic allocation performance, out of sample. The panels report wealth indexes and drawdowns for the selected dynamic allocations and the 60/40 benchmark under the INDEPENDENT and COUPLED approaches. Wealth is indexed to 100 at the start of each evaluation sample, and returns are net of 10 bps one-way transaction costs.

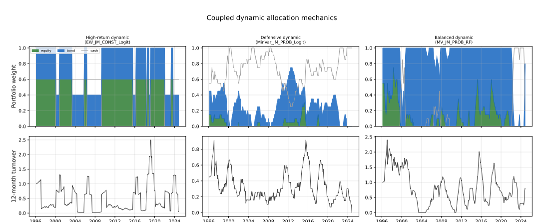
The wealth and drawdown paths show when the table results are earned. Defensive allocations cut losses during adverse forecast regimes. High-return allocations keep more risky exposure during favourable forecast regimes. The paths show when forecast-driven exposure changes coincide with drawdown control or faster compounding.

5.2 Portfolio-weight behaviour

Figure 3 reports how the selected forecasts change equity, bond, and cash weights. Defensive allocations reduce risky exposure when adverse regimes receive higher forecast probability. Higher-return allocations keep larger risky positions during favourable forecasts. Low-turnover and balanced allocations reduce the size or frequency of these changes. Full allocation diagnostics are reported in Appendices H and I.



(a) INDEPENDENT



(b) COUPLED

Figure 3: Portfolio weights and turnover, out of sample. The panels report equity, bond, and cash weights together with turnover for the selected dynamic allocations under the INDEPENDENT and COUPLED approaches.

5.3 Sensitivity

The sensitivity checks evaluate whether the allocation roles depend on narrow parameter choices. Appendix H and Appendix I report the full sweeps over risk aversion, turnover penalties, transaction costs, probability smoothing, and moment shrinkage. The main pattern is economically consistent: higher risk aversion reduces risky exposure, smoothing and turnover penalties reduce trading, and higher transaction costs lower net performance. The reported headline strategies are therefore selected from stable exposure–cost trade-offs, not from a single isolated parameter setting.

6 Discussion and conclusion

The results support a benchmark-relative role for regime forecasts in monthly stock–bond allocation. The value of the forecasts is not a single performance number. It appears through the margin chosen by the allocator: lower drawdowns, higher risk-adjusted returns, lower trading intensity, or higher terminal wealth.

The regime labels are statistical representations of market environments. The diagnostics evaluate whether they are economically separated, persistent enough for monthly trading, predictable from lagged information, and valuable when their forecast probabilities enter a portfolio rule. This sequence gives the teacher, student, and allocation stages their economic order.

The central economic implication is that the bond leg changes role across states. In disinflationary equity drawdowns, bonds can still hedge equity risk. In inflation-led sell-offs, bond exposure can become part of the adverse state. This distinction explains why asset-level regime probabilities and joint stock–bond regime probabilities are both informative. The INDEPENDENT representation gives separate signals for equity and bond exposure; the COUPLED representation gives a direct signal for the diversification environment.

The allocation evidence is strongest when read as a set of trade-offs against 60/40. Defensive rules use regime probabilities to reduce losses in adverse states. Low-turnover and balanced rules preserve part of that risk control with less rebalancing. High-return rules use favourable forecasts to keep larger risky exposure. Regime information therefore adds value when it changes exposure in the states where the static benchmark is least suited to the stock–bond environment.

A Data Sources and Feature Construction

Table A.5: Teacher features by approach. The INDEPENDENT teacher uses daily equity features and monthly bond features. The COUPLED teacher uses a monthly joint stock–bond construction. Horizons are reported in trading days for daily equity features and in months for monthly features.

Approach	Feature	Definition	Horizons
INDEPENDENT (equity)	Exponentially weighted return; Sortino ratio	Daily equity excess-return features based on recent mean performance and downside-risk-adjusted performance	21, 63, 126, 252 trading days
INDEPENDENT (bond)	Exponentially weighted return; downside risk; Sharpe ratio; volatility	Monthly bond excess-return features based on recent mean performance, negative-return magnitude, rolling risk-adjusted performance, and volatility	1, 3, 6, 12 months
COUPLED (equity, bond)	Sortino ratio	Recent average excess return over downside risk (EWMA of squared negative excess returns)	1, 2, 4, 12 m
COUPLED (equity, bond)	Drawdown level / term structure	Log downside deviation; short minus long log downside deviation	6 m; 2 vs 12 m
COUPLED (equity, bond)	MA crossover / trend	Short minus long average excess return, scaled by downside risk; EWMA-standardised trend	3 vs 12 m; 12, 24 m

A.1 Time series

Time series	Frequency used	Source
S&P 500 Index level	Daily	Bloomberg
Bloomberg US Aggregate Bond Index level	Monthly	Bloomberg
Fama–French market excess return, size, value, profitability, investment, momentum, and risk-free return	Monthly	WRDS FF
Six Fama–French size and book-to-market portfolio returns: small low, small medium, small high, big low, big medium, and big high	Monthly	WRDS FF
CRSP common-stock returns, prices, shares outstanding, SIC classifications, share classifications, and exchange identifiers	Monthly	WRDS CRSP
CRSP Treasury security returns, total amount outstanding, and maturity dates	Monthly	WRDS Tsy
CRSP fixed-term Treasury nominal yields at 1 month, 3 months, 1 year, 2 years, 5 years, 7 years, 10 years, 20 years, and 30 years	Monthly	WRDS Tsy
Equal-weight commodity portfolio excess return from *Commodities for the Long Run*	Monthly	AQR
National Financial Conditions Index	Monthly	FRED
Adjusted National Financial Conditions Index	Monthly	FRED
Investment-grade corporate bond fund adjusted close	Monthly	Yahoo
High-yield corporate bond fund adjusted close	Monthly	Yahoo
One-year Treasury yield	Monthly	FRED
Moody's Baa and Aaa corporate bond yields	Monthly	FRED
Sticky-price core consumer inflation	Monthly	FRED
CBOE VIX Index	Monthly	FRED
Chicago Fed National Activity Index	Monthly	FRED
University of Michigan Consumer Sentiment Index	Monthly	FRED
CRSP value-weighted equity market return	Monthly	WRDS CRSP
S&P 500 return	Monthly	WRDS CRSP
Aggregate bond return	Monthly	Bloomberg
Ten-year Treasury bond return	Monthly	WRDS Tsy

A.2 Teacher-feature definitions

Common return definitions.

$$\alpha_h = 1 - \exp\left(\frac{\log(0.5)}{h}\right), \tag{9}$$

$$\mathrm{EWM}_{h,t}(r) = \alpha_h r_t + (1-\alpha_h)\,\mathrm{EWM}_{h,t-1}(r), \tag{10}$$

$$r_t^- = \min\{r_t, 0\}, \tag{11}$$

$$D_{h,t}(r) = \left[\mathrm{EWM}_{h,t}\big((r_t^-)^2\big)\right]^{1/2}. \tag{12}$$

Equity teacher features. Daily equity excess return: r_t^E. Horizons: $h \in \{21, 63, 126, 252\}$.

$$\mathrm{EWMRet}^E_{h,t} = \mathrm{EWM}_{h,t}(r^E), \tag{13}$$

$$\mathrm{Sortino}^E_{h,t} = \frac{\mathrm{EWM}_{h,t}(r^E)}{D_{h,t}(r^E) + 10^{-12}}. \tag{14}$$

Bond teacher features. Monthly bond excess return: r_t^B.

$$\mathrm{EWMRet}^B_{h,t} = \mathrm{EWM}_{h,t}(r^B), \qquad h \in \{3, 6, 12\}, \tag{15}$$

$$\mathrm{Sharpe}^B_{h,t} = \frac{\mathrm{mean}(r^B_{t-h+1}, \dots, r^B_t)}{\mathrm{sd}(r^B_{t-h+1}, \dots, r^B_t) + 10^{-12}}, \qquad h \in \{3, 6, 12\}, \tag{16}$$

$$\mathrm{LogDownside}^B_{h,t} = \log\bigl(D_{h,t}(r^B) + 10^{-12}\bigr), \qquad h \in \{1, 3, 6, 12\}, \tag{17}$$

$$\mathrm{Vol}^B_{6,t} = \mathrm{sd}(r^B_{t-5}, \dots, r^B_t). \tag{18}$$

A.3 Student base-feature definitions

Rolling scalar features.

$$\mathrm{Sum}_{h,t}(x) = \sum_{j=0}^{h-1} x_{t-j}, \tag{19}$$

$$\mathrm{Mean}_{h,t}(x) = \frac{1}{h} \sum_{j=0}^{h-1} x_{t-j}, \tag{20}$$

$$\mathrm{Vol}_{h,t}(x) = \mathrm{sd}(x_{t-h+1}, \dots, x_t). \tag{21}$$

Return-block features. For $x_{1,t}, \dots, x_{K,t}$,

$$\mathrm{Disp}_t = \mathrm{sd}(x_{1,t}, \dots, x_{K,t}), \tag{22}$$

$$\mathrm{BMW}_t = \max_k x_{k,t} - \min_k x_{k,t}. \tag{23}$$

Correlation-matrix features. For the 12-month rolling correlation matrix C_t,

$$\overline{\rho}_t = \mathrm{mean}_{i<j}(C_{ij,t}), \tag{24}$$

$$\overline{|\rho|}_t = \mathrm{mean}_{i<j} |C_{ij,t}|, \tag{25}$$

$$\mathrm{PC1}_t = \frac{\lambda_{\max}(C_t)}{\sum_j \lambda_j(C_t)}, \tag{26}$$

$$\mathrm{Rotation}_t = \|C_t - C_{t-1}\|_F. \tag{27}$$

Fisher correlation changes.

$$\rho^c_t = \min\{\max\{\rho_t, -0.999\}, 0.999\}, \tag{28}$$

$$F_t = \frac{1}{2} \log\left(\frac{1 + \rho^c_t}{1 - \rho^c_t}\right), \tag{29}$$

$$\Delta F_t = F_t - F_{t-1}. \tag{30}$$

Non-commutativity features.

$$\Delta\Sigma_t = \Sigma_t - \Sigma_{t-1}, \tag{31}$$

$$\Lambda_t^{\Sigma} = \log\left(1 + 100\frac{\|\Sigma_t\Delta\Sigma_t - \Delta\Sigma_t\Sigma_t\|_F}{\|\Sigma_t\|_F\|\Delta\Sigma_t\|_F}\right). \tag{32}$$

The same formula defines Λ_t^C after replacing Σ_t by C_t. Here $\|\cdot\|_F$ denotes the Frobenius norm.

Differences.

$$\Delta x_t = x_t - x_{t-1}, \tag{33}$$

$$\Delta \log x_t = \log(x_t) - \log(x_{t-1}). \tag{34}$$

A.4 Student-feature transformations

Causal winsorisation.

$$L_t = Q_{0.01}(x_{t-120}, \dots, x_{t-1}), \tag{35}$$

$$U_t = Q_{0.99}(x_{t-120}, \dots, x_{t-1}), \tag{36}$$

$$\widetilde{x}_t = \min\{\max(x_t, L_t), U_t\}. \tag{37}$$

Window: 120 months. Minimum prior observations: 60.

Causal z-score.

$$z_t = \frac{\widetilde{x}_t - \mathrm{mean}(\widetilde{x}_{t-60}, \dots, \widetilde{x}_{t-1})}{\mathrm{sd}(\widetilde{x}_{t-60}, \dots, \widetilde{x}_{t-1}) + 10^{-12}}. \tag{38}$$

Window: 60 months. Minimum prior observations: 36.

Smoothed z-score.

$$z_t^s = \mathrm{EWM}_{12,t}(z). \tag{39}$$

Half-life: 12 months.

Student feature vector. For base features $j = 1, \dots, 134$,

$$Z_t = \big(z_{1,t}, z_{1,t}^s, \dots, z_{134,t}, z_{134,t}^s\big). \tag{40}$$

A.5 Coupled approach: return definitions and features

For half-life h, let $\mathrm{EWM}_h(\cdot)$ denote the exponentially weighted moving average with that half-life. For an excess-return series x_t, define downside deviation and the Sortino ratio as

$$\mathrm{dd}_{h,t}(x) = \Big[\mathrm{EWM}_h\big(\min(x_t, 0)^2\big)\Big]^{1/2}, \tag{41}$$

$$\mathrm{Sortino}_{h,t}(x) = \frac{\mathrm{EWM}_h(x)}{\mathrm{dd}_{h,t}(x) + \varepsilon}, \qquad h \in \{1, 2, 4, 12\}. \tag{42}$$

With $m_h = \mathrm{EWM}_h(x)$, the coupled teacher uses four additional return-based features:

$$\mathrm{dd\_lvl}_t = \log\big(\mathrm{dd}_{6,t}(x) + \varepsilon\big), \quad \mathrm{dd\_diff}_t = \log\big(\mathrm{dd}_{2,t} + \varepsilon\big) - \log\big(\mathrm{dd}_{12,t} + \varepsilon\big), \tag{43}$$

$$\mathrm{ma\_cross}_t = \frac{m_3 - m_{12}}{\mathrm{dd}_{6,t} + \varepsilon}, \qquad \mathrm{trend}_t = \mathrm{EWM}_6\left(\frac{x_t - \mathrm{EWM}_{12}(x)}{\mathrm{EWM\text{-}sd}_{24}\big(x_t - \mathrm{EWM}_{12}(x)\big) + \varepsilon}\right). \tag{44}$$

The market-derived block for the coupling logistic (5) uses the trailing 12-month equity–bond correlation $\rho^{\text{eq,bd}}_{12,t}$, together with each leg's trend_t and dd_lvl_t.

Table A.7: Coupled teacher feature families. Each family is estimated under both the L^1 and L^2 specifications.

Family	Features
all	$\text{Sortino}_{1,2,4,12}$, dd_lvl, dd_diff, ma_cross
sortino	$\text{Sortino}_{1,2,4,12}$
sortino_short	$\text{Sortino}_{1,2,4}$
dd	dd_lvl, dd_diff
trend	ma_cross
risk	dd_lvl, Sortino_4
sortino_trend	$\text{Sortino}_{2,4}$, ma_cross
dd_trend	dd_lvl, ma_cross
sortino_dd	$\text{Sortino}_{2,4,12}$, dd_lvl, dd_diff
core	Sortino_4, dd_lvl, ma_cross
mom	Sortino_2, ma_cross

Crossing the eleven feature families with the L^1 and L^2 specifications gives 22 coupled teacher members. At each month, features are z-scored against the trailing 120-month window after winsorising that window at the 1%/99% quantiles. The current observation is clipped to the same bounds before scaling, so the transformation is strictly causal. Figures D.15 and D.16 plot the resulting standardised inputs.

B Teacher Label Estimation Details

B.1 Teacher panels and refitting schedule

Table B.8: Teacher panels

Label	Teacher features	Frequency	Initial sample
Equity regime label	Daily equity return and Sortino features	Daily	12 years
Bond regime label	Monthly bond return and risk features	Monthly	12 years

$$x^c_{t,j} = \min\{\max(x_{t,j}, \mu_j - 3\sigma_j), \mu_j + 3\sigma_j\}, \tag{45}$$

$$z_{t,j} = \frac{x^c_{t,j} - \mu_j}{\sigma_j}. \tag{46}$$

Daily equity models are refitted at month ends. Monthly bond models are refitted each month. Each refit uses only observations available at the refit date. In each refit, μ_j and σ_j are computed from that training sample.

B.2 Jump-model objective

For standardised teacher features z_t, the two-state objective is

$$\min_{s_1,\dots,s_T,\, c_0,c_1} \sum_{t=1}^{T} d(z_t, c_{s_t}) + \lambda \sum_{t=2}^{T} \mathbf{1}\{s_t \neq s_{t-1}\}, \qquad s_t \in \{0,1\}. \tag{47}$$

B.3 L^2 jump model

$$\mathcal{T}_k = \{t : s_t = k\}, \qquad k \in \{0,1\}, \tag{48}$$

$$c_k = \frac{1}{|\mathcal{T}_k|} \sum_{t\in\mathcal{T}_k} z_t, \tag{49}$$

$$d_2(z_t, c_{s_t}) = \frac{1}{2}\|z_t - c_{s_t}\|_2^2. \tag{50}$$

B.4 L^1 medoid jump model

$$\widetilde{z}_k = \mathrm{median}\{z_t : t \in \mathcal{T}_k\}, \tag{51}$$

$$m_k = \arg\min_{z_i :\, i\in\mathcal{T}_k} \|z_i - \widetilde{z}_k\|_1, \tag{52}$$

$$d_1(z_t, m_{s_t}) = \|z_t - m_{s_t}\|_1 = \sum_{j=1}^{p} |z_{t,j} - m_{s_t,j}|. \tag{53}$$

B.5 Penalty grids and optimisation

$$\lambda^E \in \{2, 4, 8, 16, 32, 64, 128\}, \tag{54}$$

$$\lambda^B \in \{0, 10^{-5}, 10^{-4}, 0.001, 0.00390625, 0.0078125, 0.015625, 0.03125, 0.0625, 0.125, 0.25, 0.5, 1, 2, 4\}. \tag{55}$$

Each fit uses five initialisations and at most 100 coordinate-descent iterations. Conditional on the state centres, the state path is updated by dynamic programming.

B.6 State ordering

For fitted raw state k, define

$$\widehat{\mu}_k = \mathrm{mean}\{r_t : s_t = k\}, \tag{56}$$

$$\widehat{\sigma}_k = \mathrm{sd}\{r_t : s_t = k\}, \tag{57}$$

$$\widehat{\mathrm{SR}}_k = \frac{\widehat{\mu}_k}{\widehat{\sigma}_k}. \tag{58}$$

The favourable state is

$$g = \arg\max_{k\in\{0,1\}} \widehat{\mathrm{SR}}_k. \tag{59}$$

If the Sharpe-ratio comparison is tied,

$$g = \arg\max_{k\in\{0,1\}} \widehat{\mu}_k. \tag{60}$$

B.7 Monthly labels

For daily equity paths, the monthly favourable-state share is

$$p_m^E = \frac{1}{N_m} \sum_{t \in m} \mathbf{1}\{s_t = g\}, \tag{61}$$

and the monthly equity label is

$$S_m^E = \mathbf{1}\{p_m^E \geq 0.5\}. \tag{62}$$

For monthly bond paths,

$$S_m^B = \mathbf{1}\{s_m = g\}. \tag{63}$$

For asset $a \in \{E, B\}$, majority-vote labels are

$$\bar{S}_m^a = \frac{1}{K_a} \sum_{k=1}^{K_a} S_{m,k}^a, \tag{64}$$

$$S_m^{a,MV} = \mathbf{1}\{\bar{S}_m^a \geq 0.5\}. \tag{65}$$

The one-month-ahead student target is

$$Y_{m+1}^a = S_{m+1}^{a,MV}. \tag{66}$$

C Teacher Label and Feature Diagnostics

This appendix reports diagnostics for the equity and bond teacher labels. The diagnostics cover individual L^2 and L^1-medoid teacher paths, jump-penalty sensitivity, majority-vote construction, transition behaviour, crisis-window behaviour, and teacher-feature state separation. These diagnostics document whether the labels are persistent, economically separated, and sufficiently stable to serve as supervised targets in the student stage. All return statistics use the monthly excess return of the labelled asset. The reported independent prediction and allocation results use the equity return–Sortino teacher label and the canonical bond teacher label; volatility/downside equity labels were candidate diagnostics and are not used in the reported allocation results.

C.1 Individual teacher paths

Figures C.4–C.7 show the fitted teacher paths across jump-penalty values. Each panel plots the labelled asset's excess-return index and shades months assigned to the favourable or unfavourable regime. The panel title reports the jump penalty, favourable-state share, number of switches, and average spell duration.

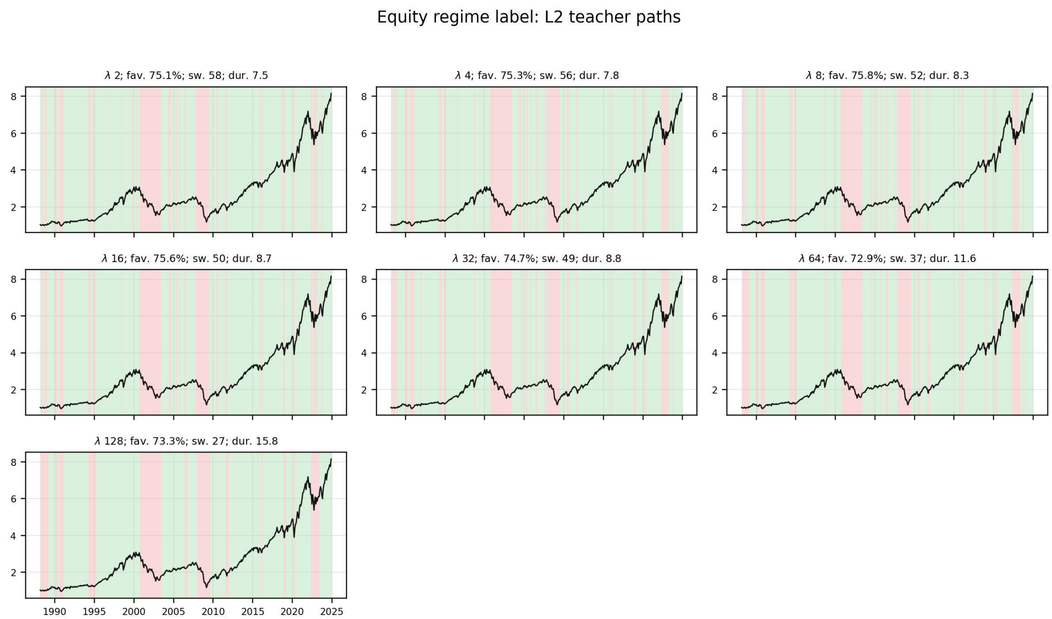


Figure C.4: Equity regime label: L^2 teacher paths.

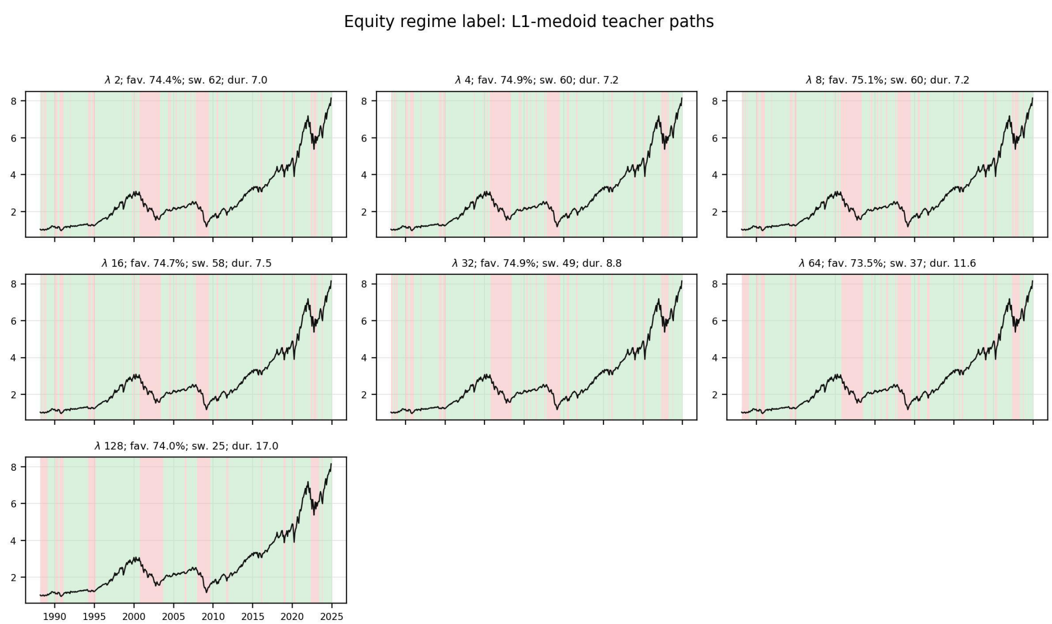


Figure C.5: Equity regime label: L^1-medoid teacher paths.

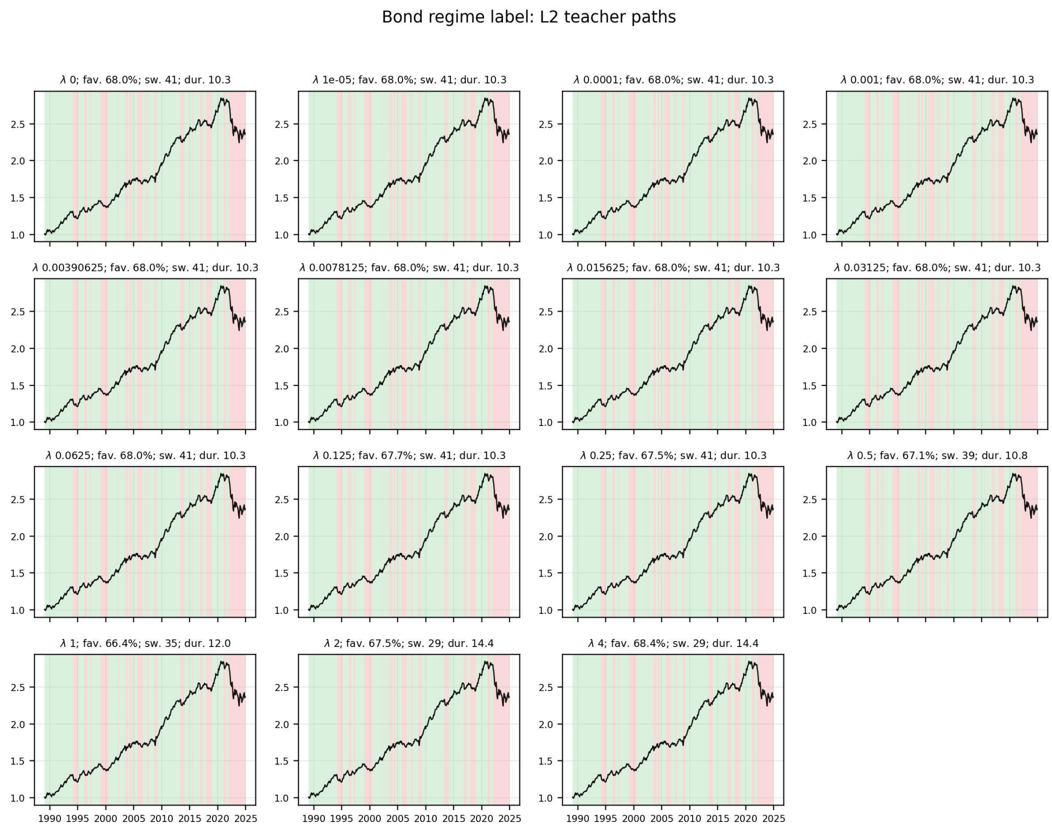


Figure C.6: Bond regime label: L^2 teacher paths.

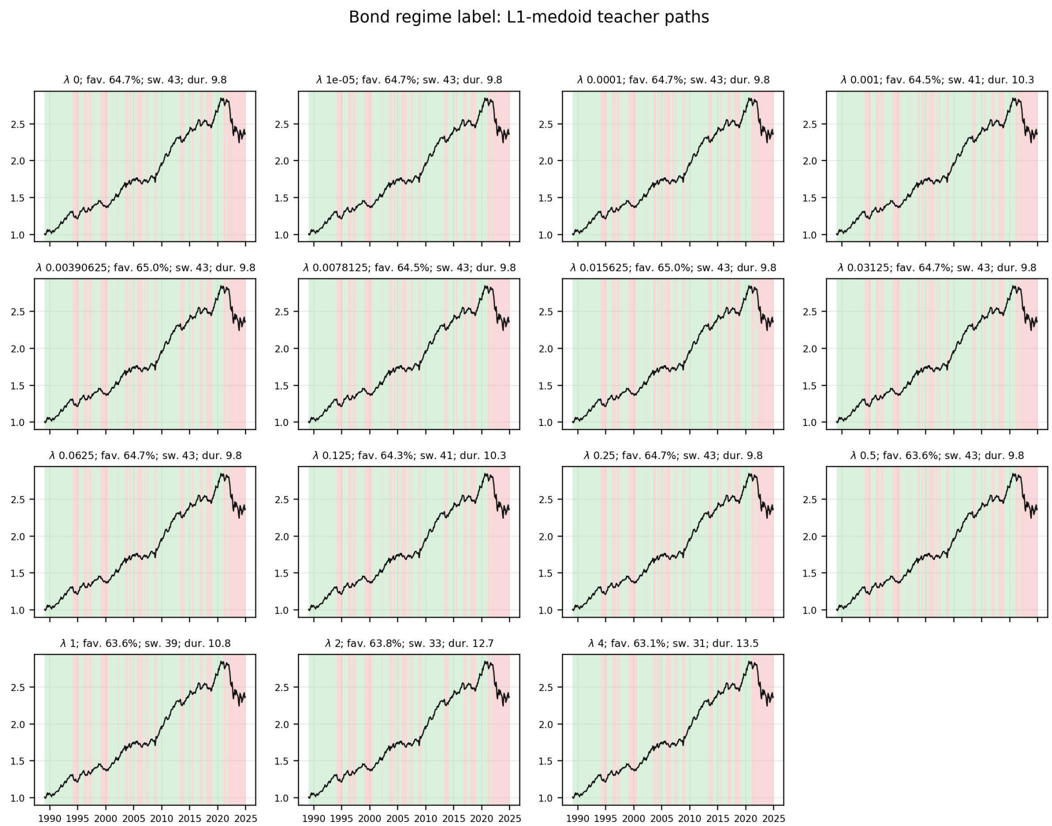


Figure C.7: Bond regime label: L^1-medoid teacher paths.

C.2 Jump-penalty diagnostics

Figures C.8 and C.9 summarise how label properties vary across the jump-penalty grid. The panels compare the L^2 and L^1-medoid teacher paths using favourable-state share, switch count, state durations, return separation, Sharpe separation, unfavourable-state volatility, and median feature-separation ratio.

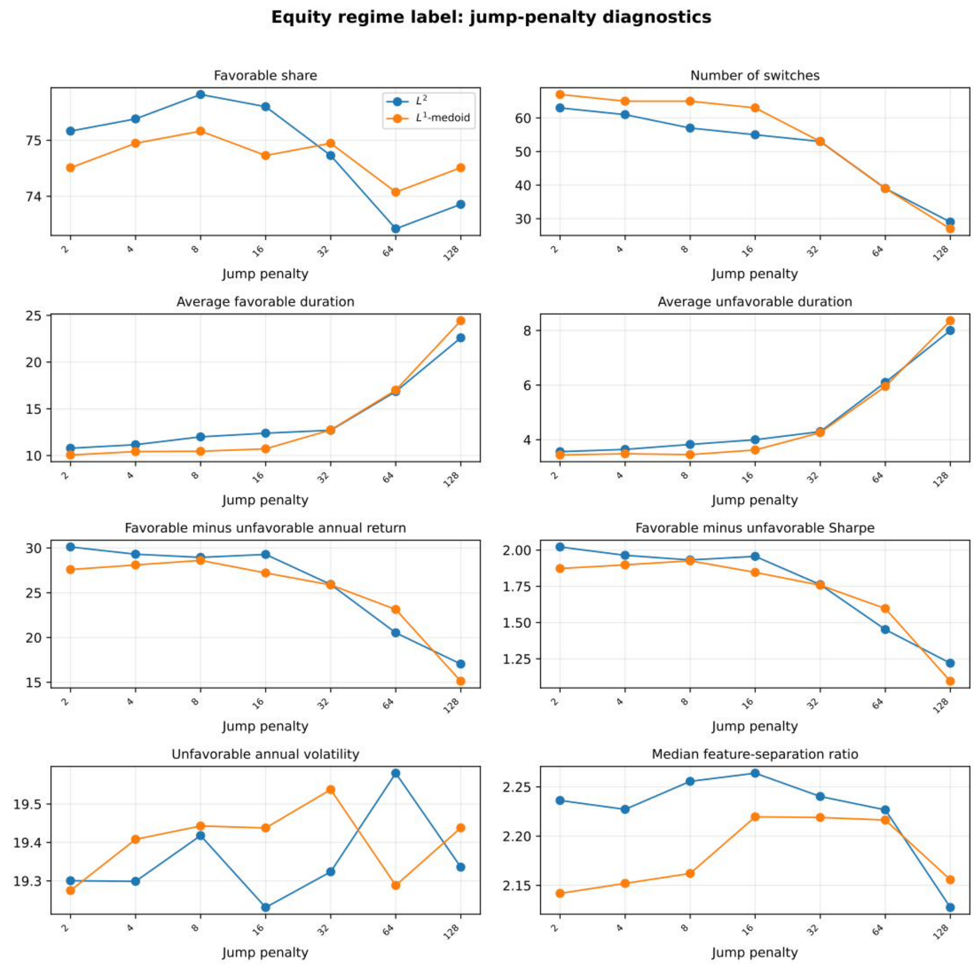


Figure C.8: Equity regime label: jump-penalty diagnostics.

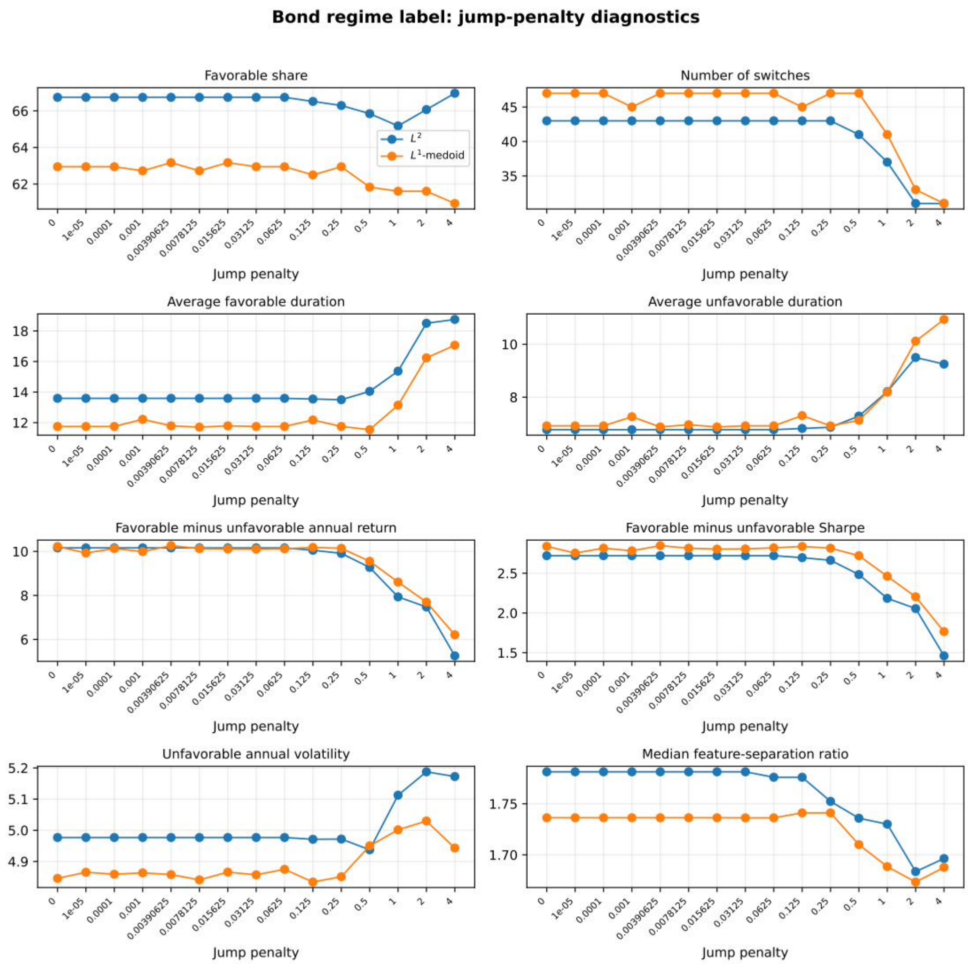


Figure C.9: Bond regime label: jump-penalty diagnostics.

C.3 Majority-vote construction

For asset a, the member vote share and final majority label are

$$\pi_m^a = \frac{1}{|\mathcal{P}_a|} \sum_{p \in \mathcal{P}_a} S_m^{a,p}, \qquad S_m^a = \mathbf{1}\{\pi_m^a \geq 1/2\}. \tag{67}$$

Figures C.10 and C.11 show how the final teacher labels are formed from the L^2, L^1-medoid, and combined teacher paths. Each panel plots the labelled asset's excess-return index, shades the majority-vote regime, and overlays the vote probability. The dashed horizontal line marks the one-half threshold used to form the majority label.

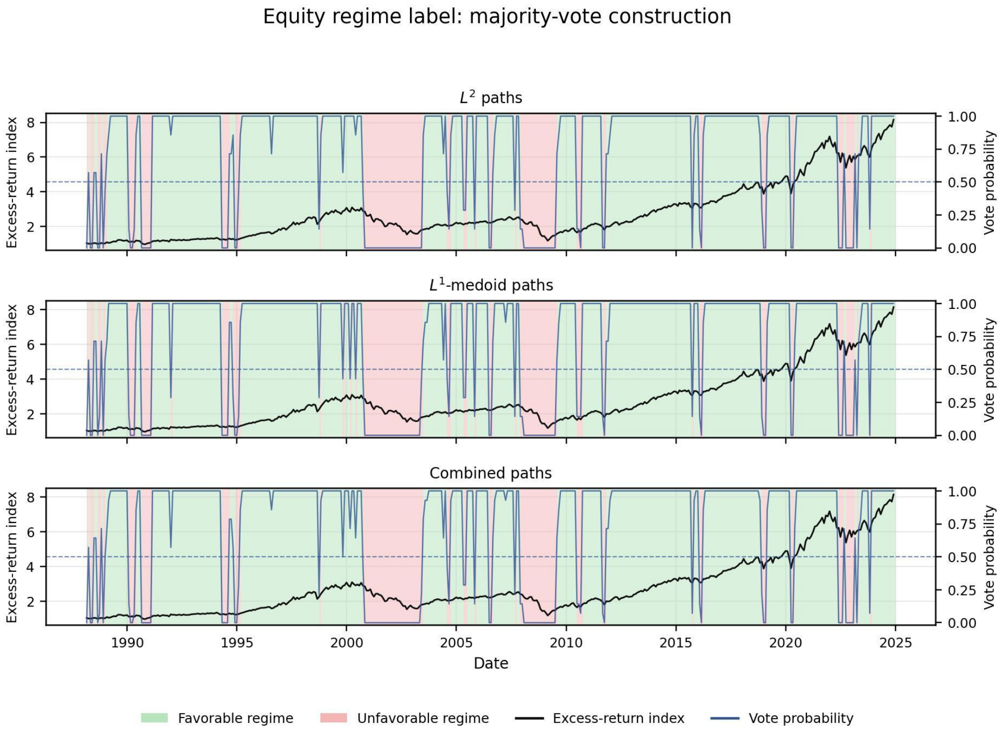


Figure C.10: Equity regime label: majority-vote construction.

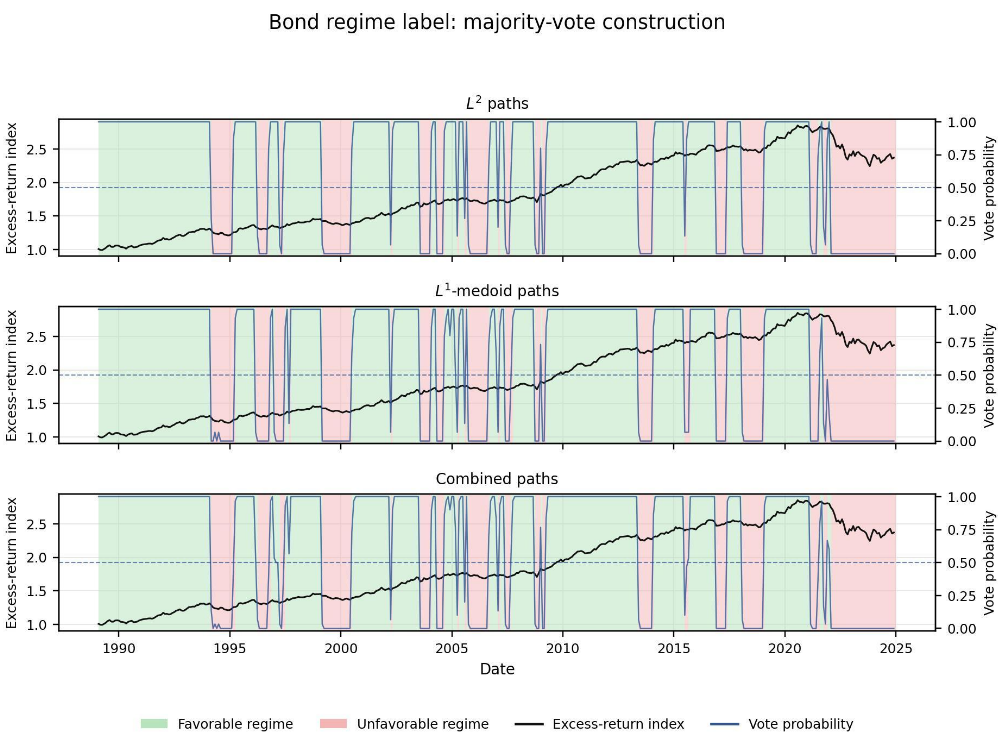


Figure C.11: Bond regime label: majority-vote construction.

C.4 Transitions, persistence, and crisis windows

Table C.9 reports transition probabilities and expected state durations for the final combined teacher labels.

Table C.9: Teacher regime label transition summary

Label	$P(U \to U)$	$P(U \to F)$	$P(F \to U)$	$P(F \to F)$	$E[D_U]$	$E[D_F]$
Equity	75.9%	24.1%	7.5%	92.5%	4.2	13.3
Bond	85.8%	14.2%	7.3%	92.7%	7.1	13.8

Notes: F denotes the favourable regime and U denotes the unfavourable regime. The transition probabilities are estimated from the final monthly combined majority-vote label for each asset. Expected duration is computed as $1/(1 - p_{ss})$ months for state s.

Table C.10 reports path-level persistence and label-conditional Sharpe statistics by asset, teacher model, and jump-penalty value.

Table C.10: Teacher path persistence summary

Label	Model	λ	Fav. share	Sw.	Avg. dur.	Fav. dur.	Unfav. dur.	Fav. Sharpe	Unfav. Sharpe
Equity	L^1-medoid	2	74.4%	62	7.0	10.3	3.6	1.1	-0.68
Equity	L^1-medoid	4	74.9%	60	7.2	10.7	3.7	1.1	-0.7
Equity	L^1-medoid	8	75.1%	60	7.2	10.7	3.7	1.1	-0.72
Equity	L^1-medoid	16	74.7%	58	7.5	11.0	3.9	1.1	-0.66
Equity	L^1-medoid	32	74.9%	49	8.8	13.2	4.4	1.1	-0.61
Equity	L^1-medoid	64	73.5%	37	11.6	17.1	6.2	1.1	-0.55
Equity	L^1-medoid	128	74.0%	25	17.0	25.2	8.8	0.86	-0.24
Equity	L^2	2	75.1%	58	7.5	11.1	3.8	1.2	-0.79
Equity	L^2	4	75.3%	56	7.8	11.5	3.9	1.1	-0.76
Equity	L^2	8	75.8%	52	8.3	12.4	4.1	1.1	-0.75
Equity	L^2	16	75.6%	50	8.7	12.8	4.3	1.1	-0.76
Equity	L^2	32	74.7%	49	8.8	13.2	4.5	1.1	-0.61
Equity	L^2	64	72.9%	37	11.6	16.9	6.3	1	-0.44
Equity	L^2	128	73.3%	27	15.8	23.1	8.4	0.92	-0.31
Bond	L^1-medoid	0	64.8%	43	9.8	12.7	6.9	2	-0.9
Bond	L^1-medoid	1e-05	64.8%	43	9.8	12.7	6.9	1.9	-0.85
Bond	L^1-medoid	0.0001	64.8%	43	9.8	12.7	6.9	2	-0.88
Bond	L^1-medoid	0.001	64.6%	41	10.3	13.3	7.3	2	-0.86
Bond	L^1-medoid	0.00390625	65.0%	43	9.8	12.8	6.9	2	-0.91
Bond	L^1-medoid	0.0078125	64.6%	43	9.8	12.7	7.0	2	-0.88
Bond	L^1-medoid	0.015625	65.0%	43	9.8	12.8	6.9	2	-0.88
Bond	L^1-medoid	0.03125	64.8%	43	9.8	12.7	6.9	2	-0.88
Bond	L^1-medoid	0.0625	64.8%	43	9.8	12.7	6.9	2	-0.88
Bond	L^1-medoid	0.125	64.4%	41	10.3	13.2	7.3	2	-0.88
Bond	L^1-medoid	0.25	64.8%	43	9.8	12.7	6.9	2	-0.89
Bond	L^1-medoid	0.5	63.7%	43	9.8	12.5	7.1	2	-0.76
Bond	L^1-medoid	1	63.7%	39	10.8	13.8	7.8	1.9	-0.63
Bond	L^1-medoid	2	63.9%	33	12.7	16.2	9.2	1.7	-0.53
Bond	L^1-medoid	4	63.2%	31	13.5	17.1	9.9	1.5	-0.33
Bond	L^2	0	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	1e-05	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.0001	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.001	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.00390625	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.0078125	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.015625	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.03125	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.0625	68.1%	41	10.3	14.0	6.6	1.8	-0.95
Bond	L^2	0.125	67.8%	41	10.3	14.0	6.6	1.8	-0.93
Bond	L^2	0.25	67.6%	41	10.3	13.9	6.7	1.8	-0.91
Bond	L^2	0.5	67.1%	39	10.8	14.5	7.1	1.7	-0.81
Bond	L^2	1	66.4%	35	12.0	15.9	8.1	1.6	-0.59
Bond	L^2	2	67.6%	29	14.4	19.5	9.3	1.6	-0.54

Label	Model	λ	Fav. share	Sw.	Avg. dur.	Fav. dur.	Unfav. dur.	Fav. Sharpe	Unfav. Sharpe
Bond	L^2	4	68.5%	29	14.4	19.7	9.1	1.2	-0.25

Table C.11 reports crisis-window behaviour for the final combined teacher labels. Favourable share is computed from the final monthly label within each window; return statistics use the labelled asset's monthly excess returns over the same window.

Table C.11: Teacher regime label crisis-window summary

Label	Window	Months	Fav. share	Ann. ret.	Ann. vol.	Sharpe	Worst	Max. DD
Equity	1990 recession	9	33.3%	0.8%	18.3%	0.045	-10.0%	-16.3%
Equity	1994 bond shock	12	50.0%	-4.9%	10.6%	-0.46	-4.8%	-9.0%
Equity	Dot-com bust	32	0.0%	-19.9%	18.9%	-1.1	-11.1%	-46.9%
Equity	GFC	18	0.0%	-29.6%	24.6%	-1.2	-17.0%	-50.7%
Equity	Euro sovereign 2011	8	75.0%	-10.6%	18.7%	-0.57	-7.2%	-15.9%
Equity	Covid crash	3	33.3%	-33.8%	47.0%	-0.72	-12.6%	-12.6%
Equity	Inflation 2022	12	41.7%	-20.4%	22.9%	-0.89	-9.5%	-21.3%
Bond	1990 recession	9	100.0%	4.5%	3.4%	1.3	-2.0%	-2.0%
Bond	1994 bond shock	12	8.3%	-6.7%	4.3%	-1.5	-2.7%	-7.8%
Bond	Dot-com bust	32	96.9%	6.5%	3.3%	1.9	-1.8%	-2.6%
Bond	GFC	18	72.2%	3.3%	5.2%	0.64	-2.4%	-4.8%
Bond	Euro sovereign 2011	8	100.0%	8.8%	2.6%	3.5	-0.3%	-0.3%
Bond	Covid crash	3	100.0%	11.0%	4.9%	2.3	-0.7%	-0.7%
Bond	Inflation 2022	12	0.0%	-14.9%	8.2%	-1.8	-4.5%	-14.5%

C.5 Teacher features and state separation

Figures C.12 and C.13 show the standardised teacher features used to fit the equity and bond teacher paths. Standardisation is used only for visual comparability across features.

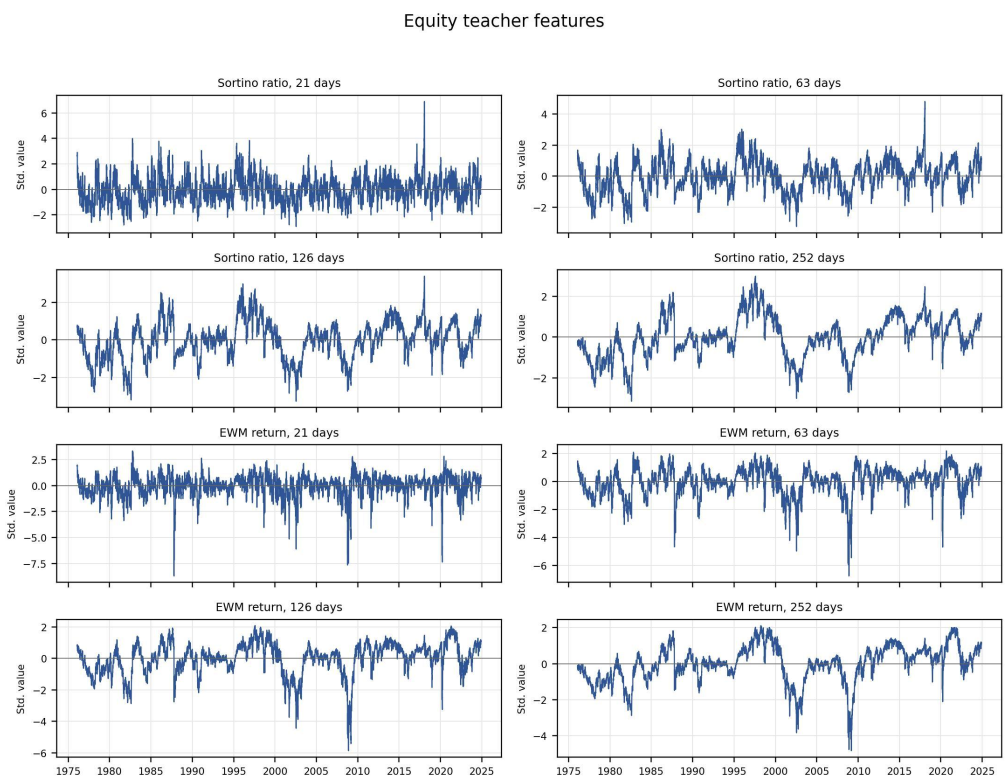


Figure C.12: Equity teacher features.

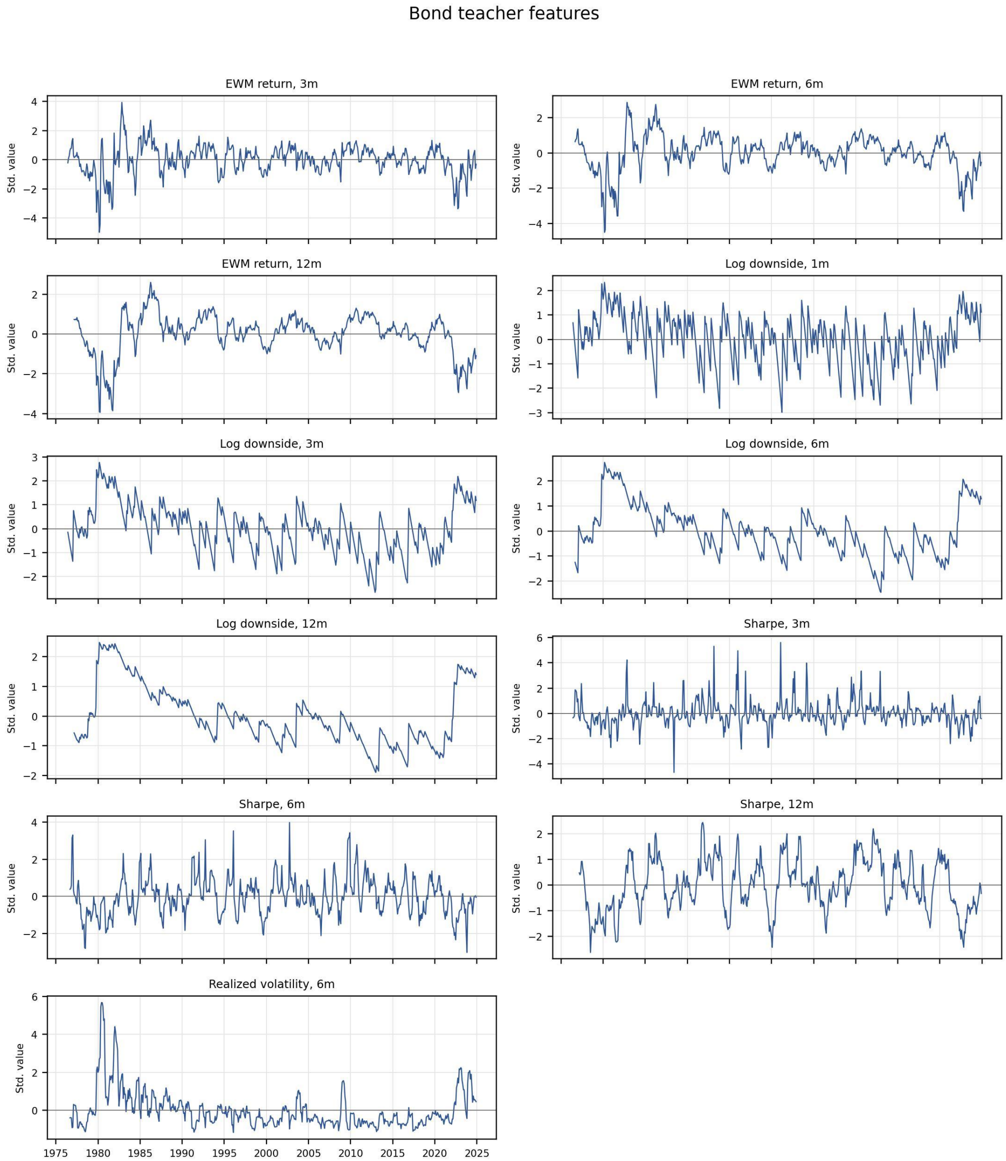


Figure C.13: Bond teacher features.

Table C.12 reports the features with the strongest separation between favourable and unfavourable teacher states.

Table C.12: Teacher feature state-separation summary

Label	Model	Feature	Median rank	Median sep.
Equity	L^1-medoid	Sortino ratio, 63 days	1.0	2.53
Equity	L^1-medoid	Sortino ratio, 126 days	2.0	2.49
Equity	L^1-medoid	EWM return, 126 days	3.0	2.39
Equity	L^1-medoid	EWM return, 63 days	4.0	2.32
Equity	L^1-medoid	Sortino ratio, 252 days	5.0	2.08
Equity	L^1-medoid	EWM return, 252 days	6.0	1.99
Equity	L^1-medoid	Sortino ratio, 21 days	7.0	1.62
Equity	L^1-medoid	EWM return, 21 days	8.0	1.44
Equity	L^2	Sortino ratio, 63 days	2.0	2.59
Equity	L^2	Sortino ratio, 126 days	1.0	2.55
Equity	L^2	EWM return, 126 days	3.0	2.44
Equity	L^2	EWM return, 63 days	4.0	2.4
Equity	L^2	Sortino ratio, 252 days	5.0	2.13
Equity	L^2	EWM return, 252 days	6.0	2.03
Equity	L^2	Sortino ratio, 21 days	7.0	1.64
Equity	L^2	EWM return, 21 days	8.0	1.46
Bond	L^1-medoid	Sharpe ratio, 12 months	1.0	2.25
Bond	L^1-medoid	Sharpe ratio, 6 months	2.0	1.98
Bond	L^1-medoid	Log downside risk, 3 months	3.0	1.94
Bond	L^1-medoid	Log downside risk, 1 month	4.0	1.93
Bond	L^1-medoid	EWM return, 6 months	5.0	1.9
Bond	L^1-medoid	EWM return, 3 months	6.0	1.74
Bond	L^1-medoid	EWM return, 12 months	7.0	1.71
Bond	L^1-medoid	Log downside risk, 6 months	8.0	1.62
Bond	L^2	Sharpe ratio, 12 months	1.0	2.14
Bond	L^2	Sharpe ratio, 6 months	2.0	2.0
Bond	L^2	Log downside risk, 3 months	3.0	1.99
Bond	L^2	Log downside risk, 1 month	4.0	1.99
Bond	L^2	EWM return, 6 months	5.0	1.91
Bond	L^2	EWM return, 3 months	6.0	1.78
Bond	L^2	EWM return, 12 months	7.0	1.69
Bond	L^2	Log downside risk, 6 months	8.0	1.65

Notes: For each fitted teacher path, features are ranked by their separation between favourable and unfavourable state centroids. Median rank is the median of that within-path rank over all jump-penalty paths in the model family. Median separation is the median standardised centroid-separation ratio over those paths; larger values mean stronger separation between the two regimes.

D Coupled teacher: estimation details and diagnostics

Objective, coupling logistic, and fitting

For standardised joint features $(z_t^{\mathrm{eq}}, z_t^{\mathrm{bd}})$ the per-month cost is (4) and the path minimises the total of emission and jump costs by Viterbi. The L^2 member uses Euclidean leg distances with mean centroids; the L^1 member uses absolute leg distances with median centroids (two centroids per leg, indexed by the benign/adverse bit). The coupling probability κ_t in (5) is fitted by L^2-penalised logistic regression ($C = 1$) with balanced class weights and clipped to $[0, 1]$. Within each monthly refit, the coupling regression is estimated on the expanding history available at the refit date. Each fit alternates: (i) re-estimate centroids and refit the coupling logistic given the provisional path; (ii) update the path by Viterbi given the centroids and κ_t. At most ten iterations, a single random restart plus a warm start from the previous month's path; the lowest-objective solution is kept. Per leg, the lower-mean state is set to adverse; a stressed-share cap then relabels to normal the stressed months whose equity moving-average crossover exceeds the 18% quantile of its trailing window, so only the weakest-trend months remain stressed. After a 132-month warm-up, each month is refit on the expanding history; (λ, η) is selected per month from $\lambda \in \{2, 3, 4, 6\}$ and $\eta \in \{1, 2, 4, 8\}$ to maximise the regime separation score subject to share, stressed-cap and minimum-run-length constraints (Table D.13).

Table D.13: Coupled teacher panel and causal refitting schedule.

Setting	Value
Legs	Equity (S&P 500 excess) and bond (aggregate excess)
Frequency	Monthly
Warm-up (initial sample)	132 months (11 years)
Refit cadence	Every month
Standardisation window	120 months, winsorised at 1%/99%
Random restarts per fit	1, plus a warm start from the previous month
Coordinate-descent iterations	up to 10
Jump-penalty grid λ	$\{2, 3, 4, 6\}$
Coupling-scale grid η	$\{1, 2, 4, 8\}$
Ensemble members	22 (11 feature families $\times$ $\{L^1, L^2\}$)
Combination	Equal-weight vote (argmax of class counts)

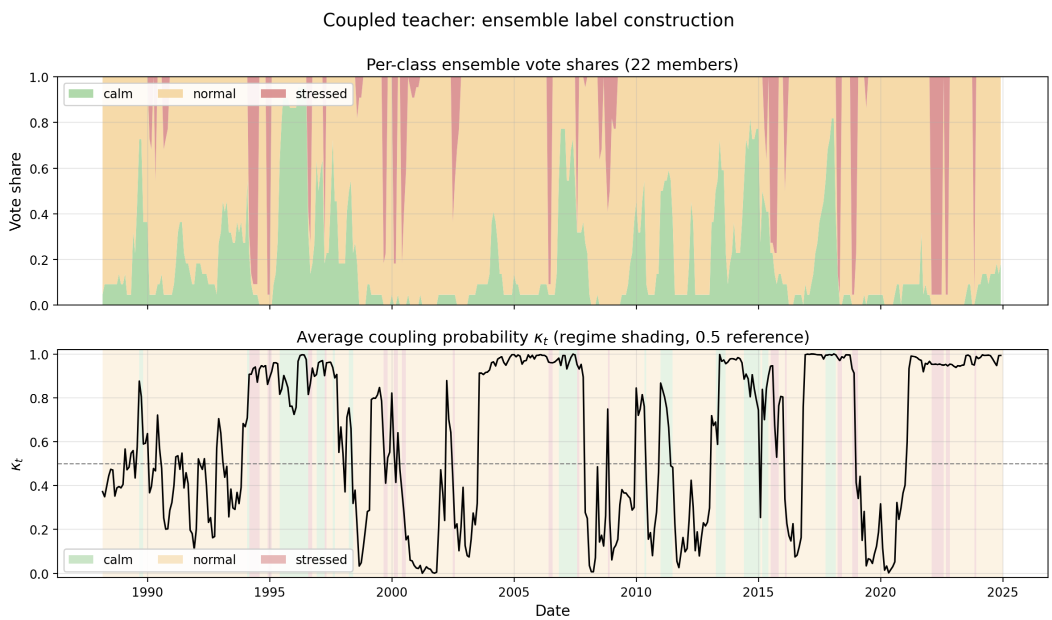


Figure D.14: COUPLED ensemble label construction. The top panel reports per-class vote shares across the 22 coupled members. The bottom panel reports the average coupling probability κ_t, with regime shading and the 0.5 reference line.

Table D.14: Coupled teacher: regime transition probabilities (%), row = from, column = to, with expected durations $1/(1 - p_{ss})$ in months.

From \ To	Calm	Normal	Stressed	E[dur]
Calm	78.7	14.8	6.6	4.7
Normal	3.8	92.1	4.1	12.6
Stressed	0.0	45.0	55.0	2.2

Table D.15 reports crisis-window behaviour for the final combined coupled labels. The stressed share is the fraction of months assigned to the stressed joint regime within each window; return statistics use each leg's monthly excess returns over the same window. This is the coupled counterpart of the independent crisis summary in Table C.11.

Table D.15: Coupled teacher: crisis-window summary by leg. Stressed share is the fraction of window months in the stressed joint regime; return statistics use the labelled leg's monthly excess return over the window.

Leg	Window	Months	Stressed sh.	Ann. ret.	Ann. vol.	Sharpe	Worst	Max. DD
Equity	1990 recession	9	0%	+0.8%	17.4%	+0.05	−10.1%	−16.4%
Equity	1994 bond shock	12	58%	−4.9%	10.1%	−0.48	−4.8%	−9.0%
Equity	Dot-com bust	32	3%	−20.0%	18.6%	−1.07	−11.1%	−46.9%
Equity	GFC	18	6%	−36.4%	23.1%	−1.58	−17.0%	−51.3%
Equity	Euro sovereign 2011	8	0%	−10.6%	17.5%	−0.61	−7.2%	−15.9%
Equity	Covid crash	3	0%	−33.9%	38.4%	−0.88	−12.6%	−12.6%
Equity	Inflation 2022	12	67%	−20.4%	21.9%	−0.93	−9.5%	−21.3%
Bond	1990 recession	9	0%	+4.5%	3.2%	+1.41	−2.0%	−2.0%
Bond	1994 bond shock	12	58%	−6.7%	4.2%	−1.61	−2.7%	−7.8%
Bond	Dot-com bust	32	3%	+6.5%	3.3%	+1.97	−1.8%	−2.6%
Bond	GFC	18	6%	+3.8%	5.2%	+0.74	−2.4%	−4.8%
Bond	Euro sovereign 2011	8	0%	+8.8%	2.4%	+3.69	−0.3%	−0.3%
Bond	Covid crash	3	0%	+11.0%	4.0%	+2.76	−0.7%	−0.7%
Bond	Inflation 2022	12	67%	−14.9%	7.9%	−1.90	−4.5%	−14.5%

Label construction and state summaries

Table D.16 records how the joint regime is assembled from the two per-leg state bits, and Table D.17 reports the label-conditional economics for both legs across the three joint regimes (the body reports a condensed version in Table 1).

Table D.16: Coupled teacher: label construction. The equity and bond per-leg state bits are estimated jointly through the coupling term, and their combination defines the three regimes.

Component	Teacher features	Frequency	Initial sample
Equity state bit	Equity Sortino, drawdown and trend features	Monthly	11 years
Bond state bit	Bond Sortino, drawdown and trend features	Monthly	11 years
Joint regime	Combination (eq bit, bd bit) mapped to calm / normal / stressed, with the coupling probability κ_t tying the two bits	Monthly	11 years

Table D.17: Coupled teacher: three-regime summary with spell counts. Share is the fraction of labelled months in each joint regime; equity and bond statistics are same-month annualised excess returns conditional on the regime; spells is the number of distinct runs.

		Equity (same-month)			Bond (same-month)				
Regime	Share	Ret.	Vol.	Sharpe	Ret.	Vol.	Sharpe	Avg. dur.	Spells
Calm	13.8%	+21.0%	8.1%	+2.60	+0.8%	3.5%	+0.22	4.7	13
Normal	77.1%	+9.9%	14.8%	+0.67	+3.9%	3.9%	+1.00	12.2	28
Stressed	9.0%	−40.9%	13.7%	−2.99	−8.4%	5.0%	−1.69	2.2	18

Member-level and feature-level diagnostics

Table D.18 reports the twenty-two members. Most of the members select the smallest jump penalty $\lambda = 2$: the persistence in the final labels comes from the coupling term and the monthly refit, not from a large λ. Members separate well (most separation scores between 4 and 7), with Sortino- and momentum-based families separating best. Table D.19 ranks the per-leg features by how strongly their standardised mean separates across the three regimes.

Table D.18: Coupled teacher: member persistence and separation, one row per feature family and distance norm (22 members), sorted by the separation score. (λ, η) is the modal selected pair.

Family	Norm	Calm	Normal	Stress.	Switches	Avg. dur.	Sep.	λ	η
sortino	L^2	0.03	0.84	0.13	59	7.4	+7.07	2	8
all	L^1	0.19	0.70	0.10	86	5.1	+6.24	2	1
sortino_trend	L^2	0.06	0.83	0.11	55	7.9	+5.96	2	4
sortino	L^1	0.19	0.70	0.11	77	5.7	+5.89	2	2
mom	L^2	0.05	0.83	0.12	55	7.9	+5.83	2	4
core	L^1	0.13	0.78	0.09	52	8.3	+5.70	2	1
sortino_dd	L^1	0.18	0.72	0.10	62	7.0	+5.69	2	1
sortino_dd	L^2	0.07	0.81	0.12	53	8.2	+5.67	2	1
sortino_short	L^1	0.19	0.71	0.09	71	6.1	+5.60	2	2
sortino_trend	L^1	0.28	0.61	0.11	77	5.7	+5.29	2	2
all	L^2	0.22	0.71	0.07	51	8.5	+5.10	2	2
mom	L^1	0.31	0.60	0.09	83	5.3	+5.04	2	1
risk	L^2	0.21	0.71	0.07	40	10.8	+4.60	2	2
dd_trend	L^1	0.11	0.81	0.08	46	9.4	+4.59	2	8
risk	L^1	0.21	0.71	0.08	39	11.1	+4.54	2	1
dd_trend	L^2	0.21	0.74	0.05	31	13.8	+4.41	2	1
core	L^2	0.25	0.67	0.08	48	9.0	+4.37	3	4
trend	L^2	0.27	0.66	0.07	53	8.2	+4.33	2	1
sortino_short	L^2	0.01	0.84	0.15	55	7.9	+4.18	2	4
dd	L^1	0.21	0.72	0.08	41	10.5	+4.07	2	4
dd	L^2	0.16	0.75	0.09	42	10.3	+3.96	2	1
trend	L^1	0.66	0.33	0.01	63	6.9	+2.15	2	8

Table D.19: Coupled teacher: feature state-separation. For each feature the standardised value is averaged within each regime; "Sep." is the spread (max − min) of the three group means.

Feature	Sep.	Calm	Normal	Stressed
eq_sortino_4	2.62	+1.72	−0.13	−0.90
eq_sortino_2	2.54	+1.72	−0.12	−0.82
eq_ma_cross	2.30	+0.89	−0.04	−1.41
eq_sortino_1	2.17	+1.56	−0.10	−0.61
eq_dd_diff	2.08	−1.05	+0.01	+1.03
eq_trend	2.02	+0.82	−0.09	−1.20
eq_sortino_l	1.99	+1.60	−0.07	−0.39
bd_trend	1.60	−0.09	+0.07	−1.52
bd_sortino_l	1.43	−0.06	+0.14	−1.30
eq_dd_lvl	1.41	−1.18	+0.24	+0.22
bd_dd_lvl	1.29	−0.45	−0.31	+0.84
bd_sortino_4	1.25	−0.13	+0.12	−1.13
bd_ma_cross	1.05	−0.11	+0.10	−0.95
bd_dd_diff	1.02	+0.05	−0.04	+0.98
bd_sortino_2	0.95	−0.17	+0.08	−0.87
bd_sortino_1	0.62	−0.23	+0.05	−0.58

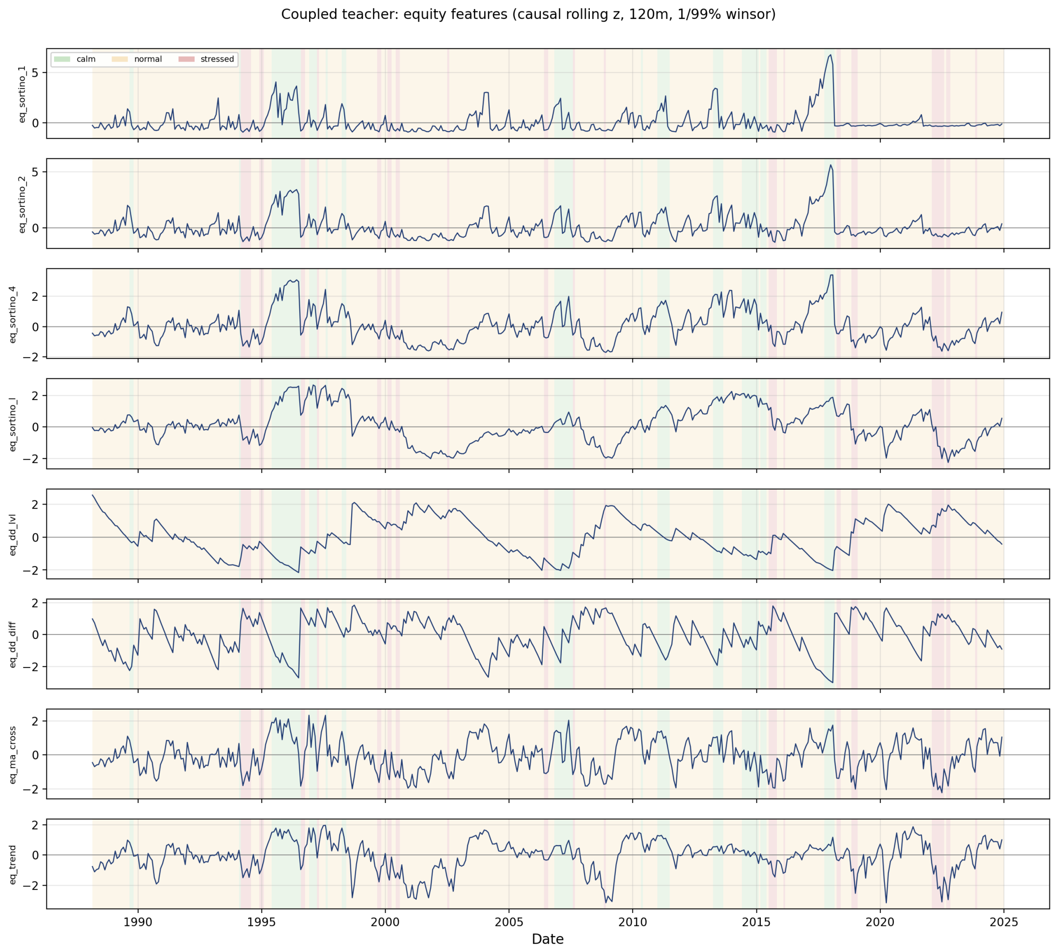


Figure D.15: Coupled teacher: equity features after the causal rolling z-score (120-month window, 1%/99% winsorisation).

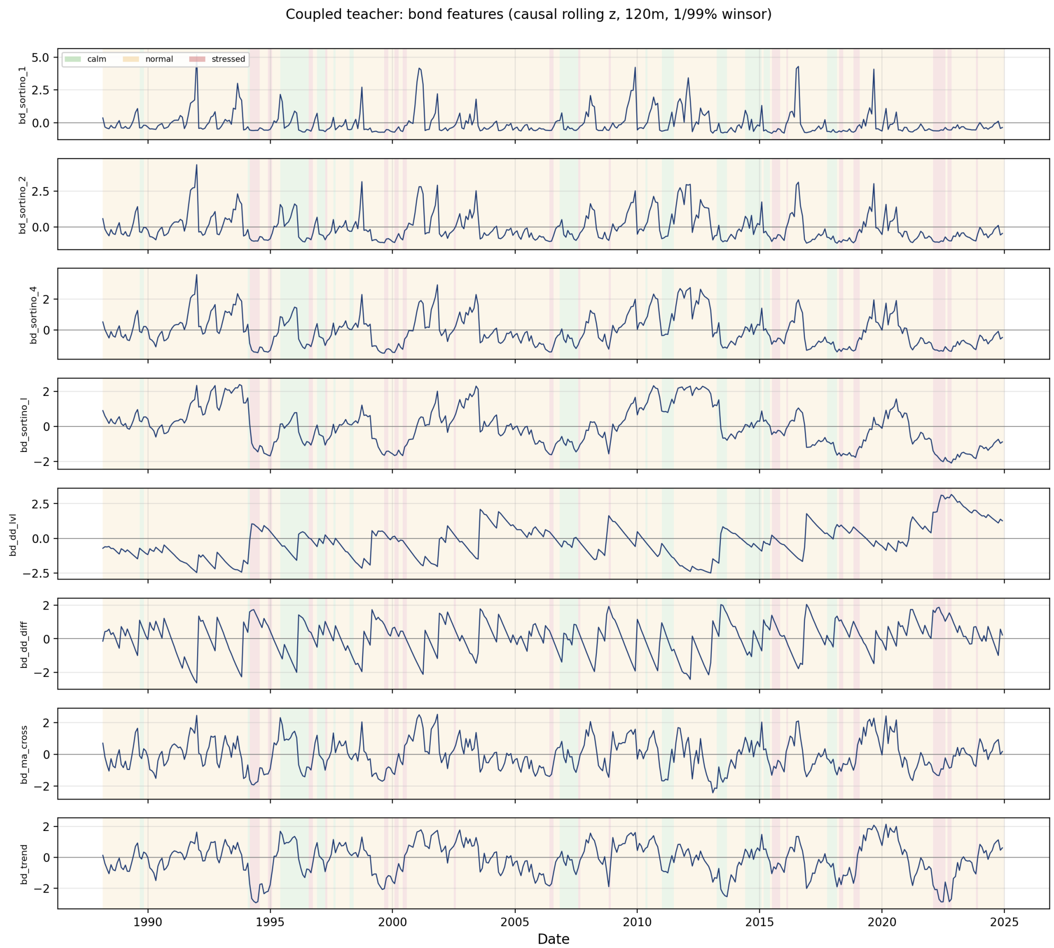


Figure D.16: Coupled teacher: bond features after the same causal rolling z-score. The flat left edge is the warm-up region before the standardisation window is populated.

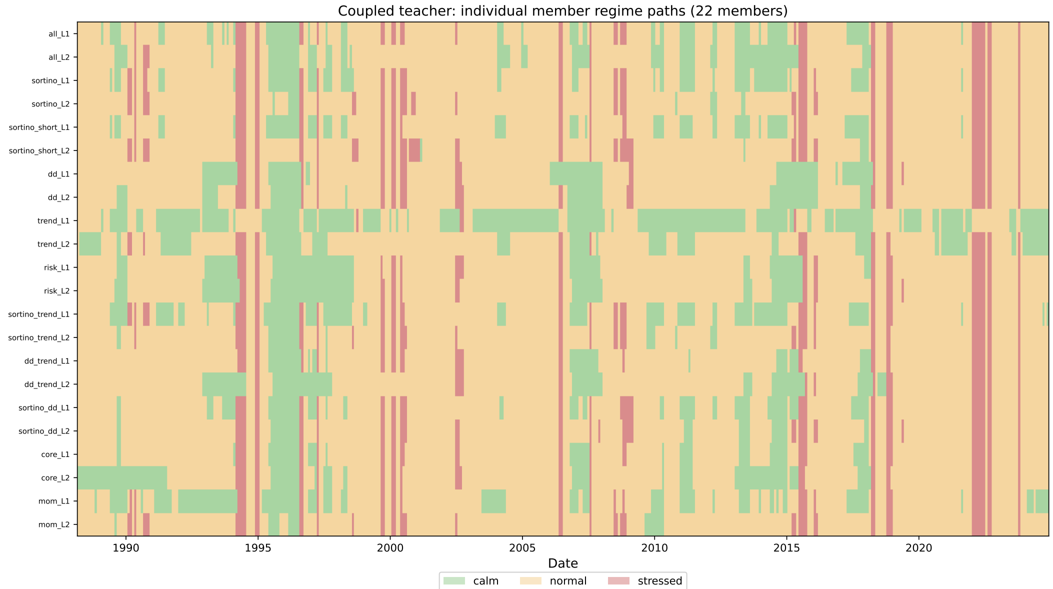


Figure D.17: Coupled teacher: per-member regime labels across time for the 22 ensemble members (rows), illustrating the agreement that produces the majority-vote label.

Figure D.18 shows how the final coupled label is formed from the candidate member paths under three scopes (L^2 members, L^1-medoid members, and all 22 combined), the coupled counterpart of the per-leg independent majority-vote construction in Figures C.10–C.11.

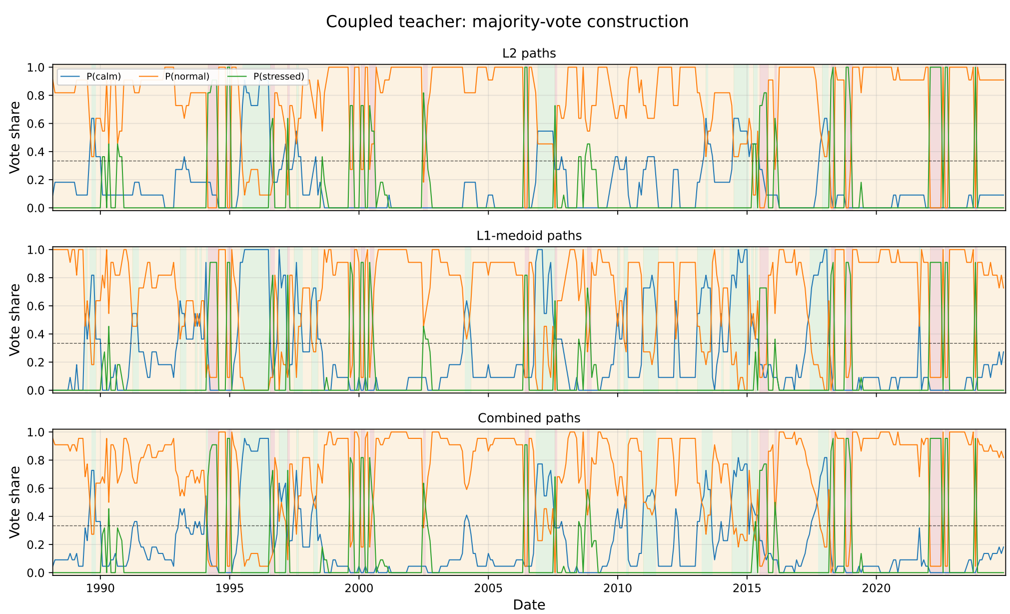


Figure D.18: Coupled teacher: majority-vote construction. Each panel restricts the ensemble to a scope (L^2 members, L^1-medoid members, or all 22 combined), shades the equity excess-return index by the scope's majority joint regime, and overlays the majority-class vote share on the right axis. The dashed line marks an absolute majority (0.5); below it the winning class is a plurality among the three regimes.

Jump-penalty diagnostics. To isolate the effect of persistence, the jump penalty is held fixed across a wider grid $\lambda \in \{2, 3, 4, 6, 8, 12, 16\}$ at $\eta = 2$ for the all-feature member under each norm (these paths form a sensitivity diagnostic, not the reported labels). A larger penalty produces longer, stickier regimes: switch counts fall and durations rise with λ while the calm share shrinks, and the separation score is highest at small penalties—consistent with the reported specification selecting $\lambda = 2$ for most of the members (Figures D.19–D.25).

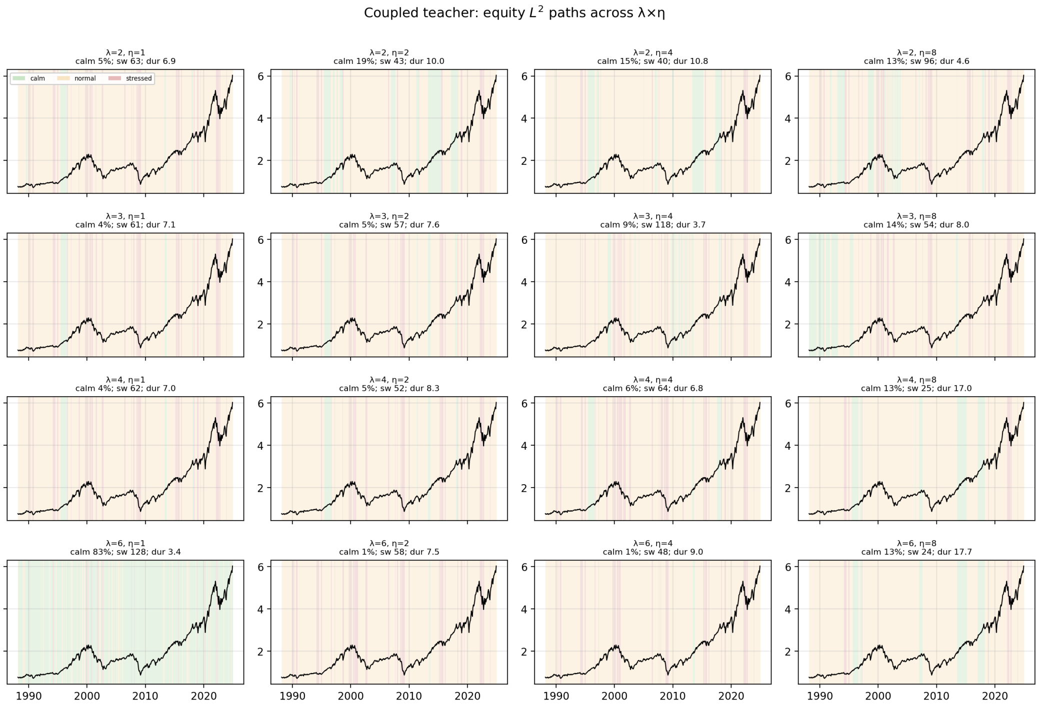


Figure D.19: Coupled teacher: equity L^2 paths across the jump penalty (each panel fixes λ, $\eta = 2$, and shades the equity excess-return index by the fitted regime).

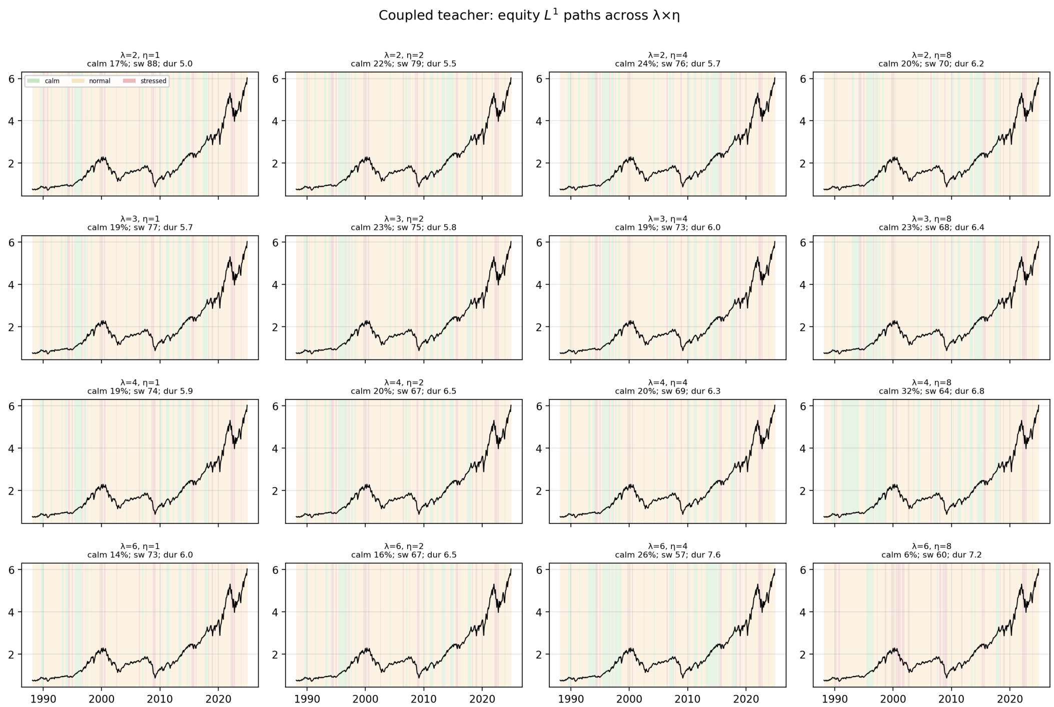


Figure D.20: Coupled teacher: equity L^1 paths across the jump penalty.

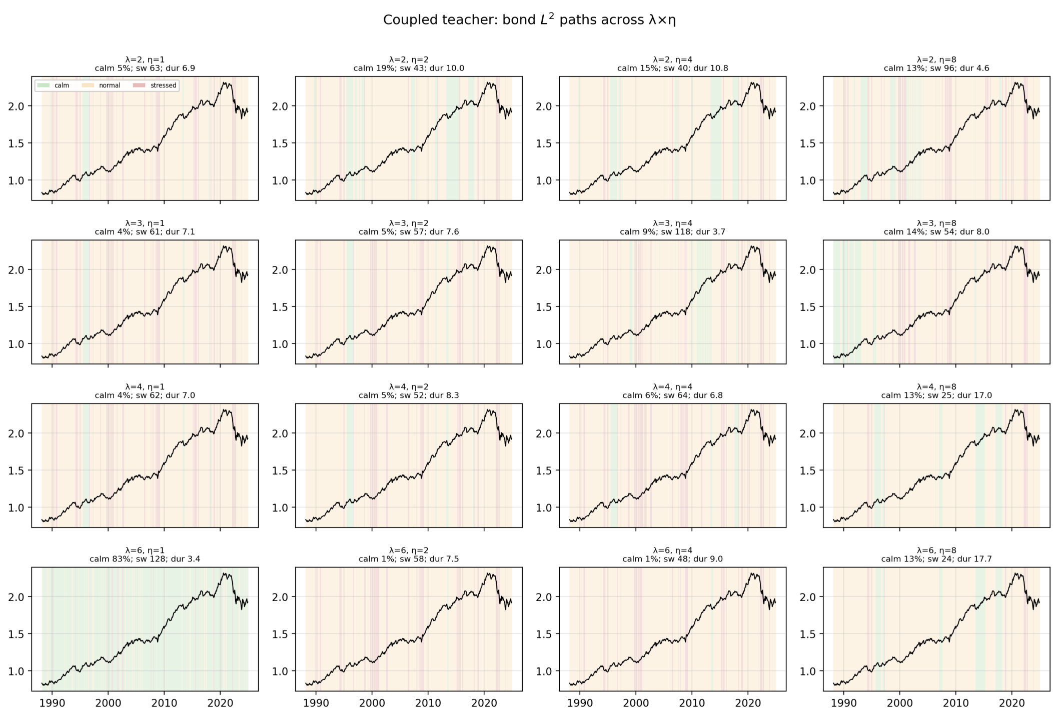


Figure D.21: Coupled teacher: bond L^2 paths across the jump penalty.

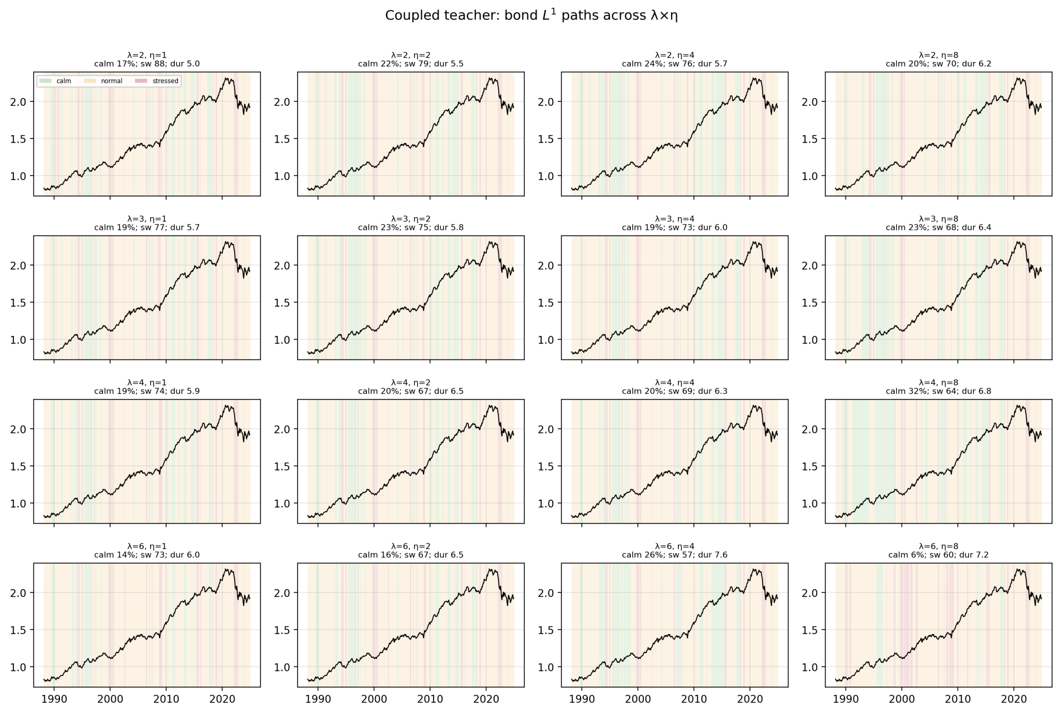


Figure D.22: Coupled teacher: bond L^1 paths across the jump penalty.

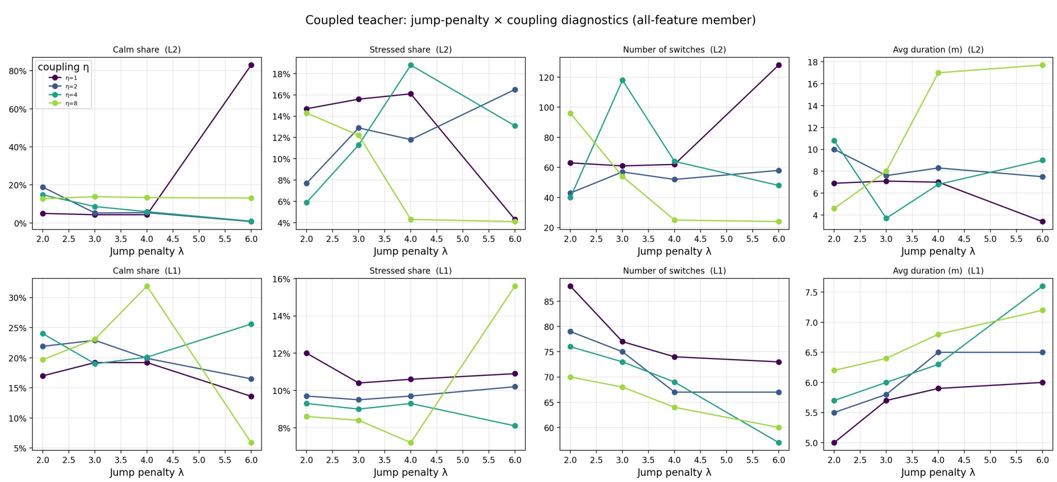


Figure D.23: Coupled teacher: jump-penalty diagnostics for the all-feature member—regime shares, switch count, average duration and separation score as functions of λ, for the L^1 and L^2 norms.

Figures D.24 and D.25 report the regime-conditional economics as a function of the jump penalty for each leg separately, the coupled counterpart of the per-leg independent jump-penalty diagnostics in Figures C.8 and C.9.

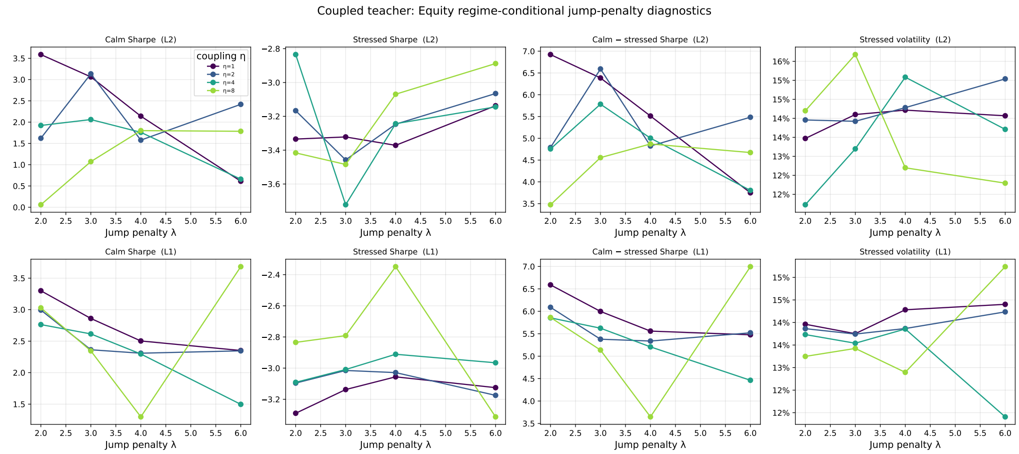


Figure D.24: Coupled teacher: equity regime-conditional jump-penalty diagnostics. For each norm (L^2, L^1-medoid) and coupling value η, the panels report the equity calm-state Sharpe, stressed-state Sharpe, calm-minus-stressed Sharpe separation, and stressed-state volatility as functions of the jump penalty λ.

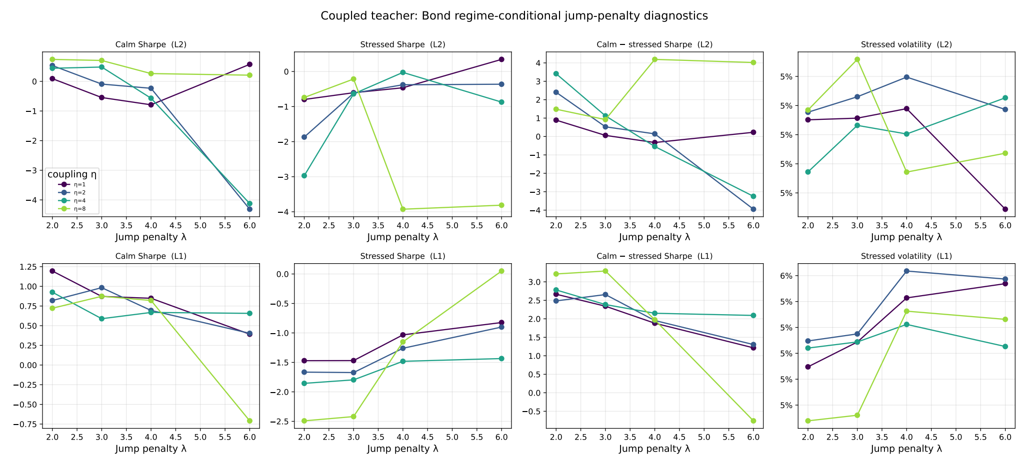


Figure D.25: Coupled teacher: bond regime-conditional jump-penalty diagnostics, defined as in Figure D.24 using the bond excess return.

E Student Prediction Details

E.1 Targets, features, and validation

The student models predict the one-month-ahead combined majority labels for the equity regime and the bond regime. The feature vector contains the z-score and smoothed z-score transformation of each monthly base predictor.

For each target and model family, prediction is performed with a rolling validation design. The initial training sample is 96 monthly observations, followed by a 36-month validation window. Candidate hyperparameters are selected on the validation data, and the selected setting is estimated on the full available sample before the forecast month. Validation ranking uses AUC, balanced accuracy, Brier score, net Sharpe ratio, drawdown improvement, volatility reduction, and turnover.

Predicted probabilities are averaged over seeds $\{101, 111, 121\}$.

E.2 Model grids

The logistic student model uses median imputation, feature standardisation, balanced class weights, and L^2 regularisation. The regularisation grid is

$$C \in \{0.03, 0.05, 0.10, 0.25, 0.50, 1.00\}. \tag{68}$$

The random-forest student model uses median imputation, 500 trees, bootstrap sampling, square-root feature subsampling, and balanced subsample class weights. The grid is

$$(d, \ell) \in \{(2, 10), (2, 20), (3, 10), (3, 20), (4, 20)\}, \tag{69}$$

where d is tree depth and ℓ is the minimum leaf size.

F Student regime prediction diagnostics

Table F.20: Out-of-sample student prediction summary

Label	Predictor set	Model	OOS sample	Fav. share	Mean $\hat{p}$	AUC	Bal. acc.	Brier	Log loss
Bond regime	z-score + smoothed z-score	Logistic regression	1996-02–2024-12	62.2%	0.594	0.879	0.841	0.125	0.515
Bond regime	z-score + smoothed z-score	Random forest	1996-02–2024-12	62.2%	0.601	0.873	0.800	0.147	0.471
Bond regime	smoothed z-score	Logistic regression	1996-02–2024-12	62.2%	0.574	0.892	0.831	0.128	0.462
Bond regime	smoothed z-score	Random forest	1996-02–2024-12	62.2%	0.590	0.849	0.784	0.153	0.484
Equity regime	z-score + smoothed z-score	Logistic regression	1995-03–2024-12	76.3%	0.707	0.929	0.846	0.088	0.296
Equity regime	z-score + smoothed z-score	Random forest	1995-03–2024-12	76.3%	0.658	0.903	0.831	0.107	0.361
Equity regime	smoothed z-score	Logistic regression	1995-03–2024-12	76.3%	0.700	0.918	0.862	0.086	0.307
Equity regime	smoothed z-score	Random forest	1995-03–2024-12	76.3%	0.660	0.902	0.837	0.105	0.355

Note. The table reports out-of-sample diagnostics from the selected rolling student models. AUC is the probability that a favourable-regime month receives a higher predicted probability than an unfavourable-regime month. Balanced accuracy is $\frac{1}{2}\{TP/(TP + FN) + TN/(TN + FP)\}$. The Brier score is $n^{-1}\sum_i(\hat{p}_i - y_i)^2$, and log loss is $-n^{-1}\sum_i\{y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\}$, where y_i is the realised teacher label and $\hat{p}_i$ is the predicted favourable-regime probability.

Equity-regime prediction is strongest for logistic regression. The logistic model with the combined z-score and smoothed z-score predictor set reaches an AUC of 0.929, balanced accuracy of 0.846, Brier score of 0.088, and log loss of 0.296. The smoothed-only logistic model gives the highest balanced accuracy for the equity label, 0.862, with a similar AUC of 0.918. Bond-regime prediction is strongest for logistic regression with the smoothed-only predictor set, which reaches an AUC of 0.892 and balanced accuracy of 0.831. Random forests also predict both labels, but their AUC and balanced-accuracy values are below the corresponding logistic-regression values in most specifications.

Figures F.26 and F.26 plot the out-of-sample probability paths. The shaded background marks the realised teacher regime, and the black line is the student-model probability assigned to the favourable regime.

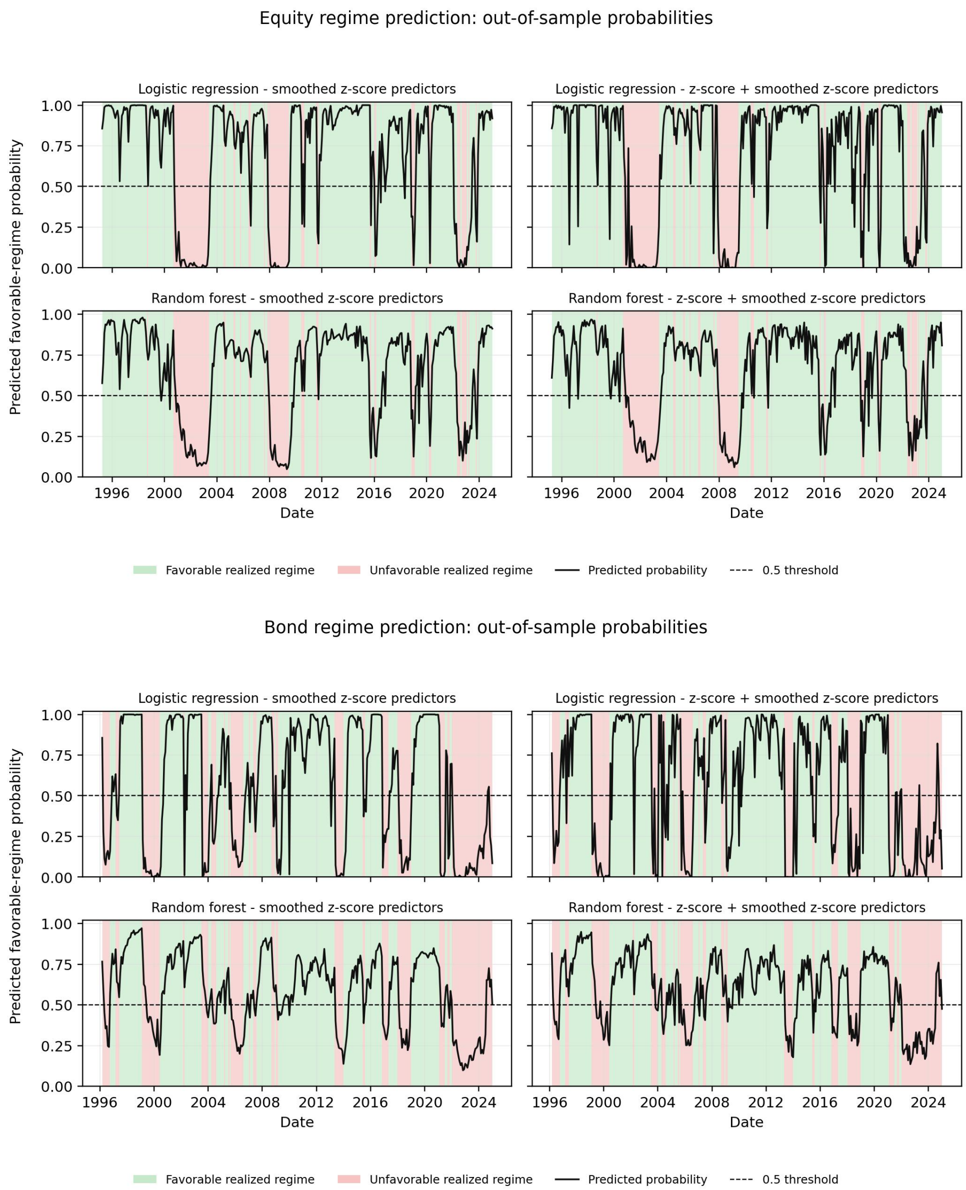


Figure F.26: INDEPENDENT student out-of-sample probability paths. The top panel reports equity favourable-regime probabilities, and the bottom panel reports bond favourable-regime probabilities. Shading marks the realised teacher label, the black line reports the predicted favourable-regime probability, and the dashed line marks the 0.5 threshold.

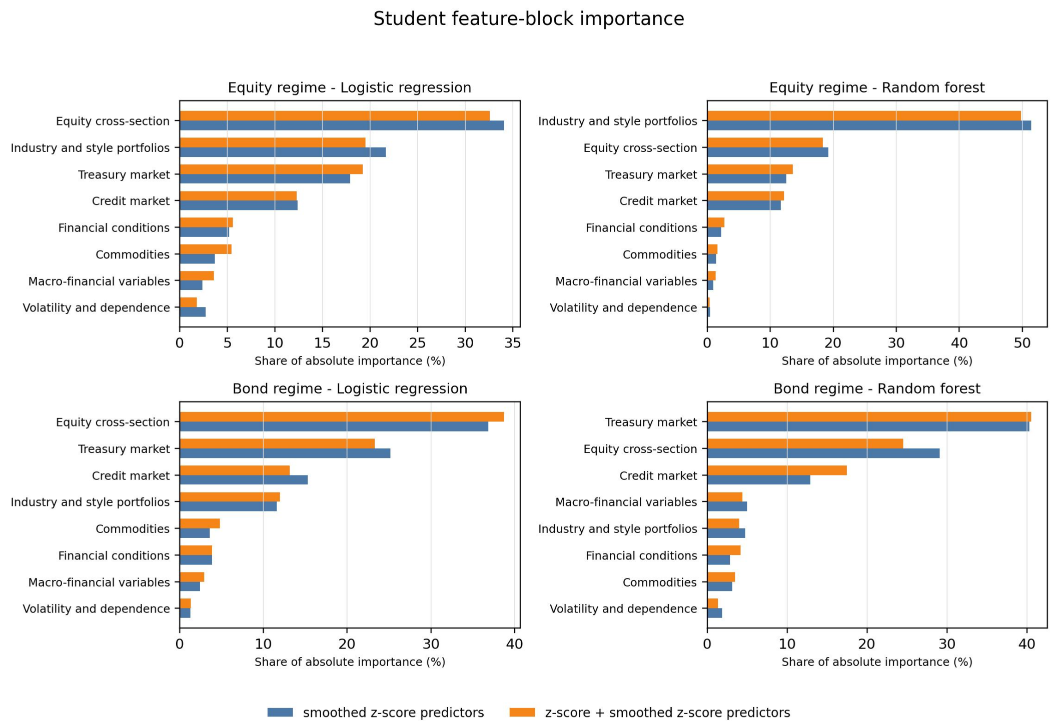


Figure F.27: INDEPENDENT student feature-block relevance. Logistic-regression relevance is measured by absolute coefficient magnitude. Random-forest relevance is measured by impurity-based feature importance. Values are aggregated within predictor blocks.

Table F.21 reports the most frequently selected hyperparameter setting over the rolling out-of-sample test months. The table reports which specification is most often selected within each target, predictor set, and model class.

Table F.21: Most frequent selected student-model specification

Label	Predictor set	Model	Most frequent setting	Sel. share	Months
Bond regime	z-score + smoothed z-score predictors	Logistic regression	$C = 0.03$, class weight=balanced	56.2%	347
Bond regime	z-score + smoothed z-score predictors	Random forest	trees=500, depth=3, leaf=10, max features=sqrt	32.3%	347
Bond regime	smoothed z-score predictors	Logistic regression	$C = 0.03$, class weight=balanced	38.6%	347
Bond regime	smoothed z-score predictors	Random forest	trees=500, depth=2, leaf=10, max features=sqrt	30.8%	347
Equity regime	z-score + smoothed z-score predictors	Logistic regression	$C = 0.03$, class weight=balanced	60.1%	358
Equity regime	z-score + smoothed z-score predictors	Random forest	trees=500, depth=3, leaf=10, max features=sqrt	67.6%	358
Equity regime	smoothed z-score predictors	Logistic regression	$C = 0.03$, class weight=balanced	56.4%	358
Equity regime	smoothed z-score predictors	Random forest	trees=500, depth=3, leaf=10, max features=sqrt	70.7%	358

Note. The table reports the most frequently selected hyperparameter setting across out-of-sample rolling test months. The selection share is the fraction of out-of-sample months in which that setting is selected for the corresponding regime label, model class, and predictor set.

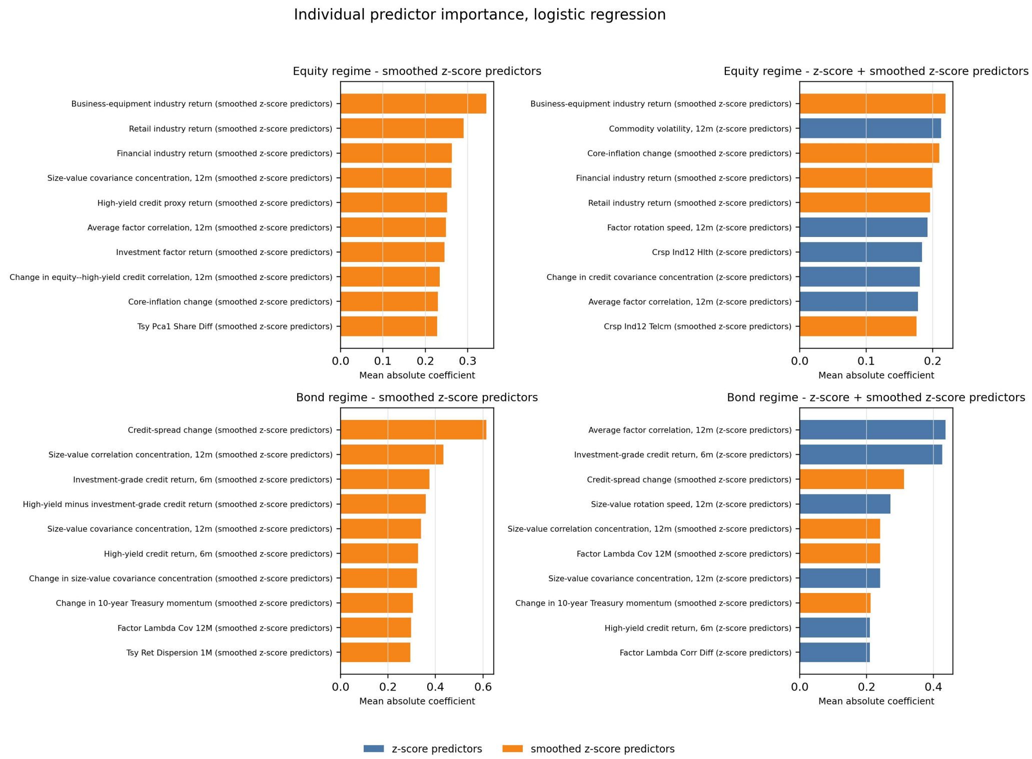


Figure F.28: Student prediction individual predictor importance.

Note. Individual predictor importance uses model-level importances from the rolling prediction exercise. For logistic regression, importance is the absolute value of the fitted coefficient after preprocessing. For random forests, importance is impurity-based feature importance. Mean absolute importance is averaged across rolling selected-model fits and seeds. When both the z-score and smoothed z-score transformations of the same predictor are available within a predictor set, the plotted value is the larger mean absolute importance for that economic predictor. Each panel shows the eight largest economic predictors by the maximum mean absolute importance across the two predictor sets for the same label and model.

The individual-predictor diagnostics are consistent with the block-level figure. Equity-regime prediction relies heavily on industry returns, style-portfolio returns, equity-factor information, and cross-market credit correlation. Bond-regime prediction relies on credit-spread changes, investment-grade and high-yield credit returns, Treasury returns, and equity-factor dependence measures. These predictors explain why the equity cross-section, Treasury market, and credit market blocks receive high importance shares in the block-level results.

The feature-block definitions are: equity cross-section contains equity factors, size-value variables, and equity dispersion or dependence measures; Treasury market contains Treasury returns, yield-curve and rate changes, bond momentum, and bond-market dependence; credit market contains investment-grade and high-yield credit returns, credit spreads, dispersion, and credit dependence; commodities contain commodity returns, momentum, volatility, Sharpe ratios, and commodity dependence; financial conditions contain NFCI, adjusted NFCI, their changes, and their spread; macro-financial variables contain activity, inflation, and sentiment variables; volatility and dependence contains volatility and remaining correlation or concentration measures; and industry and style portfolios contain remaining industry and style-portfolio returns.

G Coupled student: estimation details

Coupled student. The target is the next-month three-class regime. At each refit the five predictors most correlated with the next-month regime over the trailing window are kept, and the current-regime indicators and run-length are appended. Two classifiers are fitted with balanced class weights: an L_2-penalised multinomial logistic regression and a random forest. Estimation is walk-forward (132-month training, 60-month validation by balanced accuracy, 6-month test), giving the out-of-sample span 1996-02 to 2024-10.

Coupled student forecast diagnostics. Table G.22 reports the out-of-sample probabilistic metrics for the two coupled students, and Table G.23 reports the most frequently selected hyper-parameter setting across the walk-forward windows. The predicted three-class probability paths are shown in Figure G.29. Figure G.30 reports block-level relevance, and Figure G.31 reports individual-predictor importance.

Table G.22: Coupled student: out-of-sample three-class forecast metrics (OOS 1996-02 to 2024-10). Balanced accuracy averages per-class recall; AUC is the macro one-vs-rest area; Brier and log loss are probabilistic scores (lower is better).

Model	Accuracy	Balanced acc.	AUC (macro OVR)	Brier	Log loss
Logistic regression	0.678	0.677	0.770	0.541	1.062
Random forest	0.748	0.595	0.776	0.383	0.686

Table G.23: Coupled student: most frequently selected hyper-parameter setting across the walk-forward out-of-sample windows, with the share of windows in which it was chosen.

Model	Most frequent setting	Selection share	OOS windows
Logistic regression	$C = 0.03$	81.0%	58
Random forest	$d = 4$, leaf $= 5$, $n = 300$	72.4%	58

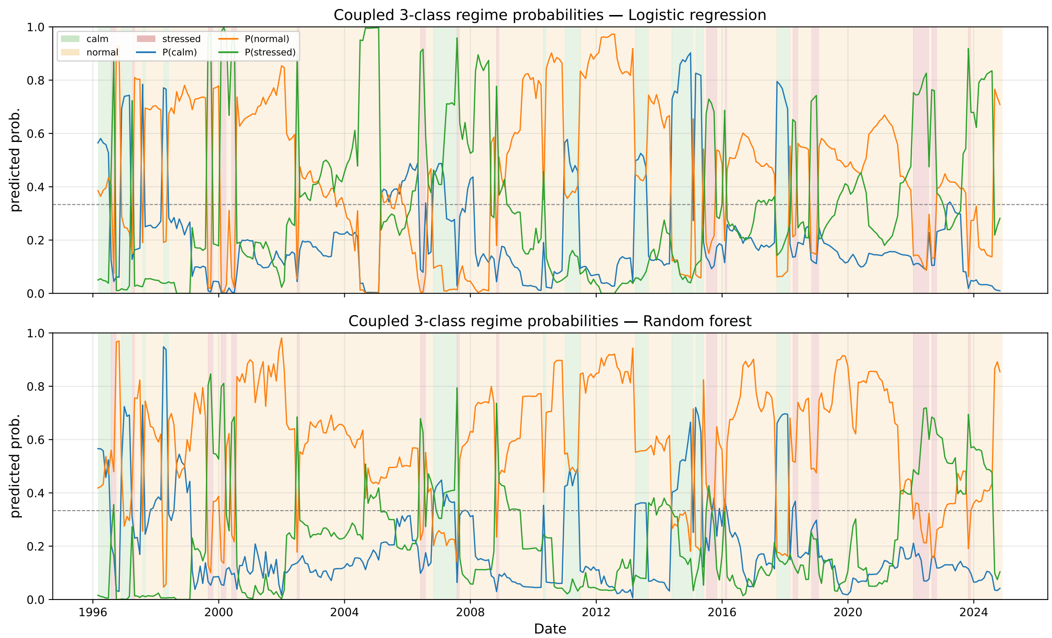


Figure G.29: COUPLED student out-of-sample probability paths for the three joint regimes (calm, normal, stressed), shown for the logistic and random-forest students. Shading marks the realised teacher regime, the black lines report the predicted regime probabilities, and the dashed line marks the 0.5 threshold.

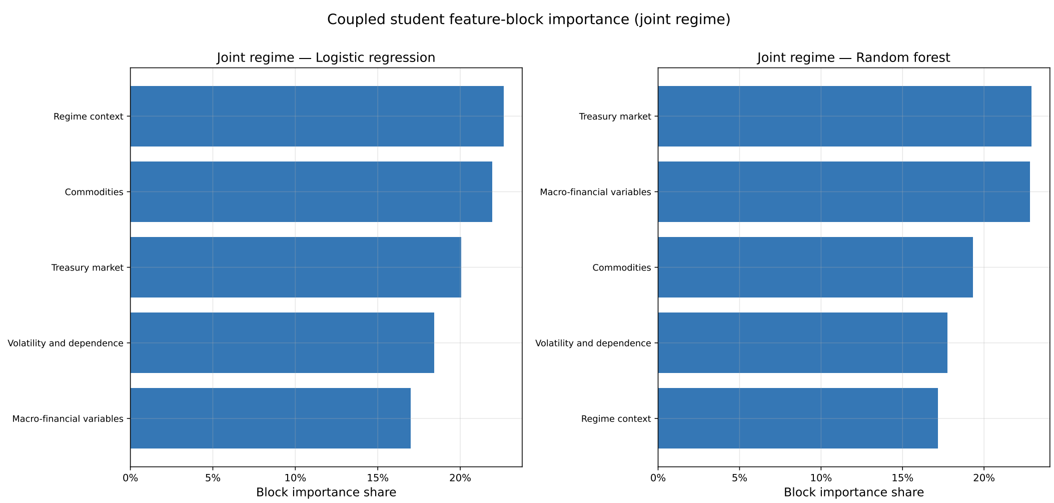


Figure G.30: COUPLED student feature-block relevance for the three-class next-month regime forecast. Logistic-regression relevance is measured by absolute coefficient magnitude. Random-forest relevance is measured by impurity-based feature importance. Values are aggregated within predictor blocks.

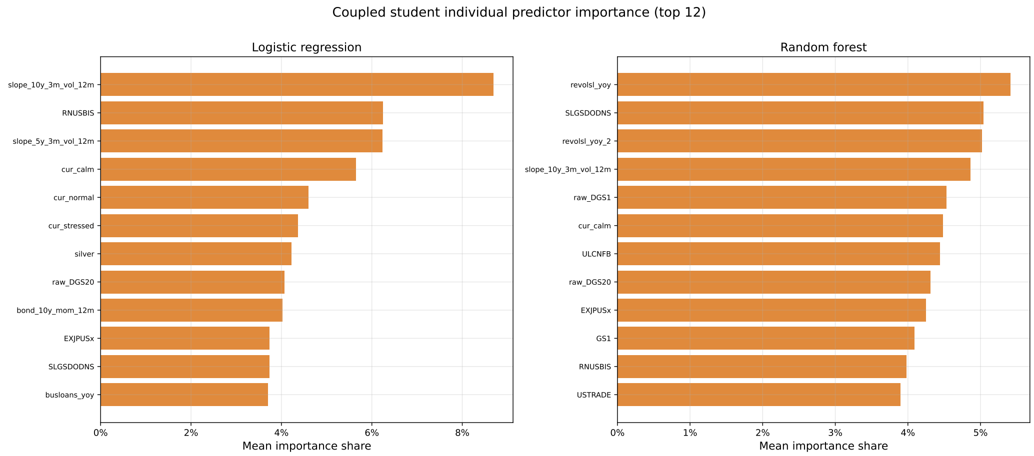


Figure G.31: Coupled student: individual-predictor importance for the next-month regime forecast.

H Allocation Details

H.1 Probability smoothing and shrinkage

Let $p_{i,m}$ be the predicted favourable-regime probability for asset $i \in \{E, B\}$. Predicted probabilities are used directly or smoothed with exponential half-life

$$h \in \{0, 3, 6\}. \tag{70}$$

For each asset and state, historical next-month excess-return moments are estimated from past observations and shrunk toward pre-prediction unconditional moments:

$$a_{i,s,m} = \frac{n_{i,s,m}}{n_{i,s,m} + 36}, \tag{71}$$

$$\tilde{\mu}_{i,s,m} = a_{i,s,m}\hat{\mu}_{i,s,m} + (1 - a_{i,s,m})\mu_{i,0}, \tag{72}$$

$$\tilde{\sigma}^2_{i,s,m} = a_{i,s,m}\hat{\sigma}^2_{i,s,m} + (1 - a_{i,s,m})\sigma^2_{i,0}. \tag{73}$$

H.2 Probability-to-moment mapping

The probability-mixture mean and variance are

$$\mu_{i,m} = (1 - p_{i,m})\tilde{\mu}_{i,0,m} + p_{i,m}\tilde{\mu}_{i,1,m}, \tag{74}$$

$$\sigma^2_{i,m} = (1 - p_{i,m})(\tilde{\sigma}^2_{i,0,m} + \tilde{\mu}^2_{i,0,m}) + p_{i,m}(\tilde{\sigma}^2_{i,1,m} + \tilde{\mu}^2_{i,1,m}) - \mu^2_{i,m}. \tag{75}$$

The equity–bond correlation is estimated from expanding historical returns and shrunk toward the pre-prediction correlation using the same shrinkage form. The covariance matrix is then formed from the conditional volatilities and the shrunk equity–bond correlation.

H.3 Portfolio grid and net returns

The reported L^1 allocation uses the cash-allowed long-only budget

$$w_E + w_B \leq 1,$$

with equity and bond weights each capped at one and selected on a 0.02 grid. The reported allocation grid varies

$$\gamma \in \{0.25, 0.5, 1, 2, 3, 5\},$$

$$c \in \{0.0005, 0.0010, 0.0025\},$$

$$h \in \{0, 3, 6\},$$

and

$$\lambda_{\text{trade}} \in \{0, 0.0005, 0.001, 0.0025, 0.005, 0.0075, 0.01, 0.015, 0.02, 0.03, 0.05\}.$$

Here h is the probability-smoothing half-life and λ_{trade} is the turnover penalty in the allocation objective. Monthly net portfolio return is

$$r^{p}_{m+1} = w_{E,m} r_{E,m+1} + w_{B,m} r_{B,m+1} - c\|w_m - w_{m-1}\|_1. \tag{76}$$

H.4 Allocation diagnostics and parameter sensitivity

This appendix reports the allocation diagnostics behind the dynamic allocation results in Section 5.1. Table H.24 reports the main independent allocation summary using 10 bps one-way transaction costs. Table H.25 gives the baseline parameter settings for the three dynamic strategies. Figure H.32 reports the corresponding predicted probabilities, portfolio weights, turnover, wealth, and drawdown.

Table H.24: Allocation performance summary

Role	Model	Predictor set	Ann. ret.	Ann. vol.	Sharpe	Max DD	Ann. turn.	Final
60/40 benchmark	--	--	4.5%	9.6%	0.47	-36.2%	19.7%	359.35
High-return dynamic	Logistic regression	smoothed z-score	7.9%	10.6%	0.74	-19.3%	152.1%	814.18
Defensive dynamic	Random forest	smoothed z-score	4.4%	5.8%	0.77	-14.9%	59.8%	341.89
Low-turnover dynamic	Logistic regression	z-score + smoothed z-score	5.0%	7.2%	0.69	-13.0%	51.4%	386.58

Note. Dynamic strategy returns are net of transaction costs. The 60/40 benchmark is rebalanced monthly and reported net of the same one-way transaction-cost convention. The benchmark and dynamic strategies are evaluated using the same monthly return, drawdown, turnover, and final-wealth definitions.

Table H.25: Selected dynamic allocation specifications

Role	Model	Predictor set	γ	τ	Cost	Sm.	Ann. ret.	Sharpe	Max DD	Turn.	Final
High-return	Logistic	smoothed z-score	0.5	0	10 bps	0	7.9%	0.74	-19.3%	152.1%	814.2
Defensive	Random forest	smoothed z-score	3	0.001	10 bps	0	4.4%	0.77	-14.9%	59.8%	341.9
Low-turnover	Logistic	z-score + smoothed z-score	3	0.001	10 bps	3	5.0%	0.69	-13.0%	51.4%	386.6

Note. Parameter settings for the three dynamic strategies shown in the main allocation figures. "Role" abbreviates High-return/Defensive/Low-turnover dynamic; "Logistic" is logistic regression; predictor sets are smoothed z-score and z-score + smoothed z-score. γ is risk aversion, τ the turnover penalty, "Sm." the probability-smoothing half-life (months), "Turn." annual turnover, "Final" terminal wealth (start = 100).

Headline-row interpretation. The three independent dynamic rows are representative points from the allocation grid. They are selected to illustrate distinct allocation trade-offs. The high-return row shows a logistic-regression specification with high terminal wealth and materially lower drawdown than the 60/40 benchmark. The defensive row shows a random-forest specification with lower volatility, lower drawdown, and a higher Sharpe ratio. The low-turnover row shows a logistic-regression specification with probability smoothing and annual turnover near 51%. The full parameter sweeps around these rows are reported below.

H.5 Additional independent benchmarks

The additional benchmarks use the same monthly excess-return columns as the independent allocation sample ending in 2024-10. Cash is treated as zero excess return because the independent allocation stage uses equity and bond excess returns but no separate cash-return column. These benchmarks are included to compare the regime-aware rules with simple dynamic or single-asset alternatives that do not use regime forecasts.

Benchmark	Ann. return	Ann. vol.	Sharpe	Max drawdown	Ann. turnover	Final wealth
60/40 benchmark	4.5%	9.6%	0.47	-36.2%	19.7%	359.35
Equity-only	6.9%	15.4%	0.44	-62.2%	0.0%	505.52
Bond-only	2.1%	4.1%	0.51	-21.3%	0.0%	177.25
Volatility-targeted 60/40	3.6%	8.9%	0.41	-35.8%	8.0%	229.92

Table H.26: Additional independent dynamic-allocation benchmarks. Returns are monthly excess returns over the independent allocation sample ending in 2024-10. Cash is treated as zero excess return. The volatility-targeted 60/40 rule uses a trailing 36-month realised volatility estimate, a 10% annual volatility target, and leverage capped at one.

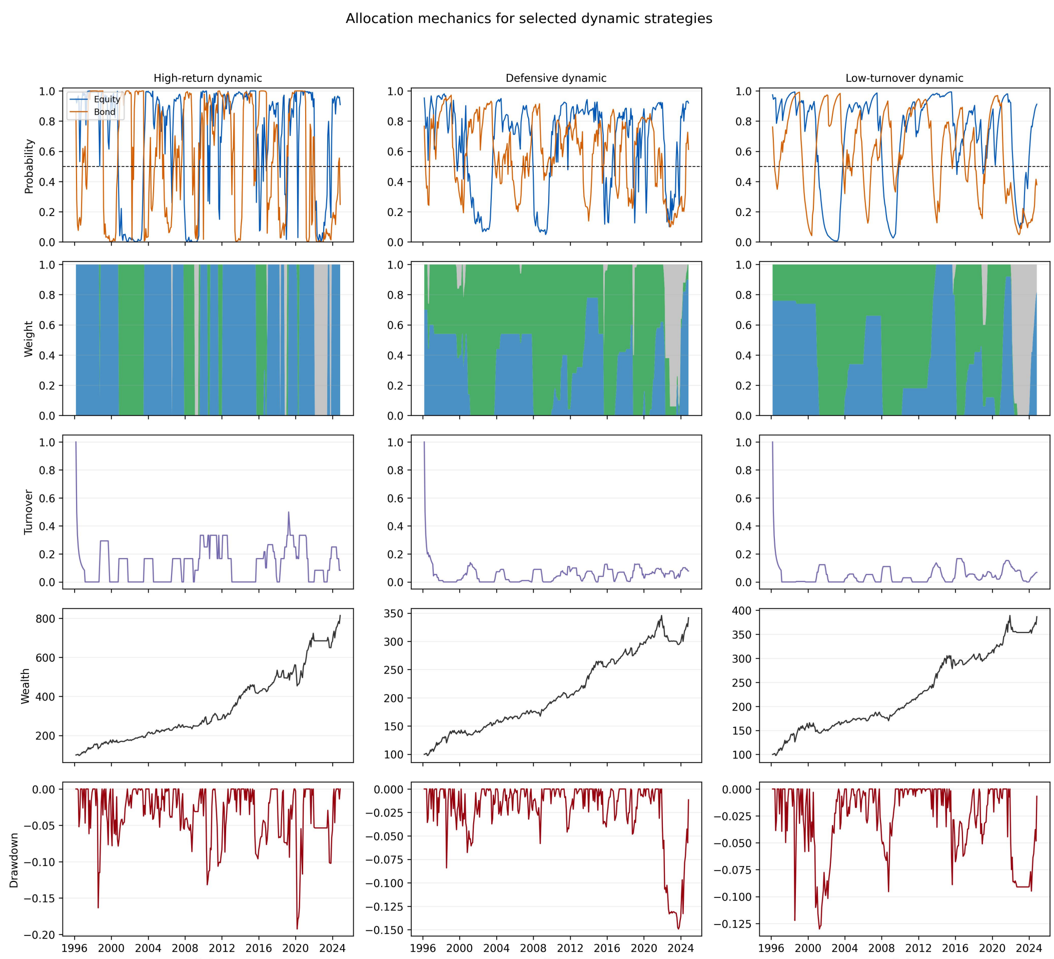


Figure H.32: Allocation mechanics for selected dynamic strategies

Note: The panels report predicted favourable-regime probabilities, portfolio weights, rolling 12-month turnover, wealth, and drawdown for the three selected dynamic strategies.

The mechanics figure shows that the three dynamic strategies use the same regime forecasts in different ways. The high-return strategy takes large equity positions and accepts higher turnover. The defensive strategy uses more bond exposure and reduces equity exposure when forecasts weaken. The low-turnover strategy changes allocations more slowly and uses probability smoothing to reduce trading.

H.5.1 Parameter sensitivity around selected strategies

The sensitivity checks vary allocation parameters around the three selected dynamic strategies. In each one-parameter sweep, the reported parameter varies and the remaining allocation parameters are fixed at the strategy-specific baseline values in Table H.25. The transaction-cost and turnover-penalty table varies these two parameters jointly and keeps the remaining parameters fixed at the same baseline values.

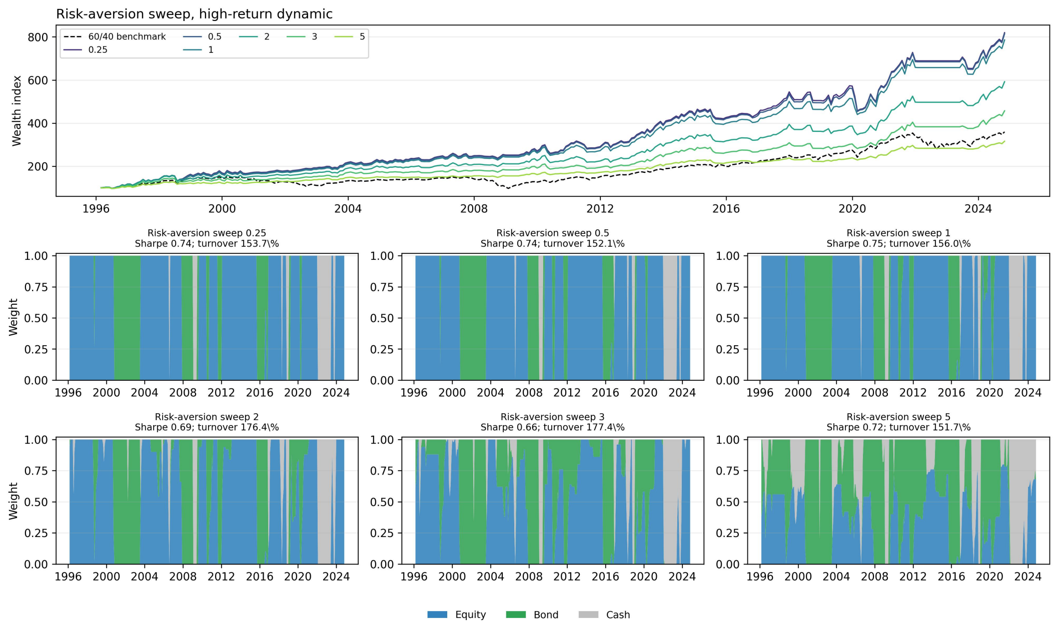


Figure H.33: Risk-aversion sweep, high-return dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.27: Risk-aversion sweep, high-return dynamic

Risk aversion	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0.25	7.9%	10.7%	0.74	-20.3%	-11.1%	819.52	153.7%
0.5	7.9%	10.6%	0.74	-19.3%	-10.8%	814.18	152.1%
1	7.7%	10.4%	0.75	-16.3%	-9.7%	785.79	156.0%
2	6.7%	9.7%	0.69	-16.3%	-9.3%	591.97	176.4%
3	5.7%	8.5%	0.66	-14.3%	-8.5%	456.98	177.4%
5	4.2%	5.8%	0.72	-8.8%	-5.6%	315.69	151.7%

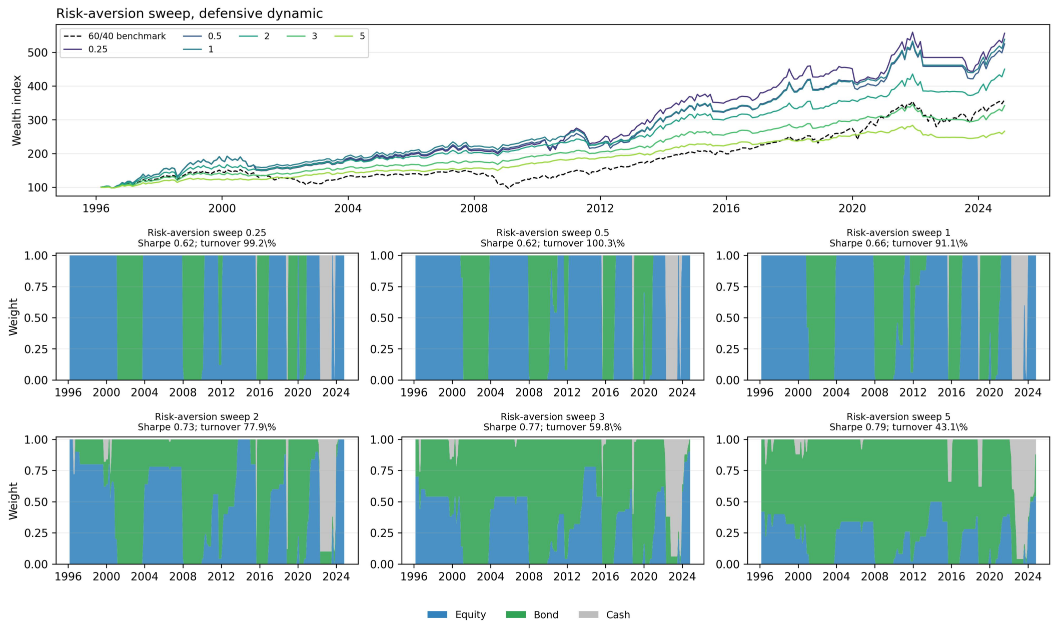


Figure H.34: Risk-aversion sweep, defensive dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.28: Risk-aversion sweep, defensive dynamic

Risk aversion	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0.25	6.6%	10.6%	0.62	-22.4%	-10.5%	556.31	99.2%
0.5	6.3%	10.2%	0.62	-20.9%	-10.0%	524.75	100.3%
1	6.3%	9.7%	0.66	-17.5%	-9.9%	538.47	91.1%
2	5.5%	7.6%	0.73	-14.6%	-7.5%	449.91	77.9%
3	4.4%	5.8%	0.77	-14.9%	-5.5%	341.89	59.8%
5	3.5%	4.4%	0.79	-13.7%	-4.2%	265.92	43.1%

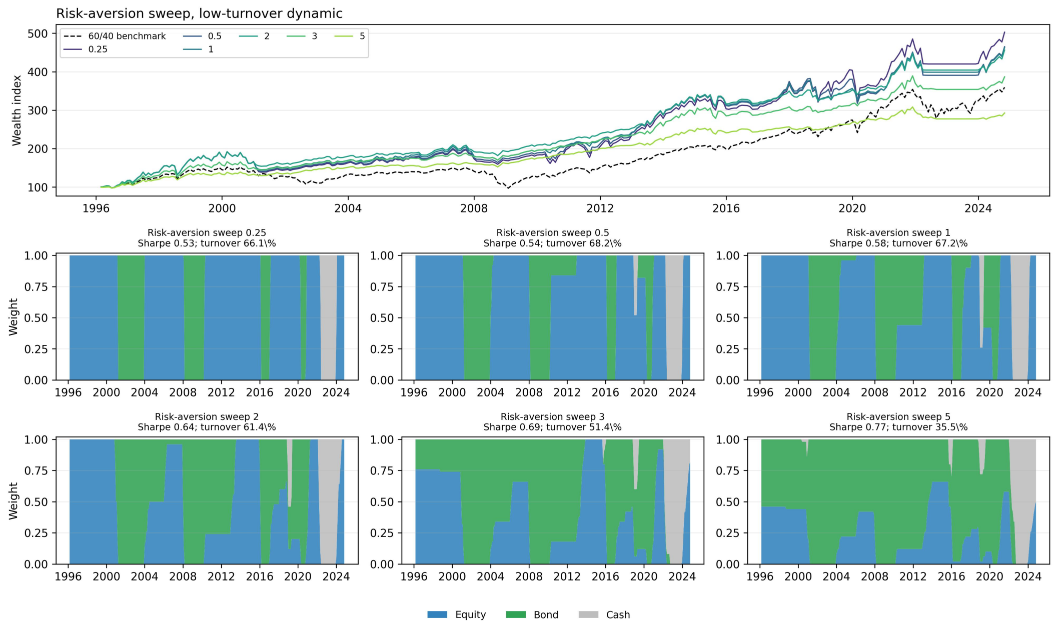


Figure H.35: Risk-aversion sweep, low-turnover dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.29: Risk-aversion sweep, low-turnover dynamic

Risk aversion	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0.25	6.3%	11.9%	0.53	-27.7%	-11.7%	502.63	66.1%
0.5	6.0%	11.1%	0.54	-26.6%	-11.1%	464.83	68.2%
1	5.8%	10.0%	0.58	-24.4%	-10.6%	462.84	67.2%
2	5.7%	8.9%	0.64	-19.7%	-9.0%	456.02	61.4%
3	5.0%	7.2%	0.69	-13.0%	-7.1%	386.58	51.4%
5	3.9%	5.0%	0.77	-10.3%	-4.5%	292.53	35.5%

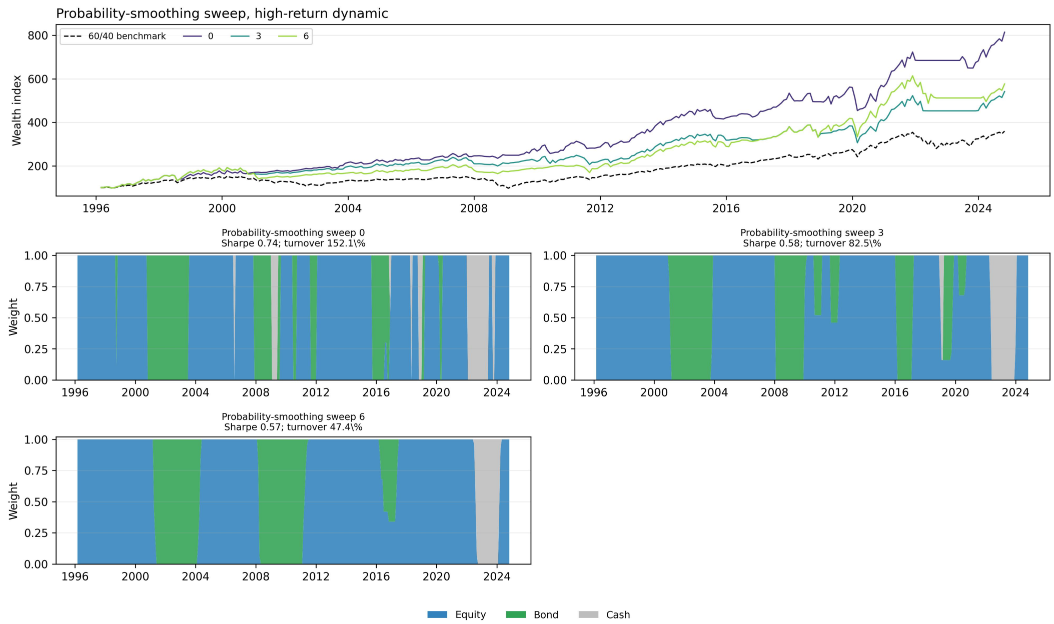


Figure H.36: Probability-smoothing sweep, high-return dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

Table H.30: Probability-smoothing sweep, high-return dynamic

Smoothing half-life	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	7.9%	10.6%	0.74	-19.3%	-10.8%	814.18	152.1%
3	6.5%	11.3%	0.58	-21.2%	-11.5%	541.38	82.5%
6	6.8%	12.0%	0.57	-27.0%	-11.7%	576.31	47.4%

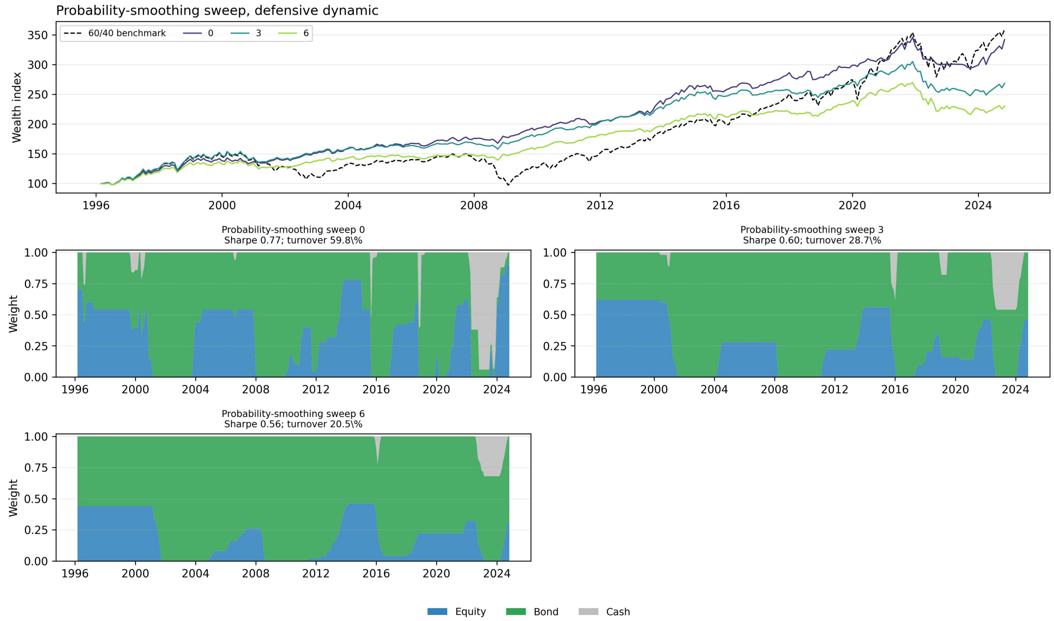


Figure H.37: Probability-smoothing sweep, defensive dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

Table H.31: Probability-smoothing sweep, defensive dynamic

Smoothing half-life	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	4.4%	5.8%	0.77	-14.9%	-5.5%	341.89	59.8%
3	3.6%	6.0%	0.60	-18.8%	-6.0%	268.16	28.7%
6	3.0%	5.4%	0.56	-20.0%	-5.2%	230.13	20.5%

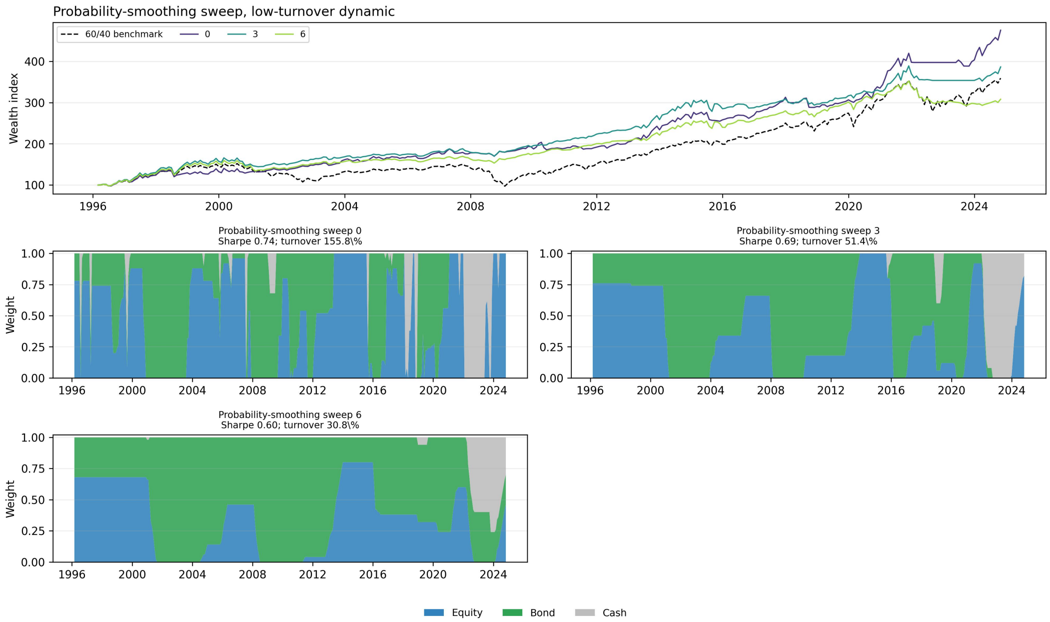


Figure H.38: Probability-smoothing sweep, low-turnover dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

Table H.32: Probability-smoothing sweep, low-turnover dynamic

Smoothing half-life	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	5.7%	7.7%	0.74	-11.9%	-7.6%	475.79	155.8%
3	5.0%	7.2%	0.69	-13.0%	-7.1%	386.58	51.4%
6	4.2%	6.9%	0.60	-16.9%	-7.0%	308.15	30.8%

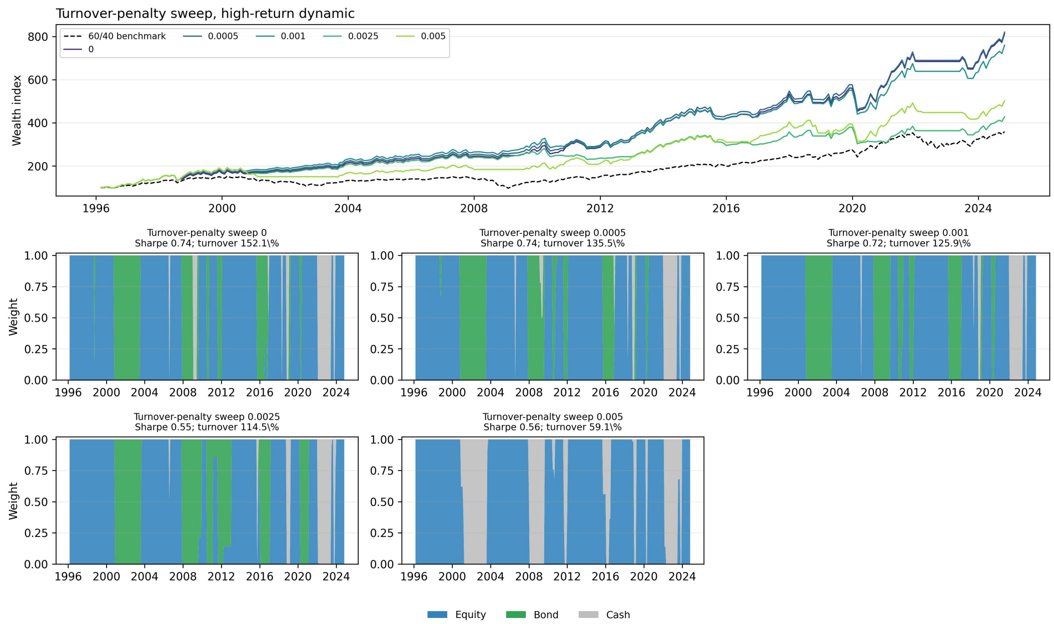


Figure H.39: Turnover-penalty sweep, high-return dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.33: Turnover-penalty sweep, high-return dynamic

Turnover penalty	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	7.9%	10.6%	0.74	-19.3%	-10.8%	814.18	152.1%
0.0005	7.9%	10.7%	0.74	-20.3%	-11.1%	820.42	135.5%
0.001	7.6%	10.7%	0.72	-20.3%	-11.1%	760.05	125.9%
0.0025	5.6%	10.2%	0.55	-20.3%	-11.2%	428.11	114.5%
0.005	6.3%	11.2%	0.56	-23.6%	-11.3%	502.79	59.1%
0.0075	4.9%	12.1%	0.40	-39.3%	-11.7%	326.37	48.1%
0.01	0.9%	5.5%	0.17	-29.5%	-5.9%	124.29	8.8%
0.015	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.02	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.03	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.05	0.0%	0.0%		0.0%	0.0%	100.00	0.0%

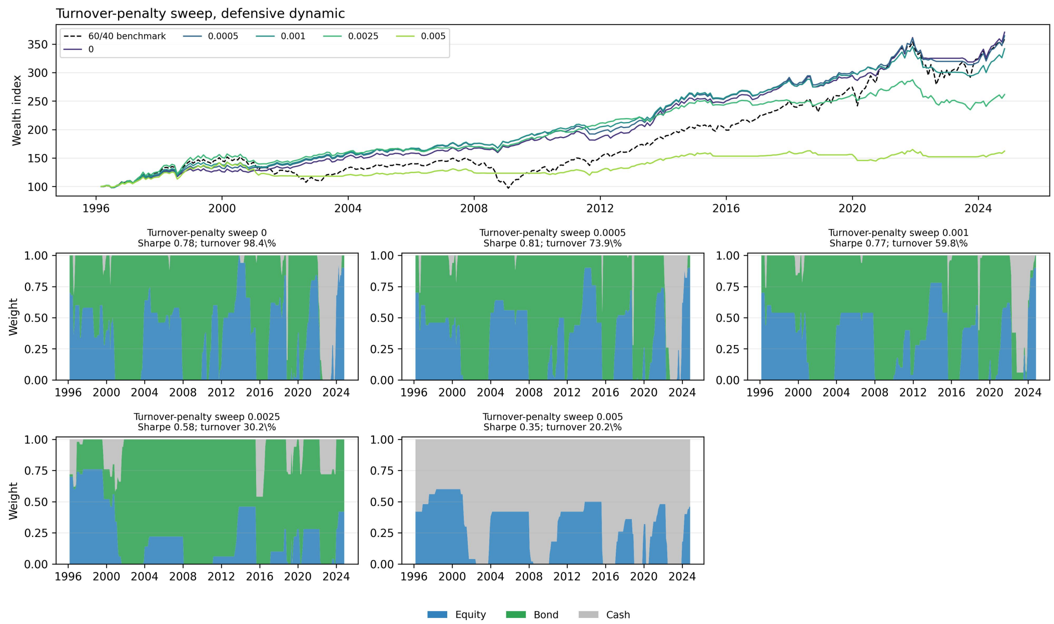


Figure H.40: Turnover-penalty sweep, defensive dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.34: Turnover-penalty sweep, defensive dynamic

Turnover penalty	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	4.8%	6.1%	0.78	-11.9%	-5.9%	370.86	98.4%
0.0005	4.7%	5.8%	0.81	-12.8%	-5.2%	364.66	73.9%
0.001	4.4%	5.8%	0.77	-14.9%	-5.5%	341.89	59.8%
0.0025	3.5%	6.1%	0.58	-18.3%	-6.4%	261.50	30.2%
0.005	1.8%	5.2%	0.35	-17.1%	-5.9%	162.19	20.2%
0.0075	0.3%	2.7%	0.12	-18.4%	-3.3%	108.88	6.3%
0.01	0.1%	0.3%	0.35	-1.7%	-0.3%	103.05	0.1%
0.015	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.02	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.03	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.05	0.0%	0.0%		0.0%	0.0%	100.00	0.0%

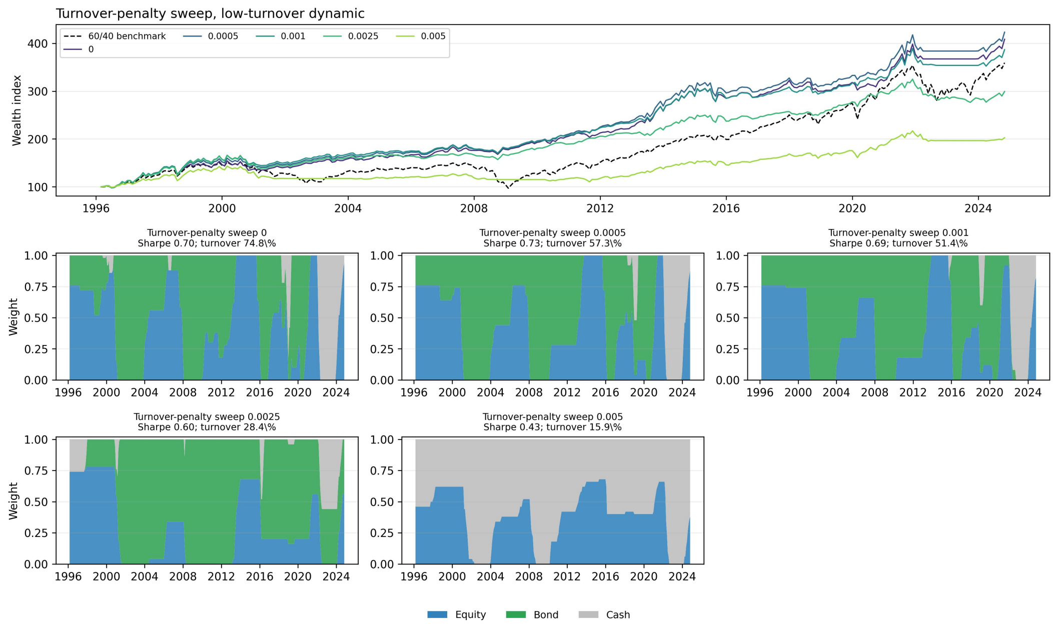


Figure H.41: Turnover-penalty sweep, low-turnover dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.35: Turnover-penalty sweep, low-turnover dynamic

Turnover penalty	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	5.2%	7.4%	0.70	-12.8%	-7.1%	408.43	74.8%
0.0005	5.3%	7.2%	0.73	-12.2%	-7.2%	423.26	57.3%
0.001	5.0%	7.2%	0.69	-13.0%	-7.1%	386.58	51.4%
0.0025	4.0%	6.8%	0.60	-16.7%	-6.9%	299.23	28.4%
0.005	2.6%	6.2%	0.43	-22.9%	-6.4%	202.56	15.9%
0.0075	1.2%	4.0%	0.30	-22.2%	-4.2%	137.74	6.7%
0.01	0.2%	0.7%	0.33	-3.4%	-0.7%	106.80	0.3%
0.015	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.02	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.03	0.0%	0.0%		0.0%	0.0%	100.00	0.0%
0.05	0.0%	0.0%		0.0%	0.0%	100.00	0.0%

Table H.36: Transaction-cost and turnover-penalty sensitivity

Strategy	Turnover penalty	5 bps	10 bps	25 bps
High-return dynamic	0.0000	0.77	0.74	0.67
High-return dynamic	0.0005	0.75	0.74	0.62
High-return dynamic	0.0010	0.75	0.72	0.53
High-return dynamic	0.0025	0.64	0.55	0.44
High-return dynamic	0.0050	0.59	0.56	0.47
High-return dynamic	0.0100	0.15	0.17	
Defensive dynamic	0.0000	0.75	0.78	0.69
Defensive dynamic	0.0005	0.79	0.81	0.61
Defensive dynamic	0.0010	0.81	0.77	0.57
Defensive dynamic	0.0025	0.63	0.58	0.43
Defensive dynamic	0.0050	0.38	0.35	0.20
Defensive dynamic	0.0100	0.15	0.35	
Low-turnover dynamic	0.0000	0.71	0.70	0.63
Low-turnover dynamic	0.0005	0.71	0.73	0.61
Low-turnover dynamic	0.0010	0.74	0.69	0.59
Low-turnover dynamic	0.0025	0.62	0.60	0.50
Low-turnover dynamic	0.0050	0.43	0.43	0.38
Low-turnover dynamic	0.0100	0.23	0.33	

Note. Entries are Sharpe ratios. For each dynamic strategy, the model, predictor set, risk aversion, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

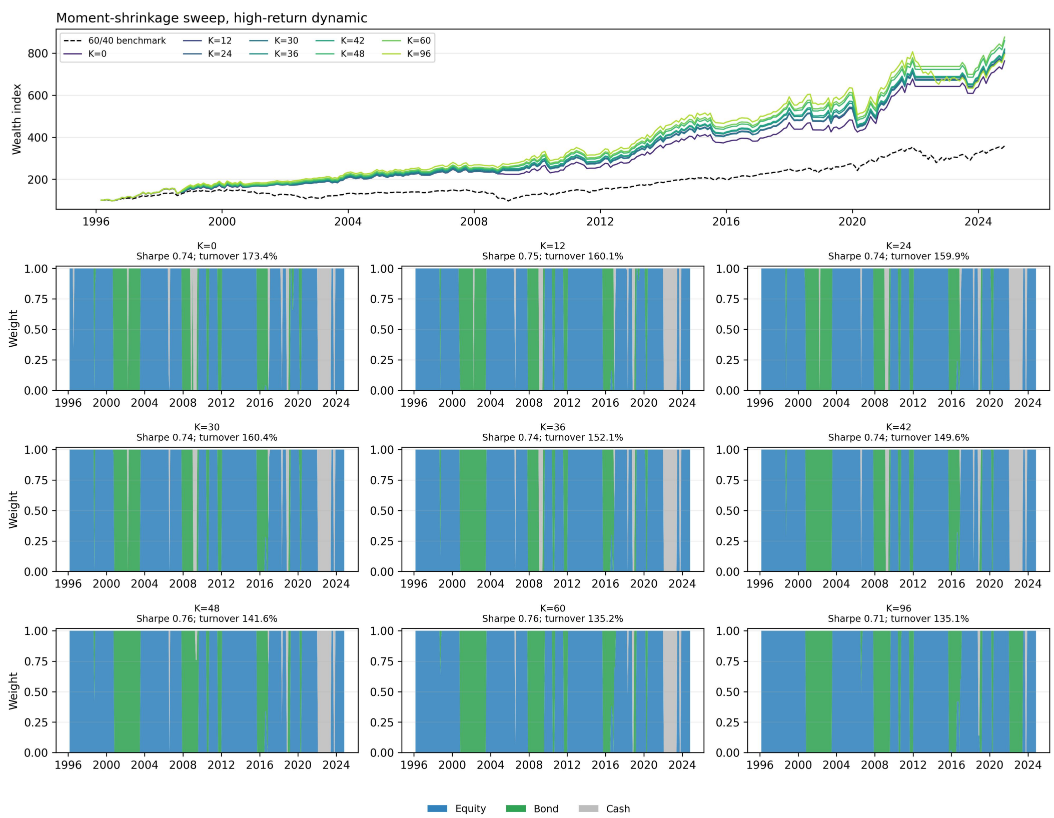


Figure H.42: Moment-shrinkage sweep, high-return dynamic

Note: The moment-shrinkage constant K varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.37: Moment-shrinkage sweep, high-return dynamic

K	Ann. ret.	Ann. vol.	Sharpe	Max DD	CVaR 1%	Final	Ann. turn.
0	7.6%	10.4%	0.74	-16.3%	-9.7%	763.30	173.4%
12	7.9%	10.5%	0.75	-16.3%	-9.7%	819.32	160.1%
24	7.8%	10.6%	0.74	-17.3%	-10.3%	804.69	159.9%
30	7.8%	10.6%	0.74	-18.2%	-10.5%	798.24	160.4%
36	7.9%	10.6%	0.74	-19.3%	-10.8%	814.18	152.1%
42	7.9%	10.7%	0.74	-20.1%	-11.0%	819.62	149.6%
48	8.1%	10.7%	0.76	-20.3%	-11.1%	859.80	141.6%
60	8.2%	10.7%	0.76	-20.3%	-11.1%	876.58	135.2%
96	7.8%	11.0%	0.71	-21.5%	-11.1%	794.32	135.1%

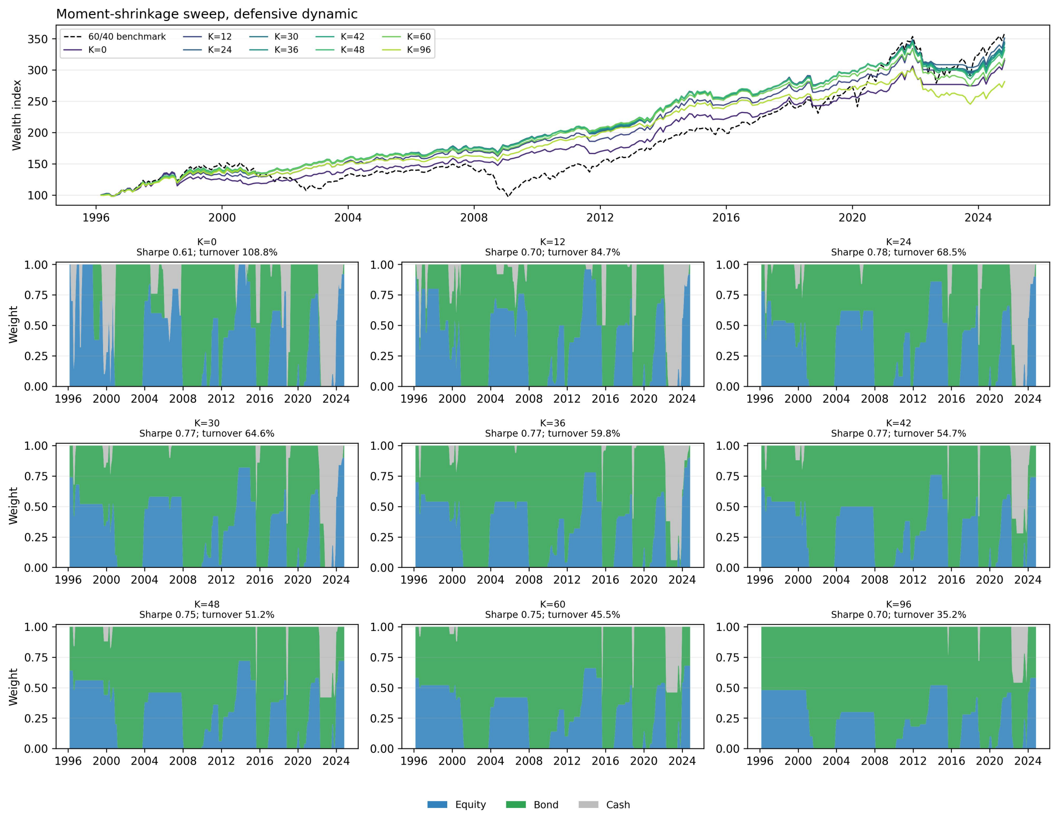


Figure H.43: Moment-shrinkage sweep, defensive dynamic

Note: The moment-shrinkage constant K varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.38: Moment-shrinkage sweep, defensive dynamic

K	Ann. ret.	Ann. vol.	Sharpe	Max DD	CVaR 1%	Final	Ann. turn.
0	4.2%	6.9%	0.61	-16.3%	-7.9%	315.35	108.8%
12	4.5%	6.4%	0.70	-12.9%	-6.7%	342.07	84.7%
24	4.5%	5.9%	0.78	-12.8%	-5.4%	351.08	68.5%
30	4.5%	5.8%	0.77	-14.1%	-5.4%	344.54	64.6%
36	4.4%	5.8%	0.77	-14.9%	-5.5%	341.89	59.8%
42	4.4%	5.7%	0.77	-16.0%	-5.5%	336.12	54.7%
48	4.3%	5.8%	0.75	-16.8%	-5.6%	331.75	51.2%
60	4.2%	5.6%	0.75	-17.7%	-5.5%	318.00	45.5%
96	3.7%	5.3%	0.70	-18.9%	-5.1%	281.36	35.2%

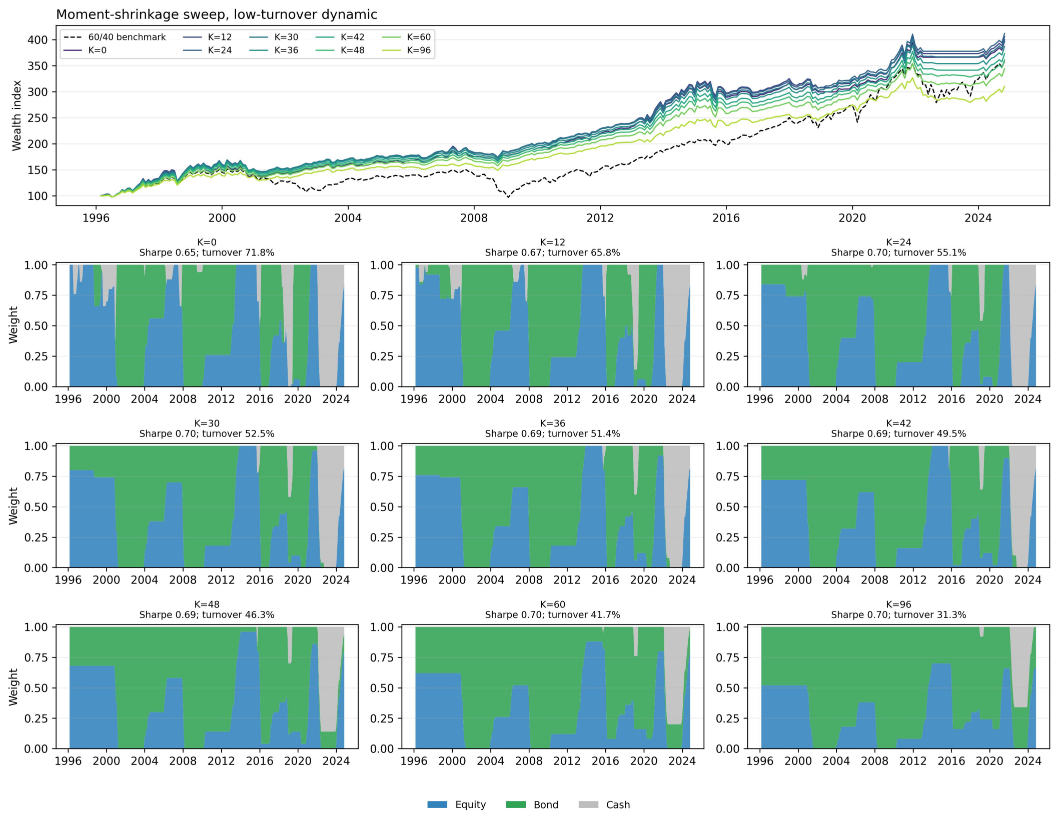


Figure H.44: Moment-shrinkage sweep, low-turnover dynamic

Note: The moment-shrinkage constant K varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

Table H.39: Moment-shrinkage sweep, low-turnover dynamic

K	Ann. ret.	Ann. vol.	Sharpe	Max DD	CVaR 1%	Final	Ann. turn.
0	5.1%	7.8%	0.65	-16.3%	-8.6%	396.71	71.8%
12	5.2%	7.7%	0.67	-15.0%	-8.2%	405.72	65.8%
24	5.2%	7.5%	0.70	-13.6%	-7.6%	411.59	55.1%
30	5.1%	7.3%	0.70	-12.9%	-7.3%	400.08	52.5%
36	5.0%	7.2%	0.69	-13.0%	-7.1%	386.58	51.4%
42	4.8%	7.0%	0.69	-13.2%	-6.9%	373.04	49.5%
48	4.7%	6.8%	0.69	-12.8%	-6.5%	360.99	46.3%
60	4.5%	6.4%	0.70	-12.0%	-6.0%	343.92	41.7%
96	4.1%	5.9%	0.70	-14.5%	-5.2%	310.10	31.3%

The sensitivity checks show how the same allocation rule changes when a single parameter is varied around each selected strategy. Higher risk aversion lowers equity exposure and reduces drawdowns in the defensive and low-turnover specifications. Probability smoothing reduces trading by slowing changes in the forecast probabilities. Higher turnover penalties reduce trading and increase cash weights, which lowers turnover but can reduce the Sharpe ratio when the penalty is too large. The moment-shrinkage sweep varies the amount of shrinkage applied to regime-conditional moments and the stock–bond correlation estimate. Larger K places more weight on unconditional pre-out-of-sample moments, while smaller K gives more weight to realised state-conditional samples.

H.6 Independent model-selection protocol

This table summarises which independent-side choices are fixed design choices, which are selected on validation or grid search, and which are used only as diagnostics. The purpose is to separate the reported independent specification from robustness checks and presentation-only diagnostics.

Component	Independent approach	Fixed in advance	Selected on validation/grid	Robustness or diagnostic only
Teacher feature set	Equity return–Sortino features and canonical bond features	Final reported feature blocks fixed before student prediction and allocation	No	Volatility/downside equity labels used only as diagnostics
Teacher jump penalty grid	Separate equity and bond jump-penalty grids	Grid fixed before final majority vote	Majority-vote label selected from fitted teacher paths	Path persistence and feature-separation plots reported as diagnostics
Distance norm	L^2 centroid and L^1-medoid teacher paths	Both norm families fixed as teacher candidates	Final label uses the combined majority vote	Separate L^2 and L^1 paths reported as diagnostics
Teacher majority vote	Final equity and bond labels combine teacher paths by majority vote	Voting rule fixed before student prediction	No refit after allocation outcomes	Vote-scope figures report candidate-path agreement
Student predictor sets	z-score predictors, smoothed z-score predictors, and their union	Transformations fixed before model fitting	Predictor-set comparison reported out of sample	Feature-importance figures are diagnostic
Student model hyperparameters	Logistic regression and random forest specifications	Model classes fixed before evaluation	Hyperparameters selected within the rolling validation design	Model-frequency table reports selected specifications
Student validation metrics	AUC, balanced accuracy, Brier score, log loss, and allocation-relevant diagnostics	Metrics fixed before reporting	Best reported specifications selected from rolling outputs	Full prediction diagnostics kept in appendix
Allocation rule	Long-only equity–bond allocation with cash allowed and an L^1 turnover penalty	Asset universe and allocation objective fixed before grid search	Candidate grid varies risk aversion, turnover penalty, transaction cost, and probability smoothing	Parameter-sensitivity appendix reports one-way and joint checks
Risk aversion	Values from the allocation grid	Grid fixed before headline reporting	Representative rows use selected grid points	Risk-aversion sweeps reported as sensitivity
Transaction cost	One-way transaction-cost grid	Grid fixed before headline reporting	Headline independent rows use 10 bps	Cost and turnover-penalty sensitivity reported in appendix
Turnover penalty	L^1 turnover penalty in the allocation objective	Grid fixed before headline reporting	Representative rows use role-specific penalty values	Turnover-penalty sweeps reported as sensitivity
Headline allocation rows	High-return, defensive, and low-turnover representative rows	Roles fixed for presentation	Rows selected as representative points from the allocation grid	Full-grid ranking rules are reported only as sensitivity diagnostics

Table H.40: Independent-side model-selection protocol. The table separates fixed design choices, validation or grid-selected choices, and robustness or diagnostic components.

I Coupled allocation: details and sensitivity

The investable universe is the equity leg, the bond leg, and cash. Weights are long-only with a gross cap of 1.0 ($w_{\text{cash}} \in [0, 1]$) and are chosen on a 0.05 grid. The COUPLED allocator uses annualised regime-conditional moments and an exponentially weighted stock–bond covariance with a 12-month half-life. This differs from the INDEPENDENT allocation grid and covariance estimator reported in Appendix H.

The CONST mapping uses the mean of the single predicted regime; the PROB mapping uses the probability mixture. Portfolios rebalance monthly and pay 10 bps on traded notional. Four static rules anchor the comparison: 60/40, 50/50, unconditional minimum-variance, and unconditional mean–variance. The regime-aware family comprises minimum-variance (γ=20) and mean–variance (γ=5) allocators under both mappings, together with a regime equal-weight overlay holding $(0.6, 0.4)$ in calm, $(0.72, 0.48)$ in normal, and $(0, 0.4)$ in stressed. The unconditional mean–variance rule uses γ=3 on an exponentially weighted mean, and the unconditional minimum-variance rule uses γ=20. Table I.41 reports the three headline strategies with the full metric set, and Table I.42 reports transaction-cost and turnover-penalty sensitivity for the balanced allocator.

Table I.41: Coupled allocation: headline results with full metrics, out of sample 1996-02 to 2024-10 (345 months). CVaR is the mean of the worst 1% of monthly excess returns; "Final" indexes 100 at the start; "Turn." is annual turnover.

Role	Strategy	Ann. exc.	Vol.	Sharpe	Max DD	$\text{CVaR}_{1\%}$	Final	Turn.
Defensive	Min-var (prob), logistic	+0.80%	1.24%	0.64	−2.0%	−1.03%	234.0	0.33
Balanced	Mean–var (prob), random forest	+2.74%	4.36%	0.63	−9.6%	−4.41%	398.5	0.82
High-return	Mean–var (prob), logistic	+4.70%	7.57%	0.62	−21.9%	−6.75%	661.2	0.49
Benchmark	60/40	+4.92%	9.61%	0.47	−33.9%	−8.57%	668.9	0.23

Table I.42: Coupled allocation: robustness of the probability-mixture mean–variance allocator on the logistic forecast to the one-way transaction cost and turnover penalty λ. Cells are out-of-sample Sharpe ratios.

λ	5 bps	10 bps	25 bps
0	0.608	0.594	0.555
0.001	0.616	0.605	0.570
0.01	0.640	0.635	0.617
0.1	0.320	0.319	0.316

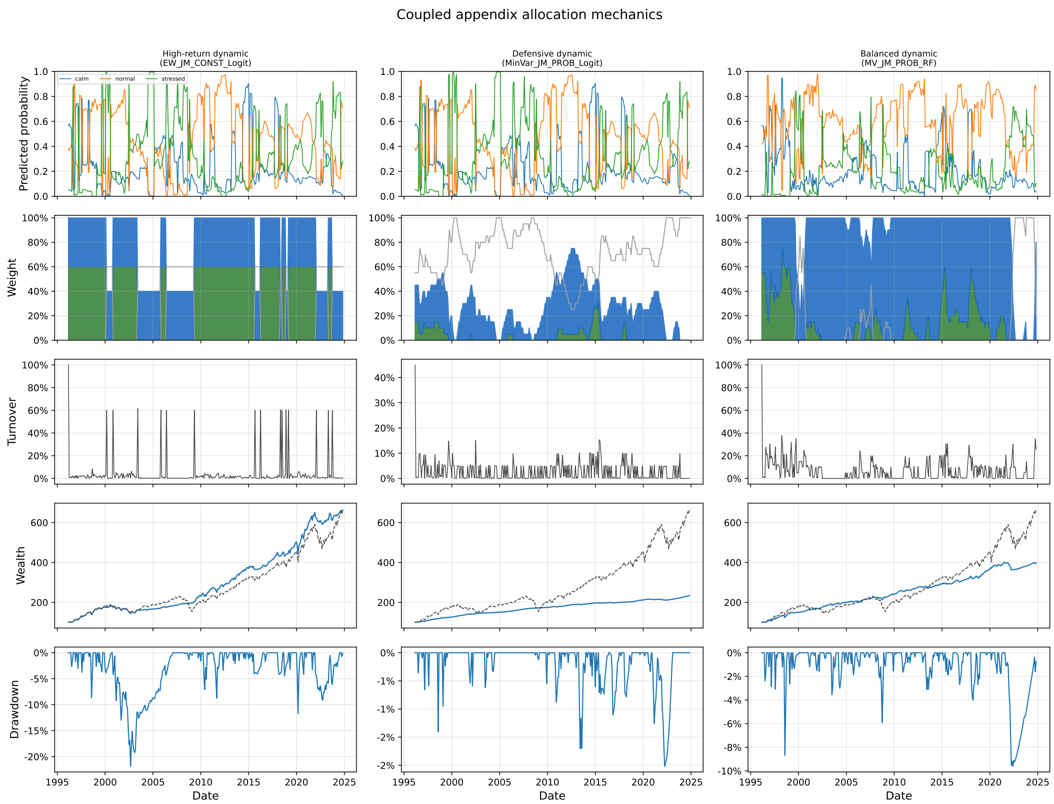


Figure I.45: Coupled allocation mechanics in detail for the three headline strategies. Rows, top to bottom: predicted regime probabilities; equity/bond/cash weights; monthly turnover; wealth index against the 60/40 (dashed); drawdown.

Figure I.46: Coupled allocation: risk-aversion sweep of the probability-mixture mean–variance allocator on the logistic forecast.

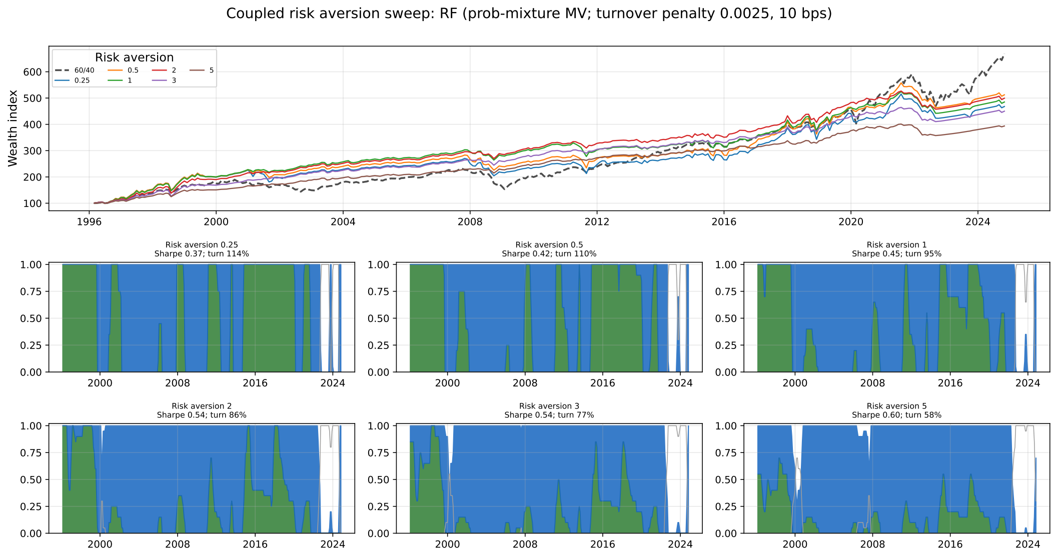


Figure I.47: Coupled allocation: risk-aversion sweep on the random-forest forecast.

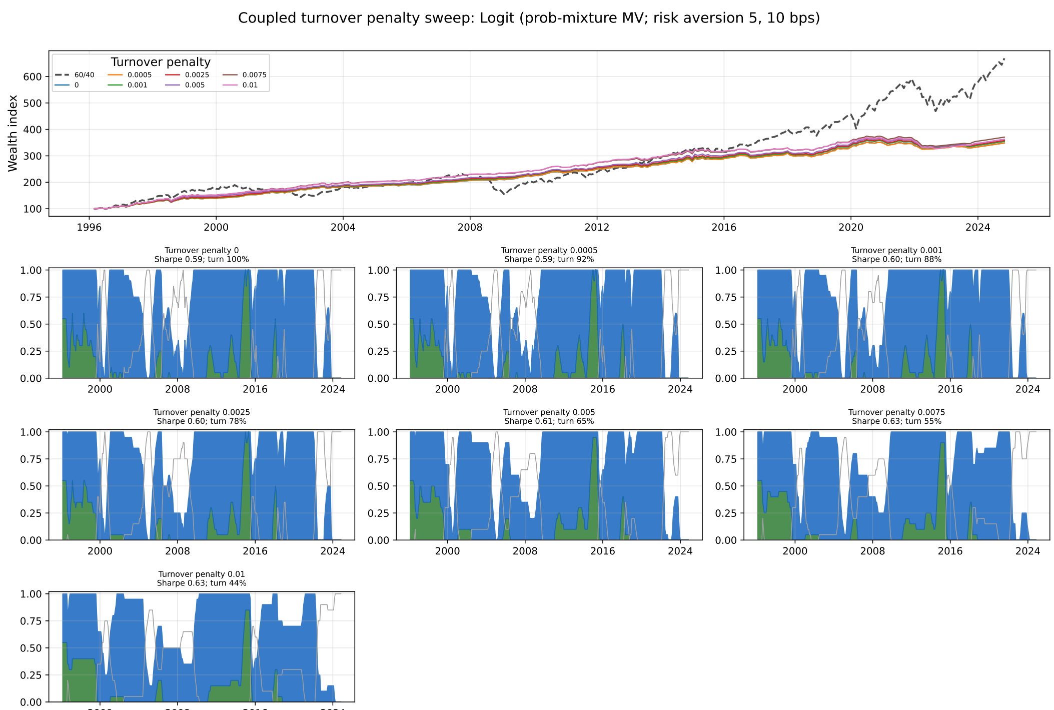


Figure I.48: Coupled allocation: turnover-penalty sweep on the logistic forecast (risk aversion fixed at 5).

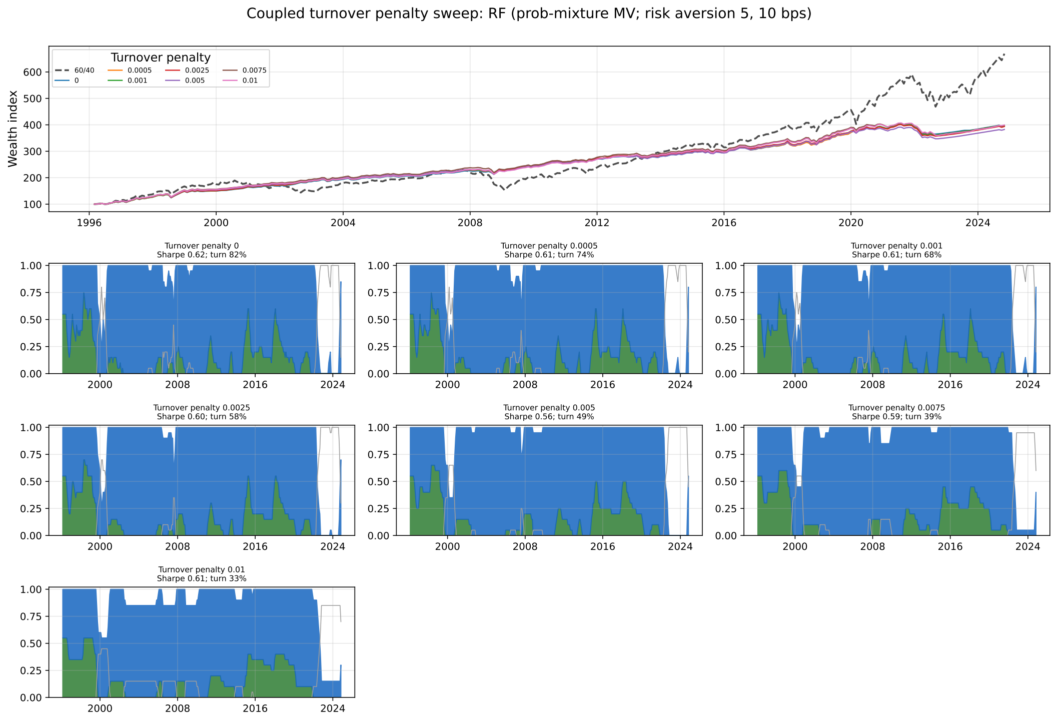


Figure I.49: Coupled allocation: turnover-penalty sweep on the random-forest forecast.

I.1 Coupled allocation: additional benchmarks and parameter sensitivity

Table I.43 reports the coupled static and unconditional benchmarks over the out-of-sample span, and Table I.44 reports the sensitivity of the probability-mixture mean–variance allocator to the moment-shrinkage constant K (the headline strategies use $K = 36$). Figures I.50–I.61 report one-parameter sweeps around the three selected strategies (high-return, defensive, low-turnover), matching the independent sweeps in Appendix H.

Table I.43: Coupled allocation: static and unconditional benchmarks, out of sample 1996-02 to 2024-10. Annualised excess returns net of 10 bps one-way costs; "Final" indexes 100 at the start; "Turn." is annual turnover.

Benchmark	Ann. exc.	Vol.	Sharpe	Max DD	Final	Turn.
60/40	+4.94%	9.61%	0.514	−33.8%	672.7	0.03
50/50	+4.46%	8.24%	0.541	−28.3%	607.9	0.03
Equity-only	+6.85%	15.41%	0.445	−52.6%	941.0	0.03
Bond-only	+2.07%	4.07%	0.509	−17.2%	330.2	0.03
Min-variance (uncond.)	−0.00%	0.00%	—	0.0%	186.6	0.00
Vol-targeted 60/40	+3.26%	6.91%	0.472	−29.8%	443.5	0.17

Table I.44: Coupled allocation: sensitivity of the probability-mixture mean–variance allocator to the moment-shrinkage constant K (annualised, net of 10 bps one-way costs). The headline strategies use $K = 36$.

Shrinkage K	Ann. return	Ann. vol.	Sharpe	Ann. turnover
0	+2.09%	3.95%	0.529	88.3%
12	+2.26%	4.00%	0.564	85.2%
24	+2.40%	4.01%	0.598	78.3%
30	+2.38%	3.91%	0.608	76.5%
36	+2.34%	3.88%	0.604	76.5%
42	+2.28%	3.87%	0.589	74.1%
48	+2.25%	3.84%	0.585	71.3%
60	+2.28%	3.71%	0.616	67.3%
96	+2.22%	3.41%	0.652	53.6%

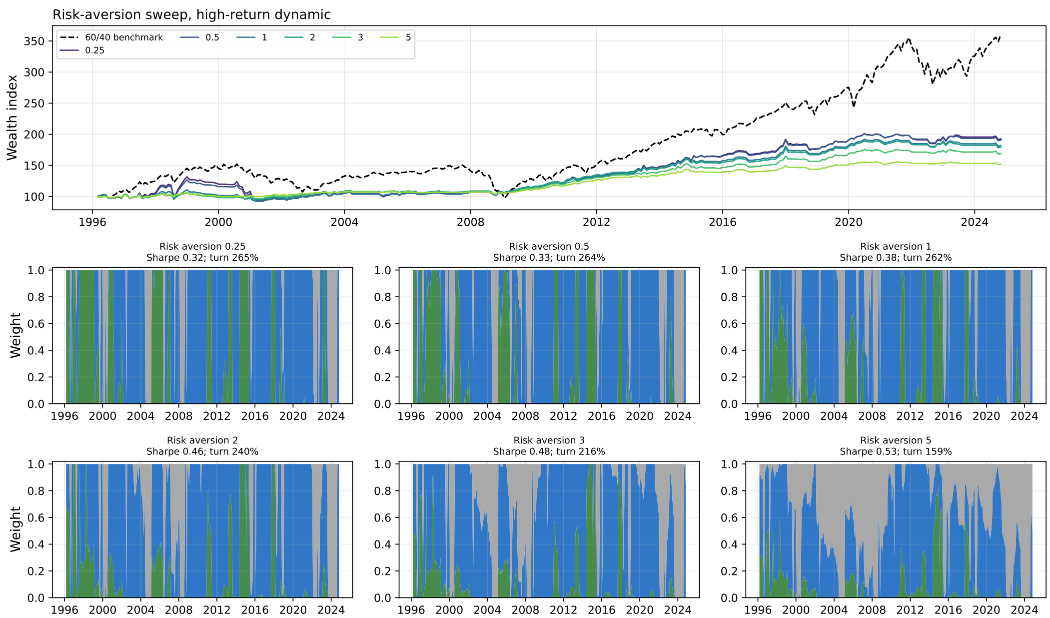


Figure I.50: Coupled allocation: risk-aversion sweep around the high-return strategy.

Figure I.51: Coupled allocation: risk-aversion sweep around the defensive strategy.

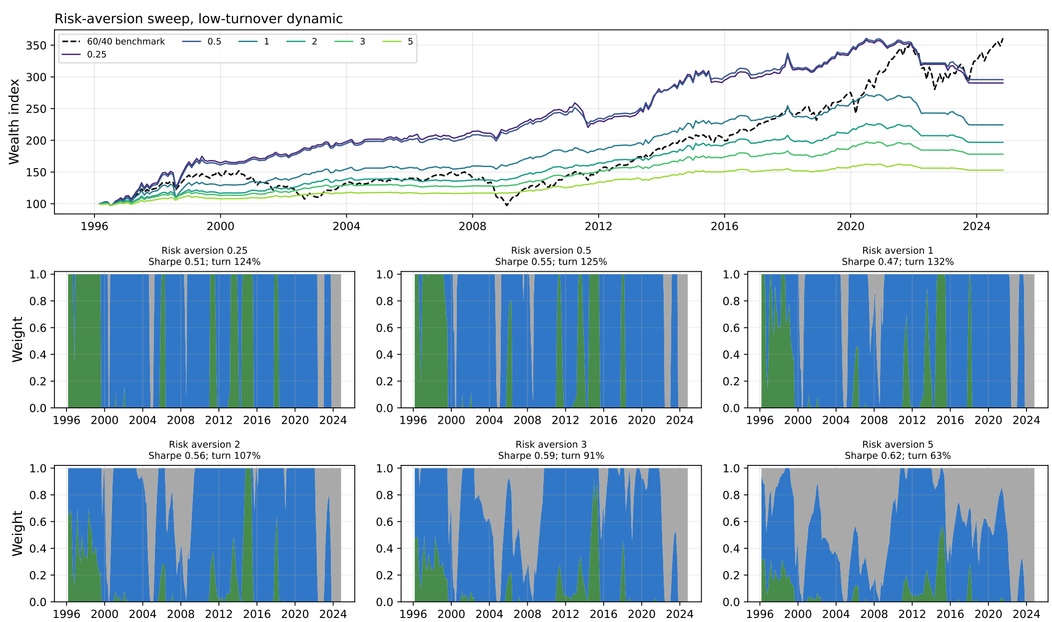


Figure I.52: Coupled allocation: risk-aversion sweep around the low-turnover strategy.

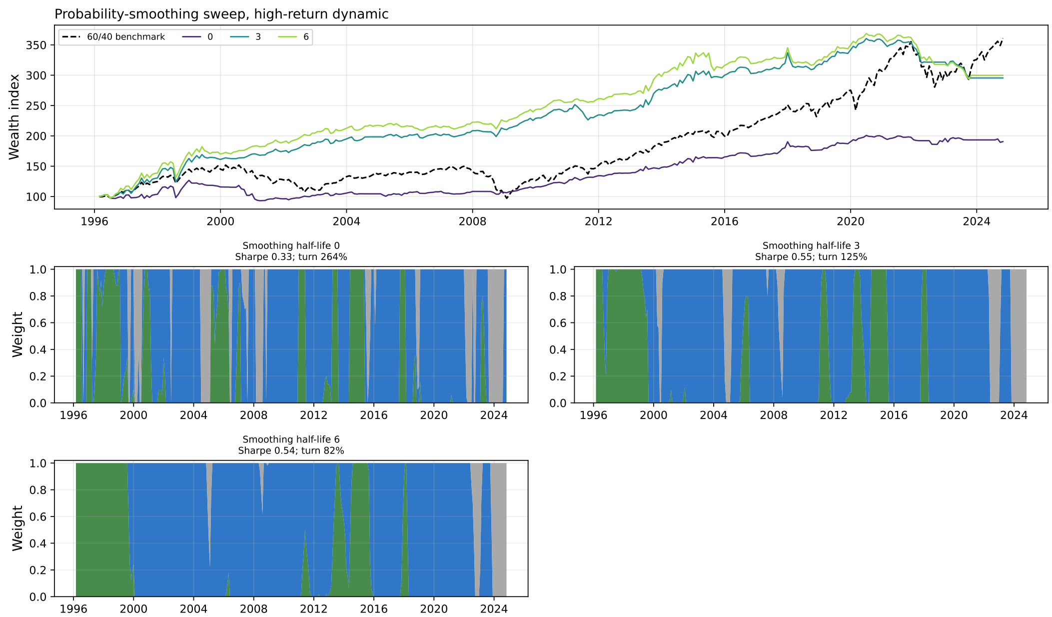


Figure I.53: Coupled allocation: probability-smoothing sweep around the high-return strategy.

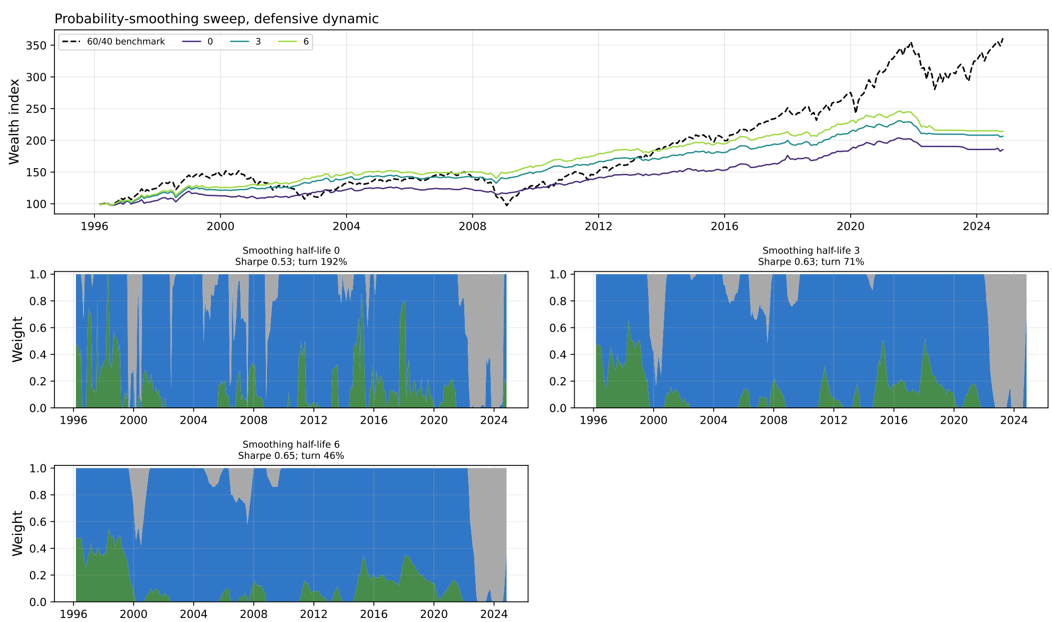


Figure I.54: Coupled allocation: probability-smoothing sweep around the defensive strategy.

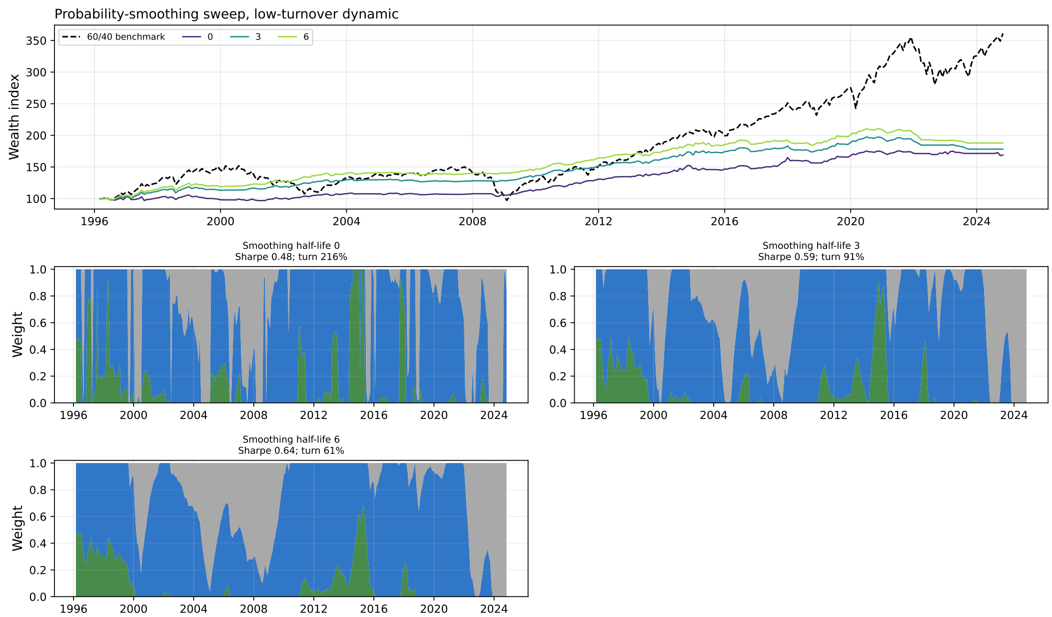


Figure I.55: Coupled allocation: probability-smoothing sweep around the low-turnover strategy.

Figure I.56: Coupled allocation: turnover-penalty sweep around the high-return strategy. The turnover-penalty axis is the coefficient on $\|\Delta w\|_1$ in the coupled objective (a τ-alone penalty), distinct from the independent $\tau \times c$ convention.

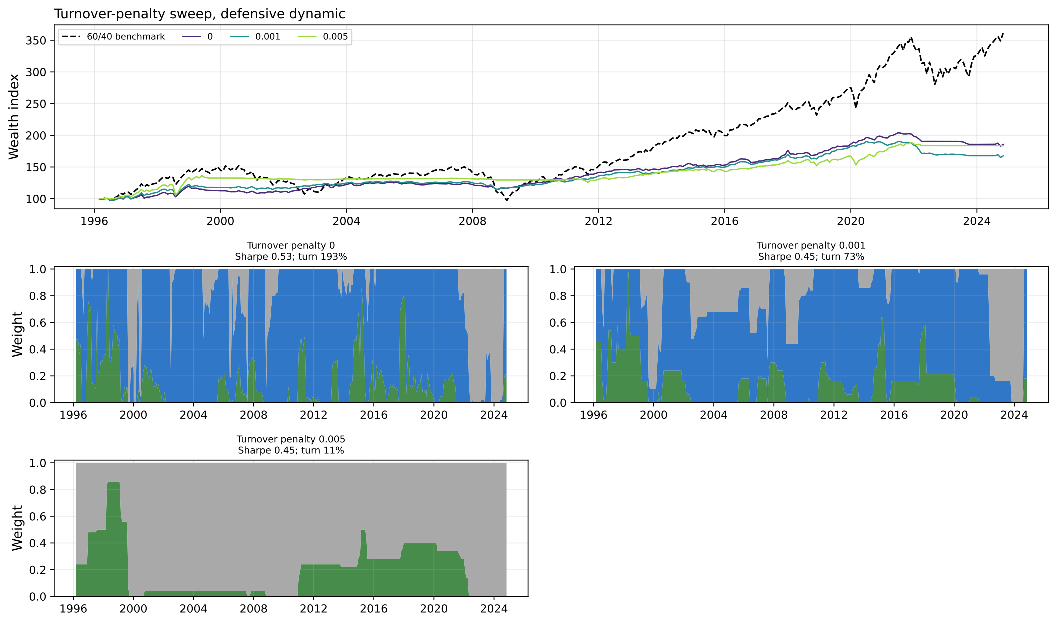


Figure I.57: Coupled allocation: turnover-penalty sweep around the defensive strategy. The turnover-penalty axis is the coefficient on $\|\Delta w\|_1$ in the coupled objective (a τ-alone penalty), distinct from the independent $\tau \times c$ convention.

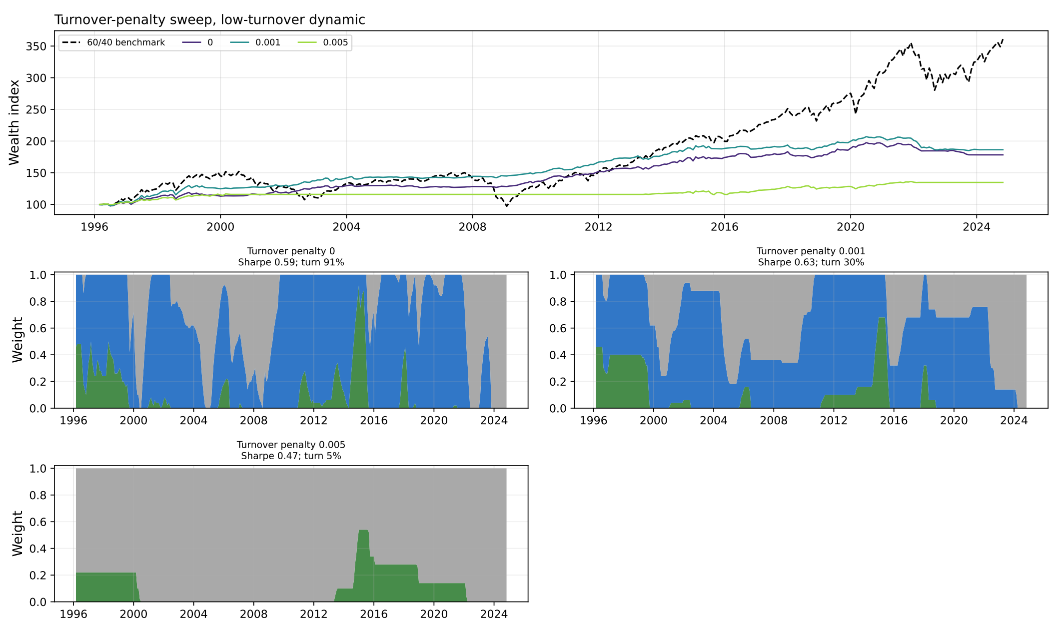


Figure I.58: Coupled allocation: turnover-penalty sweep around the low-turnover strategy. The turnover-penalty axis is the coefficient on $\|\Delta w\|_1$ in the coupled objective (a τ-alone penalty), distinct from the independent $\tau \times c$ convention.

Figure I.59: Coupled allocation: moment-shrinkage sweep around the high-return strategy.

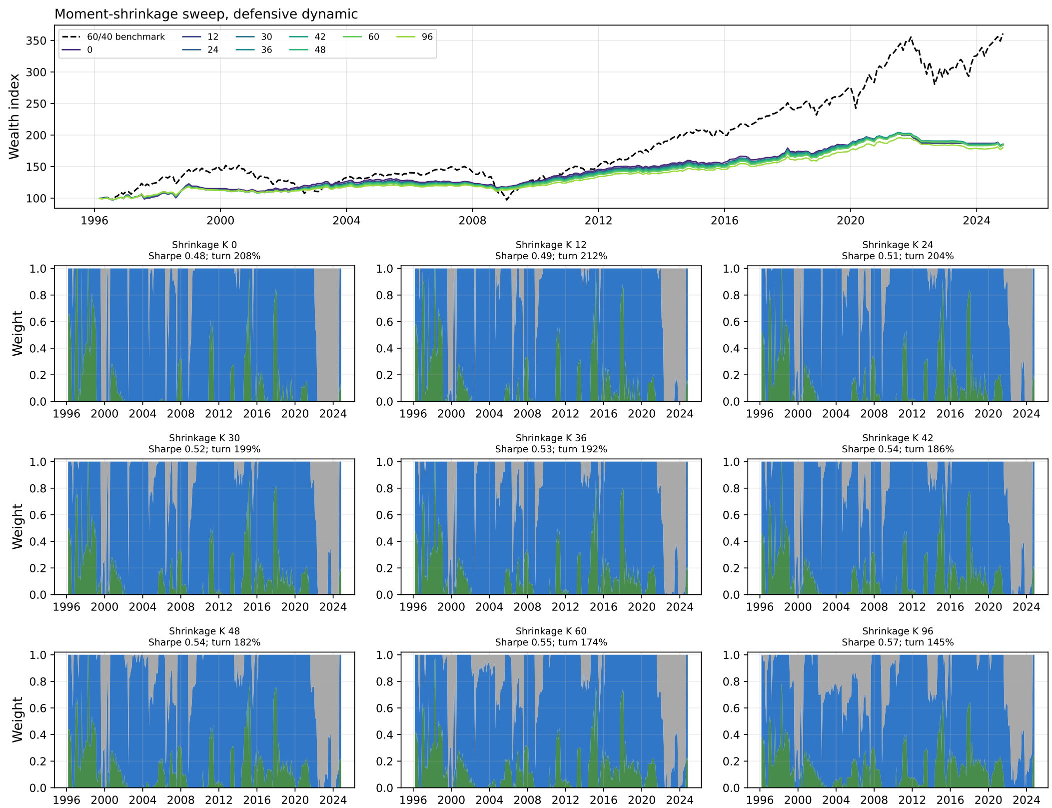


Figure I.60: Coupled allocation: moment-shrinkage sweep around the defensive strategy.

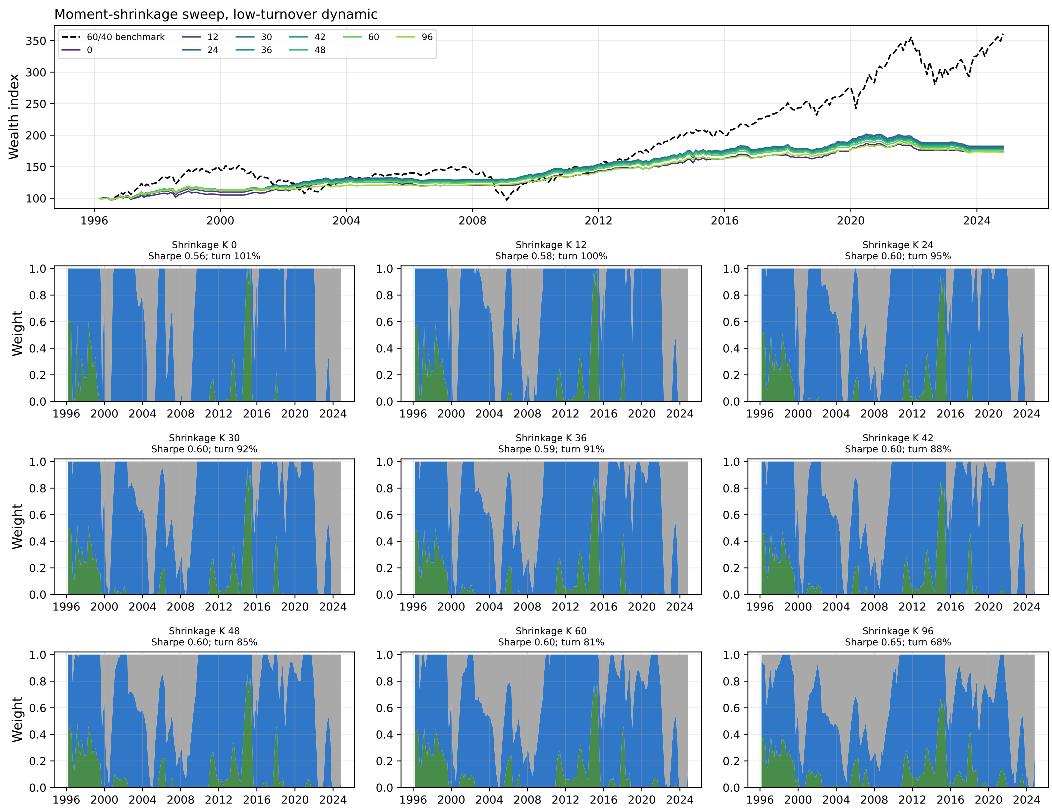


Figure I.61: Coupled allocation: moment-shrinkage sweep around the low-turnover strategy.

I.2 Coupled model-selection protocol

This table summarises which coupled-side choices are fixed design choices, which are selected on validation or grid search, and which are used only as diagnostics. As with the independent protocol in Table H.40, the purpose is to separate the reported coupled specification from robustness checks and presentation-only diagnostics. The coupled teacher produces a single joint three-state label (calm, normal, stressed), and the student forecasts that joint label directly.

Component	Coupled approach	Fixed in advance	Selected on validation/grid	Robustness or diagnostic only
Teacher feature set	Joint equity and bond return-based features feeding one three-state label	Final reported feature blocks fixed before student prediction and allocation	No	Per-leg feature-separation summaries used only as diagnostics
Teacher state structure	Three joint regimes (calm, normal, stressed) built from the two per-leg state bits	Three-state joint structure fixed before estimation	No; the mapping from per-leg bits to the joint label is fixed	Per-leg state economics reported as diagnostics
Teacher jump penalty grid	Shared jump-penalty grid λ with coupling parameter η linking the legs	Grid fixed before final majority vote	Majority-vote label selected from fitted teacher paths	Jump-penalty diagnostics reported across λ and η
Distance norm	L^2 centroid and L^1-medoid teacher paths	Both norm families fixed as teacher candidates	Final label uses the combined majority vote across members	Separate L^2 and L^1 paths reported as diagnostics
Teacher ensemble and majority vote	Final joint label combines the 22 ensemble members by majority vote	Voting rule fixed before student prediction	No refit after allocation outcomes	Member raster and vote-scope figures report candidate-path agreement
Student predictor sets	Lagged macro-financial predictors for the next-month joint regime	Transformations fixed before model fitting	Single reported predictor design evaluated out of sample	Individual-predictor importance figures are diagnostic
Student model hyperparameters	Three-class logistic regression and random forest specifications	Model classes fixed before evaluation	Hyperparameters selected within the rolling validation design	Most-frequent-specification table reports selected settings
Student validation metrics	Macro one-vs-rest AUC, balanced accuracy, multiclass Brier score, and log loss	Metrics fixed before reporting	Best reported specifications selected from rolling outputs	Full three-class prediction diagnostics kept in appendix
Allocation rule	Long-only equity–bond allocation with cash allowed and an L^1 turnover penalty, mapping regime probabilities to moments	Asset universe and allocation objective fixed before grid search	Candidate grid varies risk aversion, turnover penalty, transaction cost, and probability smoothing	Parameter-sensitivity appendix reports one-way checks
Probability-to-moment mapping	CONST mapping (single predicted regime) and PROB mapping (probability mixture)	Both mappings fixed as candidates	Headline rows use the PROB mapping	CONST mapping retained as a diagnostic comparison
Risk aversion	Minimum-variance (γ=20) and mean–variance (γ=5) allocators from the grid	Grid fixed before headline reporting	Representative rows use selected grid points	Risk-aversion sweeps reported as sensitivity
Transaction cost and turnover penalty	One-way transaction-cost grid with an L^1 turnover penalty (a τ-alone penalty)	Grid fixed before headline reporting	Headline coupled rows use 10 bps	Cost and turnover-penalty sensitivity reported in appendix
Headline allocation rows	Defensive, balanced, and high-return representative rows	Roles fixed for presentation	Rows selected as representative points from the allocation grid	Full-grid ranking rules are reported only as sensitivity diagnostics

Table I.45: Coupled-side model-selection protocol. The table separates fixed design choices, validation or grid-selected choices, and robustness or diagnostic components, similar to the independent protocol in Table H.40.